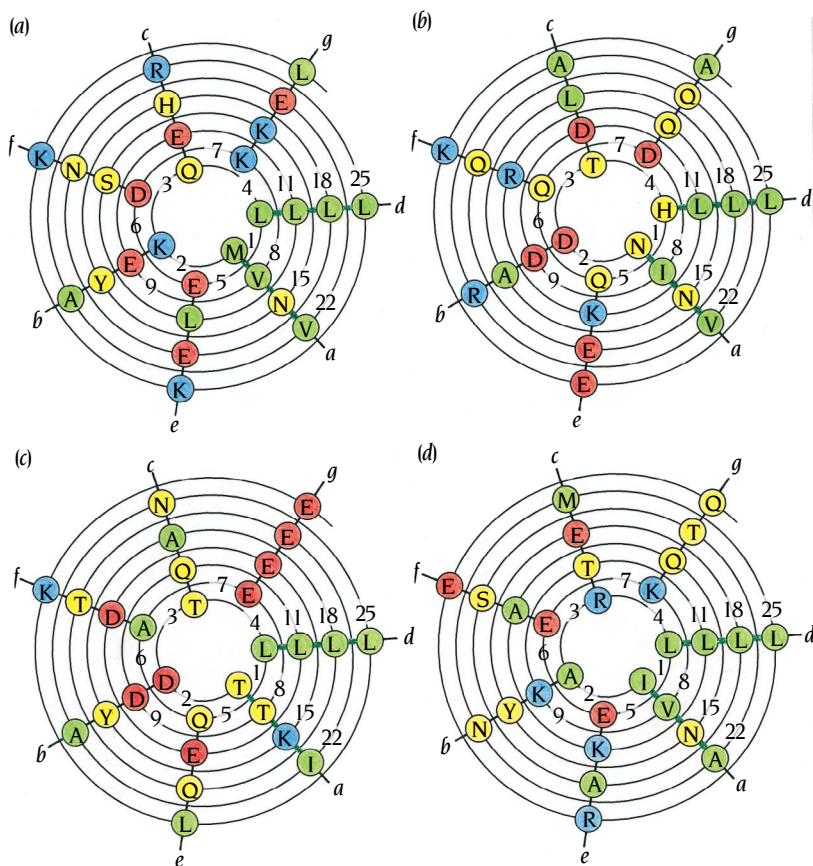
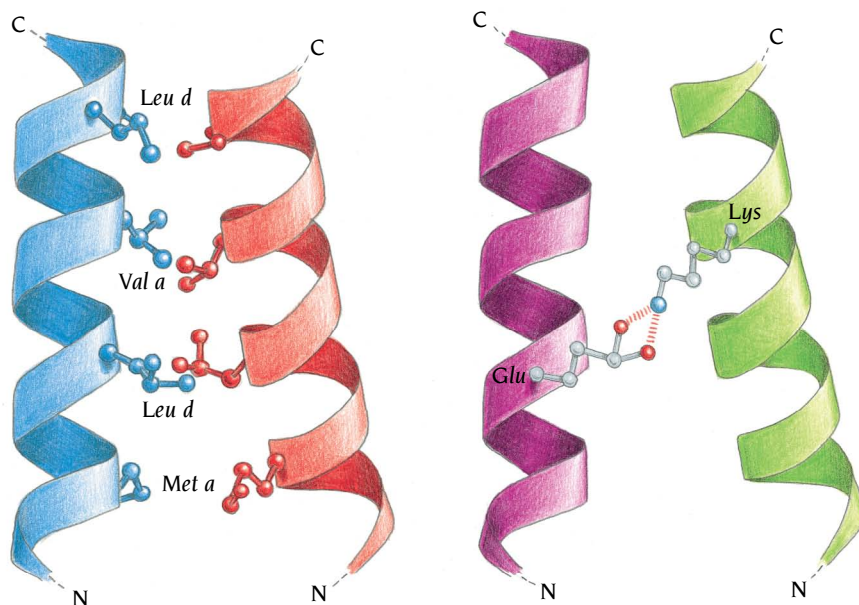




**Figure 10.17** Amino acid sequences, represented as  $\alpha$  helical wheels with 3.5 residues per turn, of a region of 28 residues from the DNA-binding domains of the transcription factors (a) GCN4, (b) Max, (c) Fos and (d) Jun. Positively charged residues are blue, negatively charged red; polar residues are yellow and hydrophobic, green. Residues with a spacing of seven residues are connected into spikes of the wheel which are labeled *a* to *g*. Side chains from each residue in a spike form a ridge on the outside of the  $\alpha$  helix which is parallel to the helical axis. Spike *d* in all four sequences forms a heptad pattern of leucine residues that is characteristic of a leucine zipper coiled-coil  $\alpha$ -helical structure.

emerges (Figure 10.17). In all of them there is a region of about 30 residues where the sequence can be arranged in modules of seven residues, and in almost all of these modules the fourth residue is leucine (position *d* in Figure 10.17), hence the name leucine zipper. In addition, the first residue of each module is frequently hydrophobic (position *a* in Figure 10.17). As was described in Chapter 3, the first structure determination of a synthetic peptide containing such modules showed that the peptide dimerizes and forms two parallel coiled-coil  $\alpha$  helices with a helical repeat of 3.5 residues per turn, so that the interaction pattern of side chains between the helices repeats integrally every seven residues (see Figure 3.3). The hydrophobic side chains in positions *a* and *d* of the heptad repeats form a hydrophobic core between the helices in this coiled-coil with the leucine residues facing each other (Figure 10.18a). This hydrophobic core is one major determinant of the stability of the dimer. The side chains that are immediately outside this core (positions *e* and *g* in Figure 10.17) are frequently charged and can either promote dimer formation by forming complementary charge interactions between the monomers, or prevent dimer formation by the repulsion of like charges (Figure 10.18b). Leucine zippers can form either homodimers or heterodimers in which the two monomers are either the same or different transcription factors, respectively.

The proto-oncogenic transcription factors Fos and Jun provide an illustrative example of differential dimer formation. These proteins are found in the transcription factor complex AP1 (active gene regulatory protein), which promotes transcription of many different genes including some that induce cell proliferation. The mammalian genome contains a family of related genes for Jun proteins, including c-Jun which is the cellular counterpart of the viral oncogene v-Jun. Members of this family can form both homodimers and heterodimers. In contrast, Fos cannot form homodimers and does not by itself bind to DNA. From the sequence of Fos in Figure 10.17c it is obvious that dimer formation is prevented by the strong charge repulsion of the five glutamic acid residues in the *e* and *g* positions with no compensating positive charge. However, Fos can form a heterodimer with Jun due to the



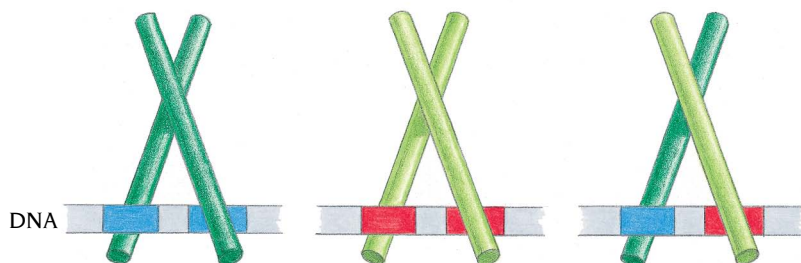
**Figure 10.18** Side-chain interactions in the leucine zipper structure. (a) The hydrophobic side chains in spikes *a* and *d* (see Figure 10.17) form a hydrophobic core between the two coiled  $\alpha$  helices. (b) Charged side chains in spikes *e* and *g* can promote dimer formation by forming complementary charge interactions between the two  $\alpha$  helices.

complementary positive charges in the *e* and *g* positions of Jun (Figure 10.17d). An x-ray structure determination of the Fos-Jun heterodimer bound to DNA by the group of Stephen Harrison has confirmed the existence of such *g/e* salt bridges (see Figure 10.18b). The heterodimer binds to DNA with the same target specificity as the Jun homodimer and with an affinity to the AP1 binding site that is 10 times higher.

The ability of the leucine zipper proteins to form heterodimers greatly expands the repertoire of DNA-binding specificities that these proteins can display. As illustrated in Figure 10.19, for example, three distinct DNA-binding specificities could, in principle, be generated from two types of monomer, while six could be created from three types of monomer and so on. This is an example of combinatorial control, in which combinations of proteins, rather than individual proteins, control a cellular process. It is one of the most important mechanisms used by eucaryotic cells to control gene expression.

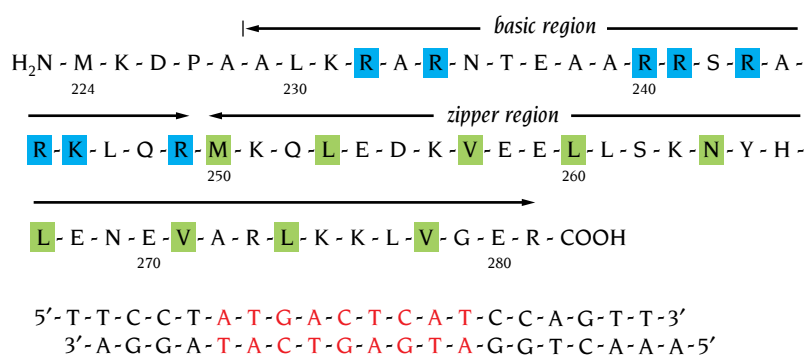
### **The GCN4 basic region leucine zipper binds DNA as a dimer of two uninterrupted $\alpha$ helices**

The yeast protein GCN4 is a typical member of the basic region-leucine zipper (b/Zip) family of transcription factors, which has more than 50 known members from yeast, mammalian and plant cells. Monomeric GCN4, which is 281 amino acids long, binds specifically to the promoter regions of more than 30 genes that specify enzymes involved in amino acid biosynthesis, inducing their transcription in response to amino acid starvation. The dimerization and DNA-binding properties reside in a C-terminal region comprising about 55 amino acids. The amino acid sequence shows that this region of GCN4 is divided into a basic region of about 20 residues followed



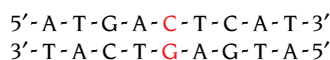
**Figure 10.19** Heterodimerization of leucine zipper proteins can alter their DNA-binding specificity. Leucine zipper homodimers bind to symmetric DNA sequences, as shown in the left-hand and center drawings. These two proteins recognize different DNA sequences, as indicated by the red and blue regions in the DNA. The two different monomers can combine to form a heterodimer that recognizes a hybrid DNA sequence, composed of one red and one blue region. (Adapted from B. Alberts et al, *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 1994.)

(a) Amino acid and DNA sequences used in the structure determination

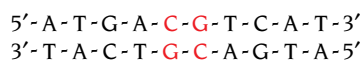


**Figure 10.20** (a) Amino acid sequence of the DNA-binding domain of the transcription factor GCN4 and nucleotide sequence of the DNA used for the x-ray structure determination of the complex. The positively charged residues arginine and lysine in the basic region that are involved in DNA binding are colored blue. The residues in the *a* and *d* positions that form the hydrophobic core of the leucine zipper region are green. The pseudosymmetric nucleotide sequence to which GCN4 binds is colored red. (b) The consensus nucleotide sequence of the *in vivo* AP1 pseudo-symmetric DNA recognition sites of GCN4. (c) A symmetric nucleotide sequence to which GCN4 can also bind.

(b) AP1-binding site



(c) Symmetric GCN4-binding site



by a C-terminal leucine zipper region (Figure 10.20a). The basic region contains eight charged residues, mainly arginines, which are involved in DNA binding and which are disordered when the protein is in solution in the absence of DNA. The transcriptional activation function is localized to a short acidic region of about 40 residues around residue 100. Surprisingly, large portions of the protein-coding sequence outside the DNA-binding and activation regions can be deleted from the gene without significantly affecting transcriptional activation.

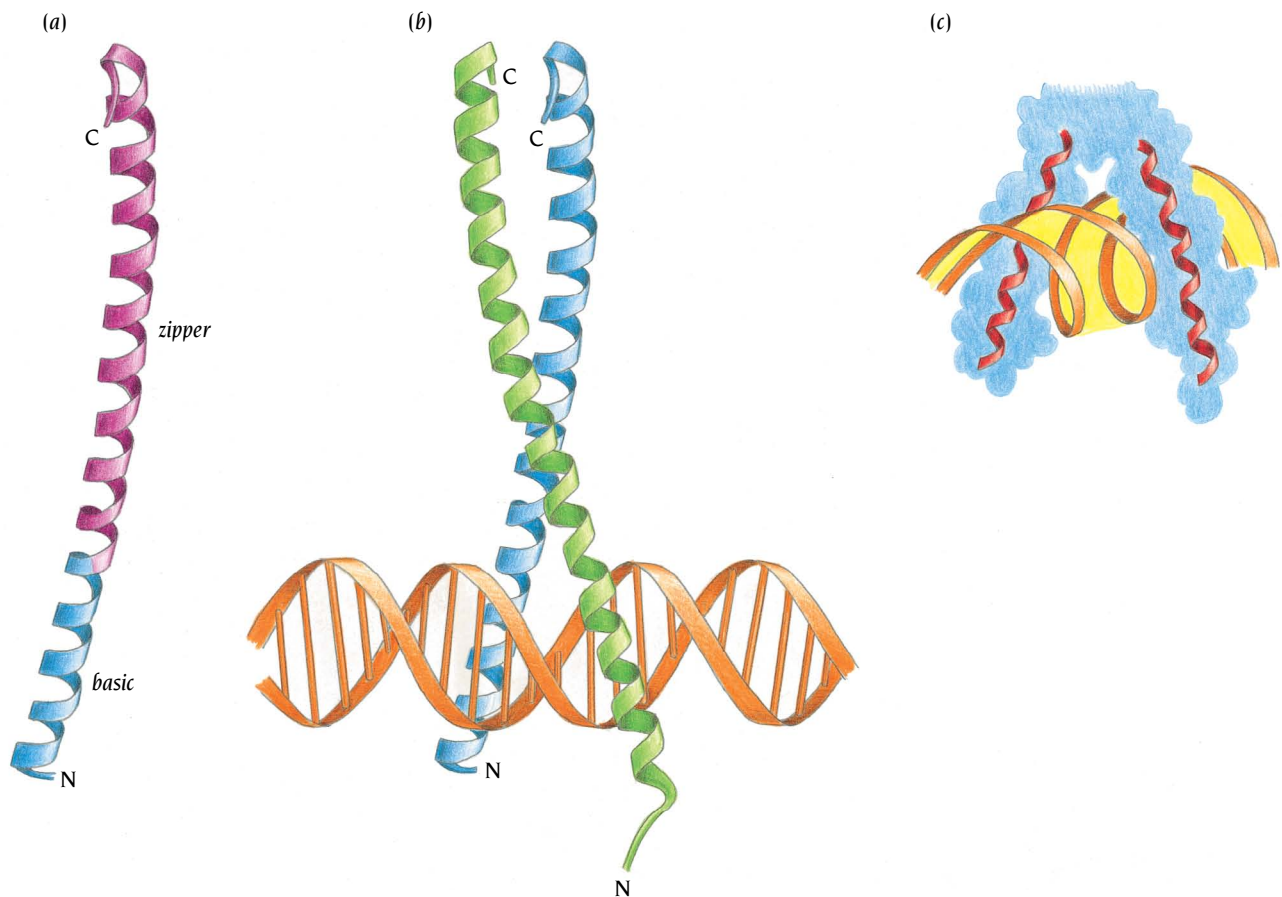
The DNA recognition sites of GCN4 *in vivo* are similar to those for the Fos/Jun heterodimer, the AP1 site, the consensus sequence of which is shown in Figure 10.20b. This sequence is pseudo-symmetric, but GCN4 also binds to a perfectly symmetric sequence, which has one base pair inserted in the middle of the recognition site (Figure 10.20c). The dimeric GCN4 molecule is therefore able to bind to the two half-sites in both these DNA recognition sequences even though they have spacer regions of different lengths. The group of Stephen Harrison at Harvard University has determined the crystal structure of the GCN4 b/zip region (the amino acid sequence shown in Figure 10.20a) complexed with a DNA fragment of 20 base pairs containing the pseudo-palindromic nucleotide sequence shown in Figure 10.20b.

The structure is in principle very simple (Figure 10.21). Each monomer of the GCN4 fragment forms a smoothly curved, continuous  $\alpha$  helix (see Figure 10.21a). The leucine zipper region of the monomers pack into a coiled coil, essentially identical to the isolated leucine zipper (see Figure 3.3). The two  $\alpha$  helices diverge from the dimer axis in a segment comprising the junction between the leucine zipper and the basic regions. This fork creates a smooth bend in each  $\alpha$  helix which displaces the basic regions away from the dimer interface so that they can pass through the major groove of DNA, with one  $\alpha$  helix on each side of the DNA (see Figure 10.21b and c). The DNA has essentially the B-DNA structure. Each basic region binds to one half-site with numerous contacts to the DNA, and the structure looks like  $\alpha$ -helical forceps gripping the major groove of DNA.

### GCN4 binds to DNA with both specific and nonspecific contacts

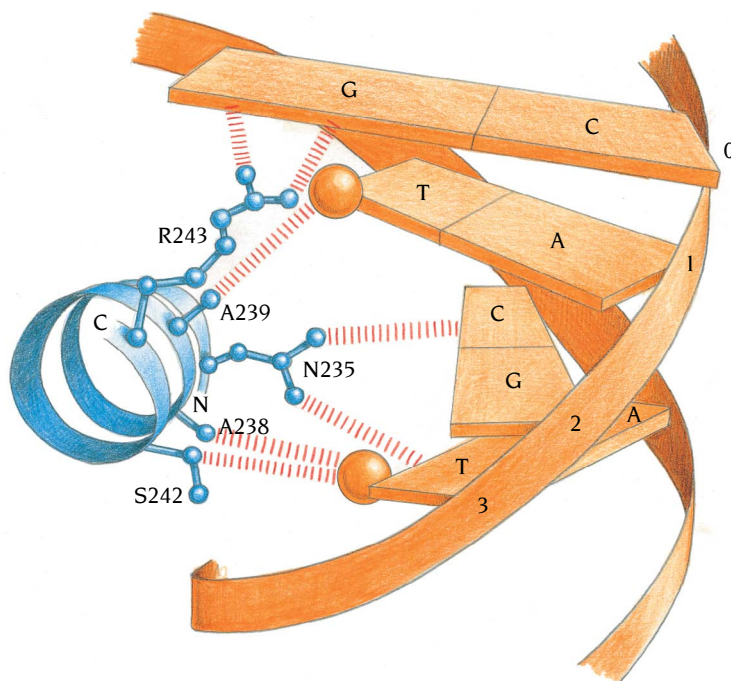
The consensus nucleotide sequence (see Figure 10.20b) used in the crystals is a symmetrized version of naturally occurring AP1 recognition sites, but GCN4 binds to this sequence with a high affinity. Each half-site in this DNA is bound to one monomer of the GCN4 dimer by both sequence-specific and





nonspecific contacts. In the same way in both half-sites, four amino acid side chains form sequence-specific contacts with bases (Figure 10.22). Asn 235, which is strictly conserved in all members of the b/zip family, is at the center of the interaction area. Its side chain forms two hydrogen bonds; the oxygen atom accepts a hydrogen bond from a nitrogen atom of the base C2, and the nitrogen atom of Asn 235 donates a hydrogen bond to an oxygen atom of T3. These contacts require that the  $\alpha$  helix of the GCN4 basic region

**Figure 10.21** The structure of a complex between the DNA-binding domain of GCN4 and a fragment of DNA. (a) Each monomer of the GCN4 domain forms a smoothly curved continuous  $\alpha$  helix comprising both the basic and the leucine zipper regions. (b) The monomers are held together in a dimer in the zipper region. They diverge from each other in the basic regions, which are bound to DNA in the major groove on opposite sides of the B-DNA fragment (c), like helical forceps gripping the DNA. (Adapted from T.E. Ellenberger et al., *Cell*, 71: 1223–1237, 1992.)



**Figure 10.22** Sequence-specific interactions between one of the  $\alpha$ -helical basic regions of GCN4 (blue) and bases in the DNA fragment (orange). The methyl groups of the thymine bases are shown as spheres.

lies deep in the major groove, and they specify two of the four base pairs in each half-site. Asn 235 lies in a hydrophobic pocket formed by the methyl side chains of alanines 238 and 239, which form hydrophobic contacts with the methyl groups of T3 and T1, respectively. In addition, the methylene group of Ser 242 contacts the methyl group of T3. These hydrophobic contacts specify the thymines of base pairs 1 and 3 of each half-site.

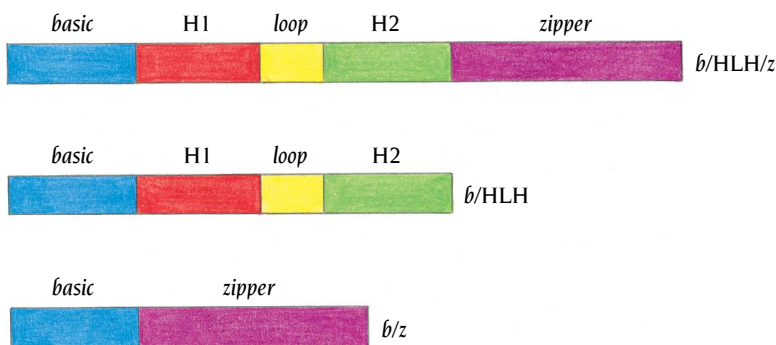
The asymmetry of the AP1 site causes GCN4 to contact the central base pair in an asymmetric manner. Arg 243 in one monomer (shown in Figure 10.22) donates hydrogen bonds to the guanine of the central G-C base pair in the bidentate manner described earlier for zinc fingers (see Figure 10.6). Arg 243 in the second monomer can not of course bind in the same way. Instead it forms nonspecific hydrogen bonds to phosphate oxygen atoms in the central region of the site. Despite this local asymmetry in the center of the binding site, the positions of the two  $\alpha$  helices in the major groove of the DNA and the side chain contacts to bases T1, C2, and T3 are the same in the two half-sites.

The position of the basic region of the protein on the DNA is also achieved by a number of basic and polar residues donating hydrogen bonds to phosphate oxygens of the DNA backbone. All the eight positively charged residues of the basic region are involved in these contacts, most of which are conserved between the half-sites, with the exception of those near the center of the binding site.

How does the GCN4 dimer bind to half-sites comprising T1, C2, and T3 when these half sites are separated by two nucleotides instead of one? The group of Tim Richmond at ETH in Zurich has answered this question by studying the crystal structure of GCN4 complexed with a DNA fragment containing the sequence shown in Figure 10.20c. They found that flexibility in both the DNA and the protein allows the GCN4 basic regions to interact with the major groove of each half-site in essentially equivalent manners in the two complexes. The basic region helices spread apart about 10 degrees, and the DNA bends towards the coiled-coil region and is inserted a bit further up into the fork. As a consequence, the actual specific protein–base pair contacts are all conserved. The combination of protein and DNA flexibility apparently allows GCN4 and other b/zip proteins to tolerate different spacings between half-sites.

### *The HLH motif is involved in homodimer and heterodimer associations*

The coiled-coil structure of the leucine zipper motif is not the only way that homodimers and heterodimers of transcription factors are formed. As we saw in Chapter 3 when discussing the RNA-binding protein ROP, the formation of a four-helix bundle structure is also a way to achieve dimerization, and the helix-loop-helix (HLH) family of transcription factors dimerize in this manner. In these proteins, the helix-loop-helix region is preceded by a sequence of basic amino acids that provide the DNA-binding site (Figure 10.23), and

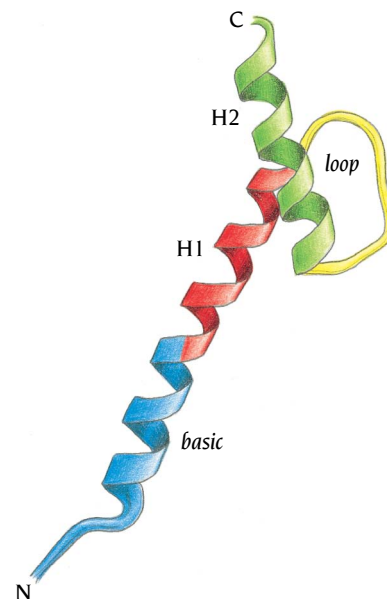


**Figure 10.23** Domain arrangement along the polypeptide chains of three families of transcription factors; b/z, b/HLH and b/HLH/z. All three have a basic region (blue) that binds DNA. Dimerization is achieved by the zipper region (purple) in the b/z family, by the H1-loop-H2 region (red-yellow-green) in the b/HLH family and by a combination of both the zipper and the HLH regions in the b/HLH/z family.

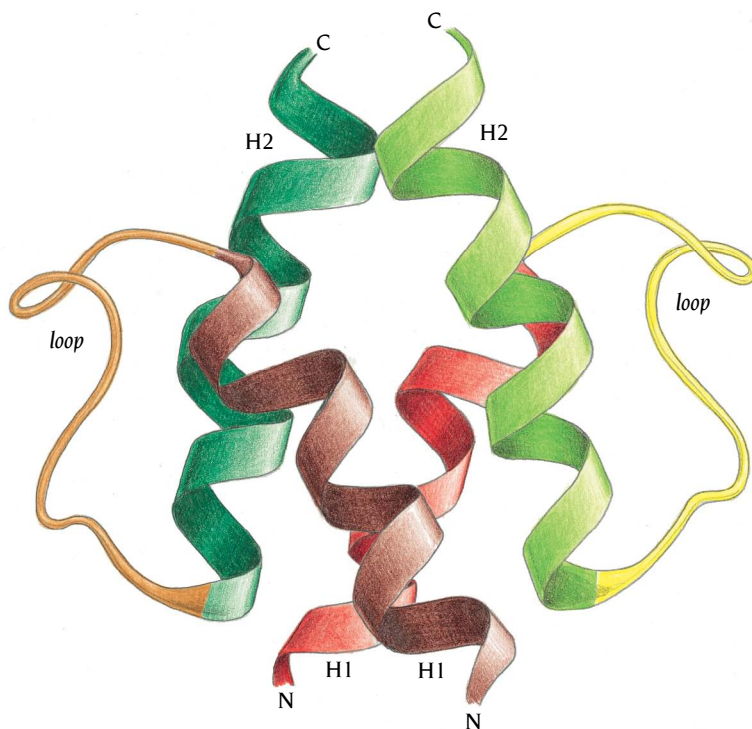
therefore these proteins are called the b/HLH transcription factors, just as members of the leucine zipper family are called the b/zip factors. Members of the b/HLH family have substantial amino acid sequence identity and they bind to the consensus DNA sequence 5'-CANNTG-3', where N is any nucleotide.

The myogenic proteins are an important class of b/HLH transcription factors that are crucially involved in the development of muscle cells. A mature muscle cell is distinguished from other cell types by a large number of characteristic proteins, including specific types of actin and myosin (see Chapter 14). The entire program of muscle differentiation can be triggered *in vitro* in cultured skin fibroblast cells by introducing any one of the myogenic proteins. Introduction of one myogenic protein activates the production of all the others, which in combination with already present but inactive HLH proteins then activate the muscle-specific genes. Thus, muscle differentiation is determined by specific combinations of these transcription factors, including heterodimers.

The crystal structure of the b/HLH domain of one myogenic protein, MyoD, complexed with a DNA fragment containing the consensus recognition sequence has been determined by the group of Carl Pabo at Massachusetts Institute of Technology, Boston. The DNA fragment, having two symmetric binding sites, binds to one dimer of the b/HLH domain. Each monomer of the protein has a very simple structure comprising two  $\alpha$  helices joined by a loop region (Figure 10.24). The first  $\alpha$  helix contains the basic region followed by the H1 region; the second helix, joined to the first by a loop, contains the H2 region of the HLH motif. The helical H1 and H2 regions of each monomer are involved in the dimer interface and participate in forming the four-helix bundle of the dimer (Figure 10.25). Conserved hydrophobic residues in both H1 and H2 form the hydrophobic core of the four-helix bundle. This core stabilizes the dimer and keeps the monomers together with the precise geometry that allows the two basic regions to interact properly with the two recognition sites of the DNA fragment, which are separated by two base pairs. These rather stringent stereochemical requirements impose constraints on the amino acid sequence of the HLH motif; this is reflected in the high degree of sequence homology that allows new proteins to be identified from their sequences as members of this family.

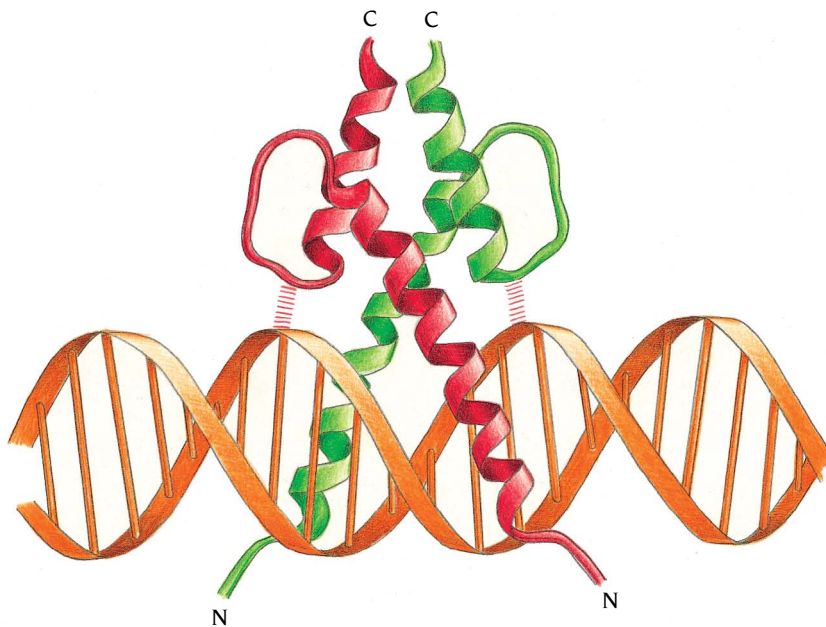


**Figure 10.24** Structure of a monomer of the DNA-binding domain of the transcription factor MyoD. The domain, which belongs to the b/HLH family, comprises two  $\alpha$  helices joined by a loop region. The basic region (blue) and the first helix H1 (red) of the helix-loop-helix region form one continuous  $\alpha$  helix. (Adapted from P.C.M. Ma et al., *Cell* 77: 451–459, 1994.)



**Figure 10.25** Structure of the dimerization region of MyoD. The  $\alpha$  helices H1 (red and brown) and H2 (light and dark green) of the two monomers form a four-helix bundle that keeps the dimer together. The loops (yellow and orange) are on the outside of the four-helix bundle. (Adapted from P.C.M. Ma et al., *Cell* 77: 451–459, 1994.)

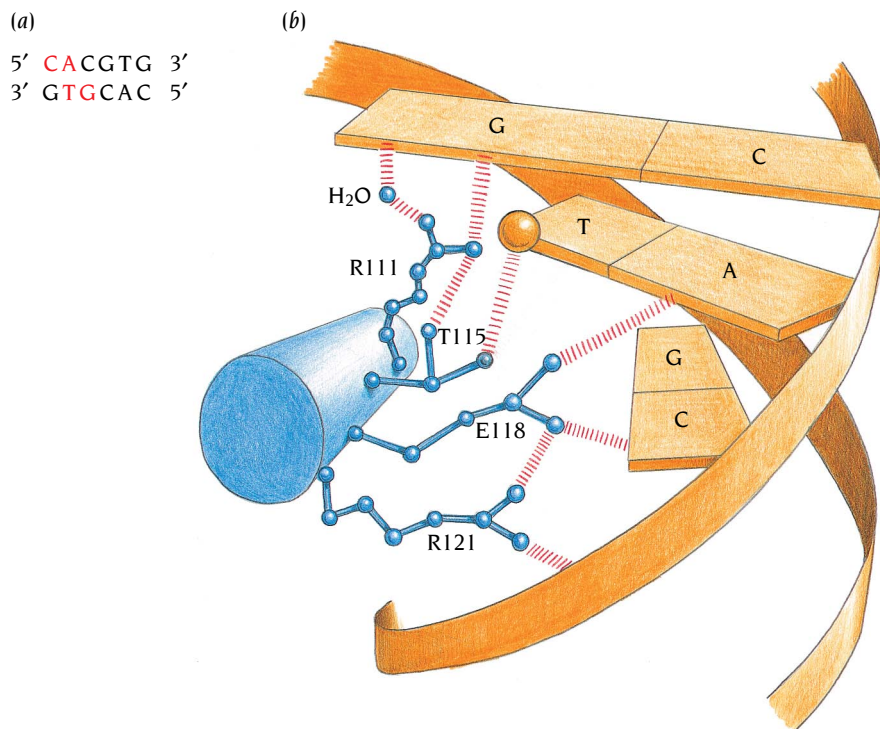




**Figure 10.26** Schematic diagram of the binding of MyoD to DNA. MyoD binds as a dimer (red and green) with the N-terminal basic regions interacting with the major groove of DNA (orange). The monomers diverge from each other after the H1 region and bind on opposite sides of DNA, as for GCN4 (see Figure 10.21). The four-helix bundle, together with contacts with phosphates of the DNA backbone, rigidify the fork and allow MyoD only to bind to enhancer elements with a specific spacing between the half-sites. The DNA structure is essentially B-DNA. (Adapted from P.C.M. Ma et al., *Cell* 77: 451–459, 1994.)

### *The $\alpha$ -helical basic region of the b/HLH motif binds in the major groove of DNA*

The helical basic regions of the MyoD dimer bind in the major groove of DNA in a very similar fashion to the basic regions of the leucine zipper protein GCN4 (Figure 10.26). The DNA structure is essentially that of B-DNA with a slightly narrower major groove and a wider minor groove. Three residues from each of the two  $\alpha$  helices make sequence specific contacts to the edges of the bases in the major groove (Figure 10.27). These contacts are



**Figure 10.27** Sequence-specific contacts between DNA and one monomer of MyoD. (a) The palindromic MyoD recognition sequence. Bases that contact one monomer of MyoD are shown in red. (b) Three residues, Glu 118, Thr 115 and Arg 111 from the basic region of MyoD, bind to the edges of the bases in the major groove. Glu 118 recognizes the first two bases, C and A, in the MyoD consensus recognition sequence.

identical in the two half-sites. Glu 118 plays an especially important role in the recognition of the specific nucleotides

5'-CAC-3'  
3'-GTG-5'

in each half site of the recognition site for the MyoD dimer (see Figure 10.27a). The side chain of Glu 118 accepts a hydrogen bond from the nitrogen atom of N4 of the first cytosine and donates a hydrogen bond to N6 of the following adenine. In addition, this side chain makes water-mediated contacts to two other bases further away in the recognition site. The side chain conformation of Glu 118 is stabilized by forming a salt bridge to Arg 121 which is located in the next turn of the  $\alpha$  helix. Thr 115 and Arg 111, whose side chains also face the major groove, form hydrogen bonds with T and G in the complementary strand (see Figure 10.27). The methyl group of Thr 115 forms a hydrophobic contact with the methyl group of thymine and Arg 111 donates a hydrogen bond to a nitrogen atom of guanine. Thus four bases of the three-base pair recognition site form sequence-specific contacts with side chains of the helical basic region of MyoD (see Figure 10.27a).

In addition to these sequence-specific contacts there are a large number of nonspecific contacts between phosphate groups of the DNA and polar or positively charged groups on the protein, notably in the basic region (where five of the eight positively charged residues are involved in such contacts), but also in the HLH domain of MyoD. The half-site spacing is invariant in the b/HLH family, in contrast to the b/zip family. This is because the small four-helix bundle in b/HLH locks the basic regions in position and prevents them spreading apart, as occurs in GCN4, for example.

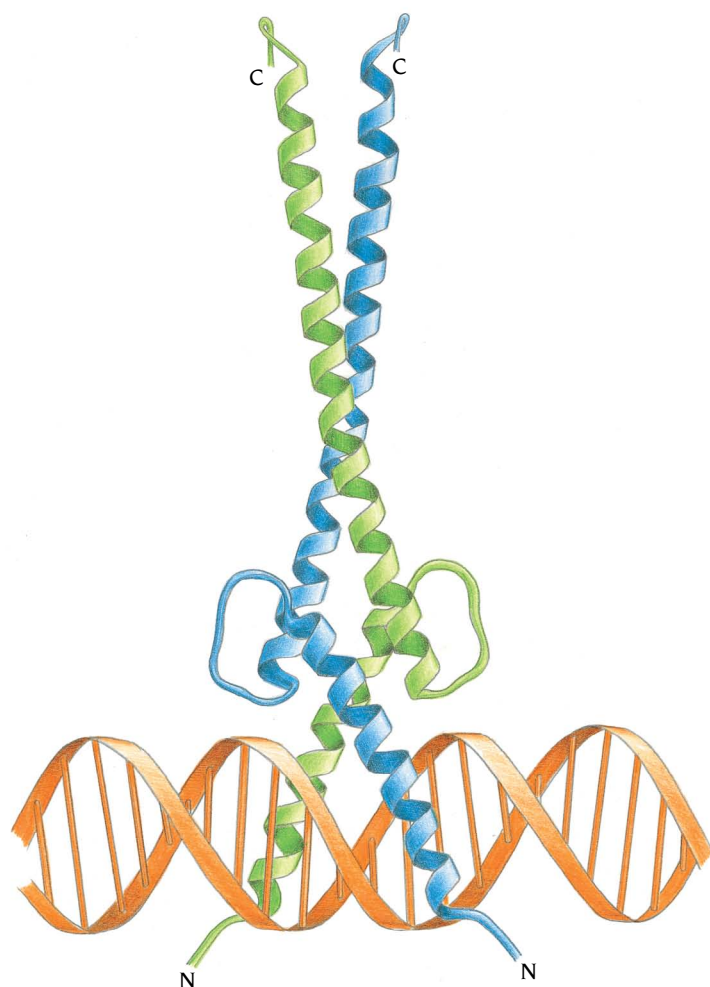
### ***The b/HLH/zip family of transcription factors have both HLH and leucine zipper dimerization motifs***

In Chapter 13 we discuss growth factors that activate phosphorylation cascades leading to changes in gene expression. Several of the genes that are first activated by growth factors, the early-response genes, code for transcription factors that in turn activate other genes, the delayed-response genes, which are more specific for the different cell types. The best-studied early-response genes are the *myc*, *fos* and *jun* oncogenes. Myc has a critical role in the normal control of cell proliferation. Cells in which Myc expression is specifically prevented will not divide even in the presence of growth factors. When Myc is overexpressed or hyperactivated by mutations, these cells will start uncontrolled proliferation leading to cancerous growth.

As we have seen, the Fos and Jun proteins belong to the b/zip family of transcription factors. The *myc* gene product also contains a leucine zipper motif but in addition has an HLH motif; it therefore belongs to the b/HLH/zip family of transcription factors (see Figure 10.23). Members of this family occur in diverse eucaryotes ranging from mammals to yeast and are important in regulating metabolism, cell differentiation and development. Some members of this family have no basic region and therefore cannot bind DNA. They can, however, form heterodimers with other members that do contain the basic region, preventing them from binding DNA and thereby acting as negative regulators of gene expression. The Myc protein does not normally form homodimers and therefore does not by itself bind DNA. However, Myc can form a heterodimer with a different member of the same family, Max, that binds to DNA under physiological conditions. The association of Max and Myc is required for malignant transformation by the *myc* oncogene. Max probably also plays an essential role in orchestrating the biological activities of many other b/HLH/zip transcription factors.

In contrast to Myc, Max can form homodimers that bind tightly to DNA. These homodimers recognize the same consensus sequence as members of the b/HLH family, 5'-CANNTG-3'. The three-dimensional structure of the b/HLH/zip domain of Max complexed with a DNA fragment containing the sequence 5'-CACGTG-3' has been determined by the group of Stephen



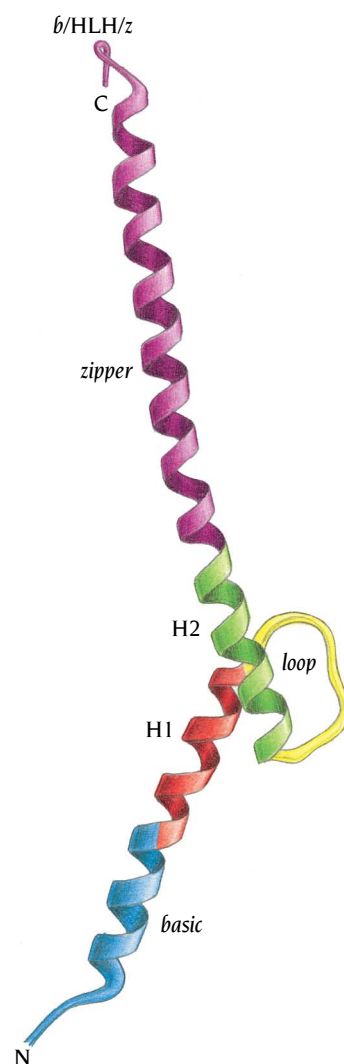


**Figure 10.28** Schematic diagram of the binding of the transcription factor Max to DNA. The two monomers of Max (blue and green) form a dimer through both the helix-loop-helix regions which form a four-helix bundle like MyoD, and the zipper regions, which are arranged in a coiled coil. The N-terminal basic regions bind to DNA in a way similar to GCN4 and MyoD. (Adapted from A.R. Ferre-D'Amare et al., *Nature* 363: 38–45, 1993.)

Burley at Rockefeller University, New York. The overall structure of the b/HLH part of the domain is very similar to that in MyoD (see Figures 10.26 and 10.28), while the leucine zippers of the dimer form a helical coiled coil very similar to that of GCN4 described earlier (see Figures 10.21b and 10.28). In the Max monomer the zipper helix is a continuation of the helical H2 region, so that H2 and zip form one long  $\alpha$  helix (Figure 10.29). The overall structure of the monomer is therefore two  $\alpha$  helices joined by a loop, where the first helix comprises the basic region followed by the H1 region of the HLH motif, and the second helix is formed by the H2 region of HLH followed by the zipper region.

The loop regions of both the b/HLH and the b/HLH/zip families are highly variable both in sequence and in length (Figure 10.30). Homologous protein sequences have loops ranging from 5 to 23 residues. Since the two helices in the HLH motif are parallel, the loop must be sufficiently long to connect one end of the four-helix bundle with the other end (see Figure 10.25). Given that the distance between successive  $\alpha$  carbon atoms in an extended polypeptide chain is 3.8 Å and that the distance between the end of H1 and the beginning of H2 in the bundle is 15 Å in these structures, it follows that the loop must be at least four or five residues long. This has been confirmed by shortening the loop of MyoD to four residues by deletion mutations, whereupon the protein loses its DNA-binding activity.

**Figure 10.29** The structure of the Max monomer is essentially built up from two long  $\alpha$  helices joined by a loop region (yellow). The basic region (blue) and H1 (red) of the helix-loop-helix region form one continuous  $\alpha$  helix, and H2 (green) and the zipper region (purple) form a second continuous  $\alpha$  helix.



|       |                   | H1 |   |   |   |   |   |   |   |   |   |   |   | loop |   |   |   |   |   |   |   |   |   |   |   | H2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |
|-------|-------------------|----|---|---|---|---|---|---|---|---|---|---|---|------|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------|
| Max   | NH <sub>2</sub> - | D  | H | I | K | D | S | F | H | S | L | R | D | S    | V | P | - | - | - | - | - | S | L | Q | G | E  | K | A | S | - | R | A | Q | I | L | D | K | A | T | E | Y | I | Q | Y | M | - | COOH |
| C-Myo | -                 | N  | E | L | K | R | S | F | F | A | L | R | D | Q    | I | P | - | - | - | - | - | E | L | E | N | N  | E | K | A | P | - | K | V | V | I | L | K | K | A | T | A | Y | I | L | S | V | -    |
| MyoD  | -                 | S  | K | V | N | E | A | F | E | T | L | K | R | C    | T | S | - | - | - | - | - | S | N | P | N | Q  | R | L | P | - | K | V | E | I | L | R | N | A | I | R | Y | I | E | G | L | - |      |
| CBF1  | -                 | E  | N | I | N | T | A | I | N | V | L | S | D | L    | L | P | - | - | - | - | - | - | V | R | E | S  | S | - | K | A | A | I | L | A | R | A | A | E | Y | I | Q | K | L | - |   |   |      |
| Pho4  | -                 | N  | R | L | A | V | A | L | H | E | L | A | S | L    | I | P | - | A | E | W | K | Q | Q | N | V | S  | A | A | P | S | - | K | A | T | T | V | E | A | A | C | R | Y | I | R | H | L | -    |

### Max and MyoD recognize the DNA HLH consensus sequence by different specific protein-DNA interactions

The helical basic region of Max binds to the major groove of DNA in the same manner as MyoD (see Figures 10.26 and 10.28). The DNA structure is essentially that of B-DNA with minor distortions as in the MyoD complex. Both Max and MyoD recognize the same consensus half-site DNA sequence

5'-CAC-3'

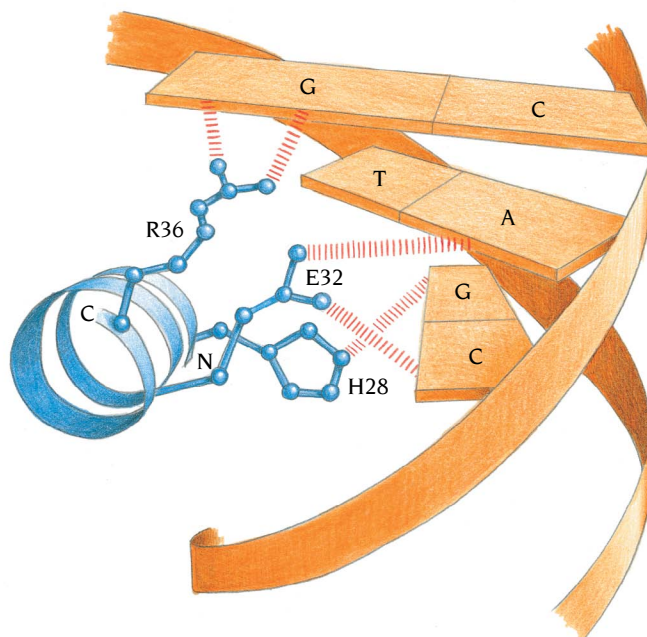
3'-GTG-5'

However, specific contacts with the bases of DNA are formed by different protein side chains in the two complexes. In both complexes a central glutamic acid residue, Glu 32 in Max and Glu 118 in MyoD, recognizes the 5' C and A bases by similar hydrogen bonds (compare Figures 10.27 and 10.31). However, in Max, His 28, which is located one turn of the  $\alpha$  helix away from Glu 32, recognizes the 5' G in the complementary strand (Figure 10.31), which is not recognized by MyoD. Conversely, the methyl groups of T recognized by Thr 115 of MyoD is not recognized by Max. Finally, in both Max and MyoD, an arginine residue, Arg 36 in the former and Arg 111 in the latter, recognizes a G residue in the half-site (Figures 10.27 and 10.31).

### Conclusion

A large group of eucaryotic transcription factors use liganded zinc atoms to stabilize a DNA-binding motif, and most classic zinc finger transcription factors contain several and sometimes over 30 concatenated zinc finger motifs. These motifs bind to a sequence of binding sites in the target DNA. Although each motif only recognizes three or four bases via a loop region at the tip of each finger, a zinc finger transcription factor containing multiple fingers, each with its own short recognition site, becomes a highly discriminating DNA-binding protein.

**Figure 10.30** Amino acid sequences of the helix-loop-helix region of some members of the b/HLH and b/HLH/zip families of transcription factors. Residues that form the hydrophobic core of the four-helix bundle are colored green and a conserved lysine residue is blue. The loop region between H1 and H2 is highly variable in length but must be at least four or five residues long.



**Figure 10.31** Sequence-specific interactions between DNA and one monomer of Max. Three residues, His 28, Glu 32 and Arg 36, form specific interactions with the edges of the bases in the major groove of DNA. Like MyoD, a Glu residue recognizes the first two bases, C and A, of the recognition sequence.

The nuclear receptors utilize pairs of zinc atoms to stabilize a DNA-binding domain. The steroid hormone receptors are homodimers with two DNA-binding sites that recognize an almost palindromic sequence in the DNA. The monomeric DNA-binding domains dimerize upon binding to DNA in such a way that the two recognition helices of the dimer can bind to target sequences of the response element with the appropriate spacer region. These dimers are symmetrical, like the dimers of the 434 repressor. Other nuclear receptors form heterodimers that are nonsymmetric and bind to response elements in which the target sequences are organized as direct repeats. These heterodimers discriminate response elements on the basis of the different lengths of their spacer region. A DNA-supported asymmetric dimerization interface located within the DNA-binding domains allows these heterodimers to distinguish between closely related response elements.

The C<sub>6</sub>-zinc cluster transcription factors utilize two zinc atoms bound by six cysteine residues to stabilize the scaffold of the monomeric subunit. Again, the active factor is a dimer with a distinct dimerization region. The dimer resembles the letter T: the dimerization region is in the upright stem and the two DNA binding motifs are at the ends of the crossbar. All members of this family recognize the same base sequence, the triplet CCG; specificity is achieved by the length of the crossbar, in other words, the separation of the CCG triplet along the DNA.

Dimerization of the C<sub>6</sub>-zinc cluster transcription factors involves an  $\alpha$ -helical coiled coil in the dimerization region. Coiled coils, often called leucine zippers, are also found in a large group of transcription factors that do not contain zinc. The leucine zipper is made up of two  $\alpha$  helices in a coiled coil with every seventh residue leucine or some other large hydrophobic residue, such as isoleucine or valine. Leucine zipper transcription factors (b/zip) include factors characterized by heterodimerization, for example Fos and Jun. The  $\alpha$ -helical DNA-binding motifs of the heterodimers recognize quite different base sequences and are continuous with the  $\alpha$  helices of the zipper.

Helix-loop-helix (b/HLH) transcription factors are either heterodimers or homodimers with basic  $\alpha$ -helical DNA-binding regions that lie across the major groove, rather than along it, and these helices extend into the four-helix bundle that forms the dimerization region. A modification of the b/HLH structure is seen in some transcription factors (b/HLH/zip) in which the four-helix bundle extends into a classic leucine zipper.

Homodimerization, the common feature of procaryotic transcription factors, is in eucaryotes extended by heterodimerization (as in the leucine zippers) and by concatenation (as in zinc fingers and POU). These devices increase the range of DNA sequences that can be recognized and the degree of binding specificity. In evolutionary terms this increase in range and specificity became necessary as genomes became larger and more complex. The real complexity of regulation of gene expression in eucaryotes has so far barely been touched upon by structural biologists, because it lies in the assembly of aggregates of general and specific transcription factors at the promoter region. The biochemical data are strong enough to conclude that the various transcription factors bound to distinct sites at a complex promoter have specific three-dimensional relationships, and so form a macromolecular assembly analogous to a ribosome, proteosome or spliceosome.

## Selected readings

### General

- Alber, T. How GCN4 binds DNA. *Curr. Biol.* 3: 182–184, 1993.
- Berg, J.M., Shi, Y. The galvanization of biology: a growing appreciation for the roles of zinc. *Science* 271: 1081–1085, 1996.
- Buratowski, S. The basics of basal transcription by RNA Polymerase II. *Cell* 77: 1–3, 1994.
- Fairman, R., et al. Multiple oligomeric states regulate the DNA binding of helix-loop-helix peptides. *Proc. Natl. Acad. Sci. USA* 90: 10429–10433, 1993.
- Freedman, L.P., Luisi, B.F. On the mechanism of DNA binding by nuclear hormone receptor: a structural and functional perspective. *J. Cell. Biochem.* 51: 140–150, 1993.
- Harrison, S.C., Sauer, R.T. (eds.) Protein–nucleic acid interactions. *Curr. Opin. Struct. Biol.* 4: 1–35, 1994.
- Kerppola, T., Currant, T. Zen and the art of Fos and Jun. *Nature* 373: 199–200, 1995.
- Pabo, C.O., Sauer, R.T. Transcription factors: structural families and principles for DNA recognition. *Annu. Rev. Biochem.* 61: 1053–1098, 1992.
- Phillips, S.E.V. Built by association: structure and function of helix-loop-helix DNA-binding proteins. *Structure* 2: 1–4, 1994.
- Philips, S.E.V., Moras, D. (eds.) Protein–nucleic acid interactions. *Curr. Opin. Struct. Biol.* 3: 1–16, 1993.
- Schwabe, J.W.R., Klug, A. Zinc mining for protein domains. *Nature Struct. Biol.* 1: 345–350, 1994.
- Tjian, R., Maniatis, T. Transcriptional activation: a complex puzzle with few easy pieces. *Cell* 77: 5–8, 1994.

### Specific structures

- Ellenberger, T.E., et al. The GCN4 basic region leucine zipper binds DNA as a dimer of uninterrupted  $\alpha$  helices: crystal structure of the protein–DNA complex. *Cell* 71: 1223–1237, 1992.
- Fairall, L., et al. The crystal structure of a two zinc finger peptide reveals an extension to the rules for zinc finger/DNA recognition. *Nature* 366: 483–487, 1993.
- Ferré-D'Amaré, A.R., et al. Recognition by Max of its cognate DNA through a dimeric b/HLH/Z domain. *Nature* 363: 38–45, 1993.
- Ferré-D'Amaré, A.R., et al. Structure and function of the b/HLH/Z domain of USF. *EMBO J.* 13: 180–189, 1994.
- Glover, J.N.M., Harrison, S.C. Crystal structure of the heterodimeric bZIP transcription factor c-Fos–c-Jun bound to DNA. *Nature* 373: 257–261, 1995.

- Hard, T., et al. Solution structure of the glucocorticoid receptor DNA-binding domain. *Science* 249: 157–160, 1990.
- Hegde, R.S., et al. Crystal structure at 1.7 Å of the bovine papillomavirus-1 E2 DNA-binding domain bound to its DNA target. *Nature* 359: 505–512, 1992.
- König, P., Richmond, T.J. The x-ray structure of the GCN4–bZIP bound to ATF/CREB site DNA shows the complex depends on DNA flexibility. *J. Mol. Biol.* 233: 139–154, 1993.
- Lee, M.S., et al. Three-dimensional solution structure of a single zinc finger DNA-binding domain. *Science* 245: 635–637, 1989.
- Luisi, B.F., et al. Crystallographic analysis of the interaction of the glucocorticoid receptor with DNA. *Nature* 352: 497–505, 1991.
- Ma, P.C.M., et al. Crystal structure of MyoD bHLH domain–DNA complex: perspectives on DNA recognition and implications for transcriptional activation. *Cell* 77: 451–459, 1994.
- Marmorstein, R., et al. DNA recognition by GAL4: structure of a protein–DNA complex. *Nature* 356: 408–414, 1992.
- Marmorstein, R., Harrison, S.C. Crystal structure of a PPR1–DNA complex: DNA recognition by proteins containing a Zn<sub>2</sub>Cys<sub>6</sub> binuclear cluster. *Genes Dev.* 8: 2504–2512, 1994.
- O'Shea, E.K., et al. Mechanism of specificity in the fos–jun oncoprotein heterodimer. *Cell* 68: 699–708, 1992.
- O'Shea, E.K., et al. X-ray structure of the GCN4 leucine zipper, a two-stranded, parallel coiled coil. *Science* 254: 539–544, 1991.
- Parraga, G., et al. Zinc-dependent structure of a single-finger domain of yeast ADR1. *Science* 241: 1489–1492, 1988.
- Pavletich, N.P., Pabo, C.O. Crystal structure of a five-finger GLI–DNA complex: new perspectives on zinc fingers. *Science* 261: 1701–1707, 1993.
- Pavletich, N.P., Pabo, C.O. Zinc finger–DNA recognition: crystal structure of a Zif268–DNA complex at 2.1 Å. *Science* 252: 809–817, 1991.
- Rastinejad, F., et al. Structural determinants of nuclear receptor assembly on DNA direct repeats. *Nature* 375: 203–211, 1995.
- Reece, R.J., Ptashne, M. Determinants of binding-site specificity among yeast C<sub>6</sub> zinc cluster proteins. *Science* 261: 909–911, 1993.
- Schwabe, J.W.R., et al. The crystal structure of the estrogen receptor DNA-binding domain bound to DNA: how receptors discriminate between their response elements. *Cell* 75: 567–578, 1993.



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# An Example of Enzyme Catalysis: Serine Proteinases

# 11

In 1946 Linus Pauling first formulated the basic principle underlying enzyme catalysis, namely, that an enzyme increases the rate of a chemical reaction by binding and stabilizing the transition state of its specific substrate tighter than the ground state. However, for many years it was not generally appreciated that the high affinity of an enzyme for the transition state of a substrate plays a major role in determining substrate specificity as well as the rate of catalysis. In the past few years, kinetic studies of site-directed mutants, combined with x-ray structures, have made it possible to identify unambiguously the role of particular amino acids in both the substrate specificity and the catalytic reaction of an enzyme as well as providing information about the energetic basis of catalysis itself. The full consequences of Pauling's principle emerged only when it was found that mutants designed to change an enzyme's catalytic rate also changed its substrate specificity and vice versa.

In this chapter we shall illustrate some fundamental aspects of enzyme catalysis using as an example the serine proteinases, a group of enzymes that hydrolyze peptide bonds in proteins. We also examine how the transition state is stabilized in this particular case.

## *Proteinases form four functional families*

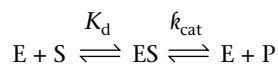
**Proteinases** are widely distributed in nature, where they perform a variety of different functions. Viral genes code for proteinases that cleave the precursor molecules of their coat proteins, bacteria produce many different extracellular proteinases to degrade proteins in their surroundings, and higher organisms use proteinases for such different functions as food digestion, cleavage of signal peptides, and control of blood pressure and blood clotting. Many proteinases occur as domains in large multifunctional proteins, but others are independent smaller polypeptide chains. *In vivo* the activity of many proteinases is controlled by endogenous protein inhibitors that complex with the enzymes and block them. The three-dimensional structures of a large number of the smaller proteinases and of their complexes with protein inhibitors have been determined, and this wealth of data allows some general conclusions to be drawn.

All the well-characterized proteinases belong to one or other of four families: serine, cysteine, aspartic, or metallo proteinases. This classification is based on a functional criterion, namely, the nature of the most prominent functional group in the active site. Members of the same functional family are usually evolutionarily related, but there are exceptions to this rule. We

have chosen two **serine proteinases**, mammalian **chymotrypsin** and bacterial **subtilisin**, as representative examples to illustrate one of the catalytic mechanisms leading to proteolysis. Before describing the structures, mechanism, and engineering of these two enzymes, however, we shall define some basic enzymological concepts.

### *The catalytic properties of enzymes are reflected in $K_m$ and $k_{cat}$ values*

Leonor Michaelis and Maud Menten laid the foundation for enzyme kinetics as early as 1913 by proposing the following scheme:



Enzyme and substrate first reversibly combine to give an **enzyme-substrate** (ES) complex. Chemical processes then occur in a second step with a rate constant called  $k_{cat}$ , or the **turnover number**, which is the maximum number of substrate molecules converted to product per active site of the enzyme per unit time. The  $k_{cat}$  is, therefore, a rate constant that refers to the properties and reactions of the ES complex. For simple reactions  $k_{cat}$  is the rate constant for the chemical conversion of the ES complex to free enzyme and products.

These definitions are valid only when the concentration of the enzyme is very small compared with that of the substrate. Moreover, they apply only to the initial rate of formation of products: in other words, the rate of formation of the first few percent of the product, before the substrate has been depleted and products that can interfere with the catalytic reaction have accumulated.

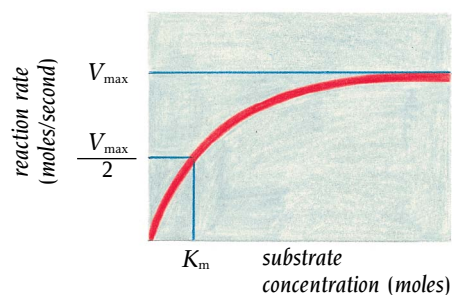
The Michaelis-Menten scheme nicely explains why a maximum rate,  $V_{max}$ , is always observed when the substrate concentration is much higher than the enzyme concentration (Figure 11.1).  $V_{max}$  is obtained when the enzyme is saturated with substrate. There are then no free enzyme molecules available to turn over additional substrate. Hence, the rate is constant,  $V_{max}$ , and is independent of further increase in the substrate concentration.

The substrate concentration when the half maximal rate, ( $V_{max}/2$ ), is achieved is called the  $K_m$ . For many simple reactions it can easily be shown that the  $K_m$  is equal to the dissociation constant,  $K_d$ , of the ES complex. The  $K_m$ , therefore, describes the affinity of the enzyme for the substrate. For more complex reactions,  $K_m$  may be regarded as the overall dissociation constant of all enzyme-bound species.

The quantity  $k_{cat}/K_m$  is a rate constant that refers to the overall conversion of substrate into product. The ultimate limit to the value of  $k_{cat}/K_m$  is therefore set by the rate constant for the initial formation of the ES complex. This rate cannot be faster than the diffusion-controlled encounter of an enzyme and its substrate, which is between  $10^8$  to  $10^9$  per mole per second. The quantity  $k_{cat}/K_m$  is sometimes called the **specificity constant** because it describes the specificity of an enzyme for competing substrates. As we shall see, it is a useful quantity for kinetic comparison of mutant proteins.

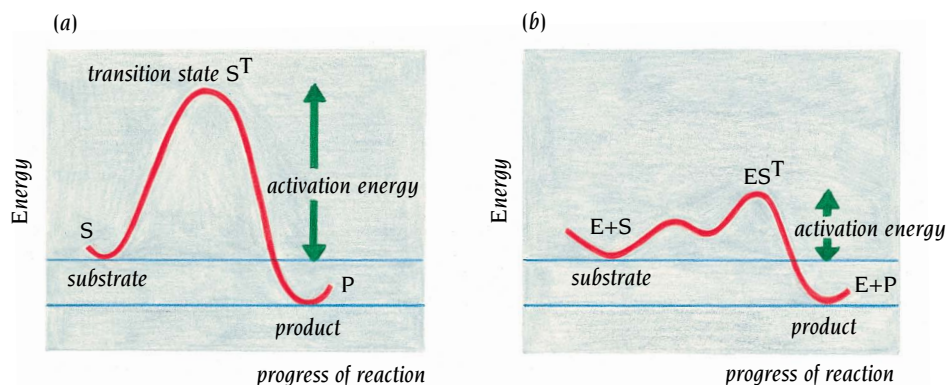
### *Enzymes decrease the activation energy of chemical reactions*

The Michaelis complex, ES, undergoes rearrangement to one or several **transition states** before product is formed. Energy is required for these rearrangements. The input energy required to bring free enzyme and substrate to the highest transition state of the ES complex is called the **activation energy** of the reaction (Figure 11.2). In the absence of enzyme, spontaneous conversion of substrate to product also proceeds through transition states that require activation energy. The rate of a chemical reaction is strictly dependent on its



**Figure 11.1** A plot of the reaction rate as a function of the substrate concentration for an enzyme catalyzed reaction.  $V_{max}$  is the maximal velocity. The Michaelis constant,  $K_m$ , is the substrate concentration at half  $V_{max}$ . The rate  $v$  is related to the substrate concentration,  $[S]$ , by the Michaelis-Menten equation:

$$v = \frac{V_{max} \times [S]}{K_m + [S]}$$



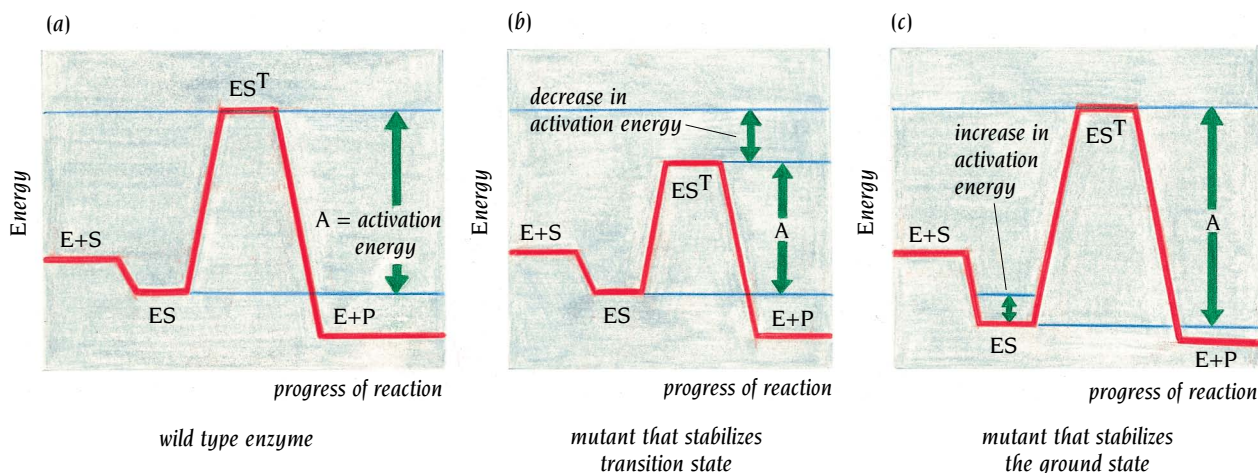
**Figure 11.2** Enzymes accelerate chemical reactions by decreasing the activation energy. The activation energy is higher for a noncatalyzed reaction (a) than for the same reaction catalyzed by an enzyme (b). Both reactions proceed through one or several transition states,  $S^T$ . Only one transition state is shown in (a), whereas the two bumps in (b) represent two different transition states.

activation energy, and the more than 1 millionfold enhancement of rate achieved by enzyme catalysis results from the ability of the enzyme to decrease the activation energy of the reaction (Figure 11.2).

This decrease in activation energy is achieved by enzymes in several different ways: for example, by providing catalytically active groups for a specific reaction mechanism, by binding several substrates in an orientation appropriate to the reaction catalyzed, and, most importantly, by using the differential binding energy of the substrate in its transition state compared with its normal state. The activation energy for the conversion of  $ES$  to  $E + P$  is lower if the enzyme binds more tightly to the transition state of  $S$  than to its normal structure (Figure 11.3). The higher affinity of the enzyme for the transition state makes the transition energetically favorable and thus decreases the activation energy. If, on the other hand, the enzyme were to bind the unaltered substrate more strongly than the transition state, the decrease in binding energy on the formation of the transition state would increase the activation energy and catalysis would not be achieved (see Figure 11.3). It is therefore catalytically advantageous for the enzyme's active site to be complementary to the transition state of the substrate rather than to the normal structure of the substrate.

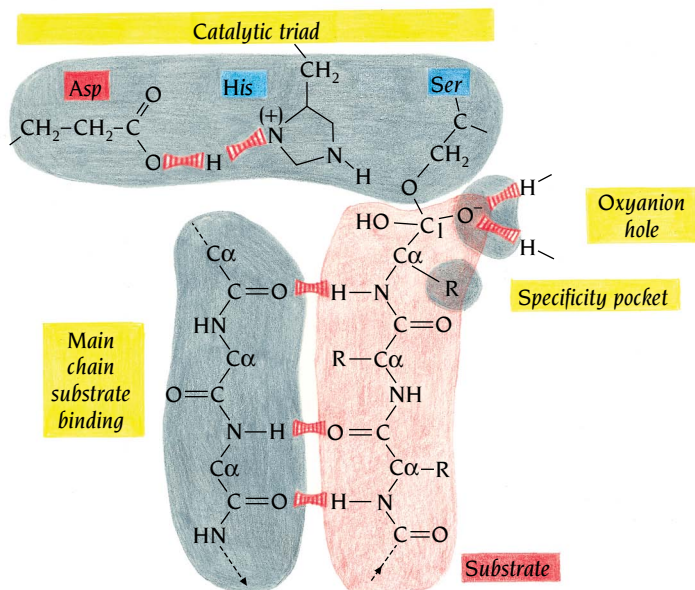
Since this differential binding energy relates to the complete substrate molecule, including groups that determine the substrate specificity, it is obvious that specificity and catalytic rate are interdependent. The importance of the differential binding energy for catalysis is nicely illustrated by the recent production of antibodies with catalytic activity. Such antibodies were raised against small hapten molecules that simulate a transition state structure for a specific chemical reaction, such as ester hydrolysis. These antibodies not only bound the transition state more tightly than the normal structure of the ester, but they also exhibited significant catalytic activity even though they had not been selected to have any catalytically competent residues in the binding site.

**Figure 11.3** One of the most important factors in enzyme catalysis is the ability of an enzyme to bind substrate more tightly in its transition state than in its ground state. The difference in binding energy between these states lowers the activation energy of the reaction. This is illustrated by energy profiles for an enzyme in its wild-type form (a), for a mutant that stabilizes the substrate in its transition state and therefore decreases the activation energy from  $ES$  to the transition state  $ES^T$  giving higher rates (b), and for a mutant that stabilizes the substrate in its ground state giving lower rates (c). (Adapted from A. Fersht, *Enzyme Structure and Mechanism*, 2nd ed. pp. 314–315. New York: W.H. Freeman, 1984.)









**Figure 11.6** A schematic view of the presumed binding mode of the tetrahedral transition state intermediate for the deacylation step. The four essential features of the serine proteinases are highlighted in yellow: the catalytic triad, the oxyanion hole, the specificity pocket, and the unspecific main-chain substrate binding.

### Four important structural features are required for the catalytic action of serine proteinases

The serine proteinases have four important structural features that facilitate this mechanism of catalysis (Figure 11.6).

1. The enzyme provides a general base, a His residue, that can accept the proton from the hydroxyl group of the reactive Ser thus facilitating formation of the covalent tetrahedral transition state. This His residue is part of a **catalytic triad** consisting of three side chains from Asp, His, and Ser, which are close to each other in the active site, although they are far apart in the amino acid sequence of the polypeptide chain (Figure 11.6).
2. Tight binding and stabilization of the tetrahedral transition state intermediate is accomplished by providing groups that can form hydrogen bonds to the negatively charged oxygen atom attached to C<sub>1</sub>. These groups are in a pocket of the enzyme called the **oxyanion hole** (see Figure 11.6). The positive charge that develops on the His residue after it has accepted a proton also stabilizes the negatively charged transition state. These features also presumably destabilize binding of substrate in the normal state.
3. Most serine proteinases have no absolute substrate specificity. They can cleave peptide bonds with a variety of side chains adjacent to the peptide bond to be cleaved (the scissile bond). This is because polypeptide substrates exhibit a nonspecific binding to the enzyme through their main-chain atoms, which form hydrogen bonds in a short antiparallel  $\beta$  sheet with main-chain atoms of a loop region in the enzyme (see Figure 11.6). One of these hydrogen bonds is long (3.6 Å) in enzyme-substrate complexes but short in complexes that simulate the transition state. This nonspecific binding therefore also contributes to stabilization of the transition state.
4. Even though these enzymes have no absolute specificity, many of them show a preference for a particular side chain before the scissile bond as seen from the amino end of the polypeptide chain. The preference of chymotrypsin to cleave after large aromatic side chains and of trypsin to cleave after Lys or Arg side chains is exploited when these enzymes are used to produce peptides suitable for amino acid sequence determination and fingerprinting. In each case, the preferred side chain is oriented so as to fit into a pocket of the enzyme called the **specificity pocket**.

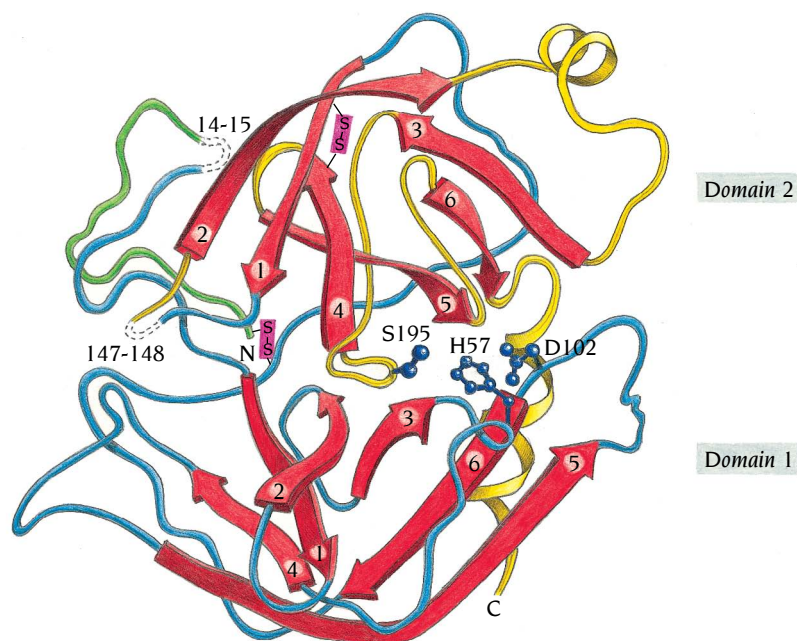
## Convergent evolution has produced two different serine proteinases with similar catalytic mechanisms

These four features all occur in an almost identical fashion in all members of the chymotrypsin superfamily of homologous enzymes, which includes among others chymotrypsin, trypsin, elastase, and thrombin. Reasonably, one might imagine that such a combination of four characteristic features had arisen only once during evolution to give an ancestral molecule from which all serine proteinases diverged. However, subtilisin, a bacterial serine proteinase with an amino acid sequence and, as we will see, a three-dimensional structure quite different from the mammalian serine proteinases, exhibits these same four characteristic features. Subtilisin is not evolutionarily related to the chymotrypsin family of enzymes; nevertheless, the atoms in subtilisin that participate in the catalytic triad, in the oxyanion hole, and in substrate binding are in almost identical positions relative to one another in the three-dimensional structure as they are in chymotrypsin and its relatives. Starting from unrelated ancestral proteins, convergent evolution has resulted in the same structural solution to achieve a particular catalytic mechanism. The serine proteinases, in other words, provide a spectacular example of convergent evolution at the molecular level, which we can best appreciate by explaining in detail the structures of chymotrypsin and subtilisin.

### The chymotrypsin structure has two antiparallel $\beta$ -barrel domains

In 1967 the group of David Blow at the MRC Laboratory of Molecular Biology, Cambridge, UK, determined the three-dimensional structure of chymotrypsin. This was one of the very first enzyme structures known at high resolution. Since then a large number of serine proteinase structures, complexed both with small peptide inhibitors and large endogenous polypeptide inhibitors, have been determined to high resolution mainly by the groups of Michael James, Edmonton, and Robert Huber, Munich.

The polypeptide chain of chymotrypsinogen, the inactive precursor of chymotrypsin, comprises 245 amino acids. During activation of chymotrypsinogen residues 14–15 and 147–148 are excised. The remaining three polypeptide chains are held together by disulfide bridges to form the active chymotrypsin molecule.



**Figure 11.7** Schematic diagram of the structure of chymotrypsin, which is folded into two antiparallel  $\beta$  domains. The six  $\beta$  strands of each domain are red, the side chains of the catalytic triad are dark blue, and the disulfide bridges that join the three polypeptide chains are marked in violet. Chain A (green, residues 1–13) is linked to chain B (blue, residues 16–146) by a disulfide bridge between Cys 1 and Cys 122. Chain B is in turn linked to chain C (yellow, residues 149–245) by a disulfide bridge between Cys 136 and Cys 201. Dotted lines indicate residues 14–15 and 147–148 in the inactive precursor, chymotrypsinogen. These residues are excised during the conversion of chymotrypsinogen to the active enzyme chymotrypsin.

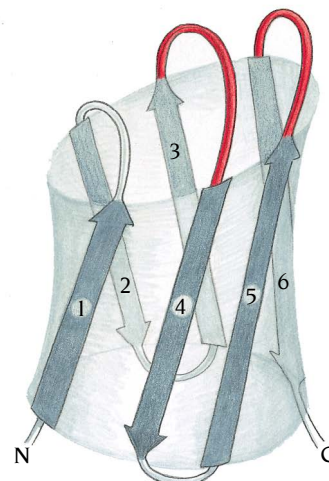
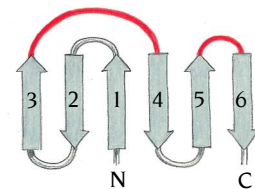
The polypeptide chain is folded into two domains (Figure 11.7), each of which contains about 120 amino acids. The two domains are both of the antiparallel  $\beta$ -barrel type, each containing six  $\beta$  strands with the same topology (Figure 11.8). Even though the actual structure looks complicated, the topology is very simple, a Greek key motif (strands 1–4) followed by an antiparallel hairpin motif (strands 5 and 6).

### *The active site is formed by two loop regions from each domain*

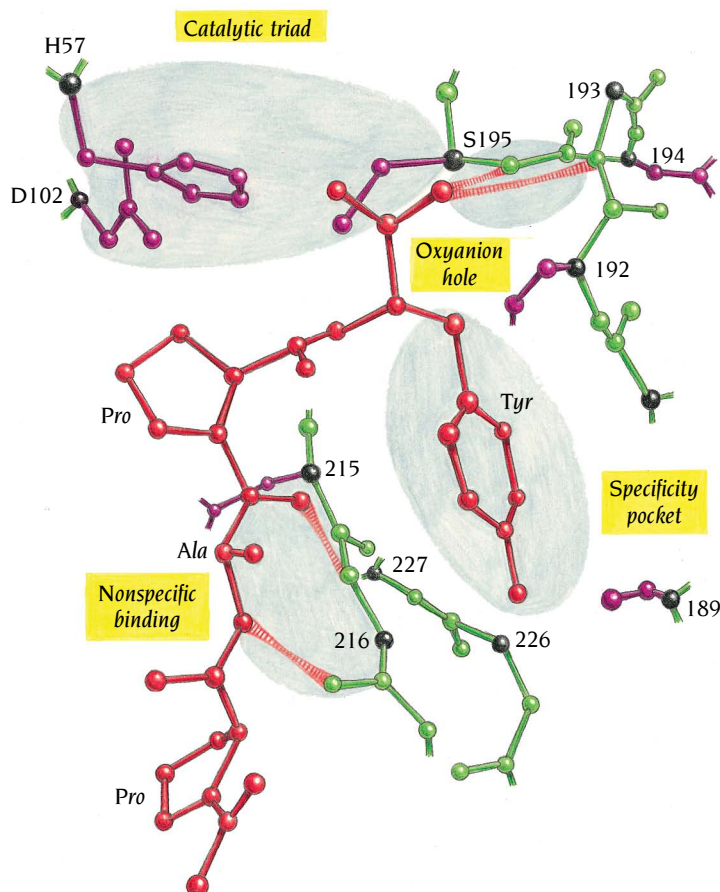
The active site is situated in a crevice between the two domains. Domain 1 contributes two of the residues in the catalytic triad, His 57 and Asp 102, whereas the reactive Ser 195 is part of the second domain (see Figure 11.7).

Inhibitors as well as substrates bind in this crevice between the domains. From the numerous studies of different inhibitors bound to serine proteinases we have chosen as an illustration the binding of a small peptide inhibitor, Ac-Pro-Ala-Pro-Tyr-COOH to a bacterial chymotrypsin (Figure 11.9). The enzyme–peptide complex was formed by adding a large excess of the substrate Ac-Pro-Ala-Pro-Tyr-CO-NH<sub>2</sub> to crystals of the enzyme. The enzyme molecules within the crystals catalyze cleavage of the terminal amide group to produce the products Ac-Pro-Ala-Pro-Tyr-COOH and NH<sub>3</sub><sup>+</sup>. The ammonium ions diffuse away, but the peptide product remains bound as an inhibitor to the active site of the enzyme.

This inhibitor does not form a covalent bond to Ser 195 but one of its carboxy oxygen atoms is in the oxyanion hole forming hydrogen bonds to the main-chain NH groups of residues 193 and 195. The tyrosyl side chain is positioned in the specificity pocket, which derives its specificity mainly from three residues, 216, 226, and 189, as we shall see later. The main chain of



**Figure 11.8** Topology diagrams of the domain structure of chymotrypsin. The chain is folded into a six-stranded antiparallel  $\beta$  barrel arranged as a Greek key motif followed by a hairpin motif.



**Figure 11.9** A diagram of the active site of chymotrypsin with a bound inhibitor, Ac-Pro-Ala-Pro-Tyr-COOH. The diagram illustrates how this inhibitor binds in relation to the catalytic triad, the substrate specificity pocket, the oxyanion hole and the nonspecific substrate binding region. The inhibitor is red. Hydrogen bonds between inhibitor and enzyme are striped. (Adapted from M.N.G. James et al., *J. Mol. Biol.* 144: 43–88, 1980.)



the inhibitor forms a short stretch of antiparallel  $\beta$  sheet with residues 215–216 of the enzyme forming hydrogen bonds to the NH and CO groups of residue 216.

A closer examination of these essential residues, including the catalytic triad, reveals that they are all part of the same two loop regions in the two domains (Figure 11.10). The domains are oriented so that the ends of the two barrels that contain the Greek key crossover connection (described in Chapter 5) between  $\beta$  strands 3 and 4 face each other along the active site. The essential residues in the active site are in these two crossover connections and in the adjacent hairpin loops between  $\beta$  strands 5 and 6. Most of these essential residues are conserved between different members of the chymotrypsin superfamily. They are, of course, surrounded by other parts of the polypeptide chains, which provide minor modifications of the active site, specific for each particular serine proteinase.

His 57 and Ser 195 are within loop 3–4 of domains 1 and 2, respectively. The third residue in the catalytic triad, Asp 102, is within loop 5–6 of domain 1. The rest of the active site is formed by two loop regions (3–4 and 5–6) of domain 2. As in so many other protein structures described previously, the barrels apparently provide a stable scaffold to position a few loop regions that constitute the essential features of the active site.

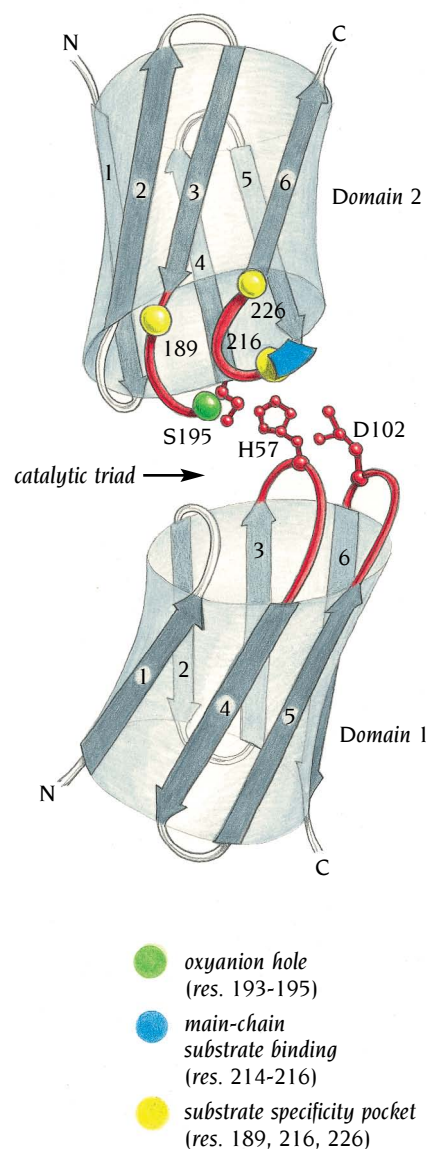
### *Did the chymotrypsin molecule evolve by gene duplication?*

Although the two domains of chymotrypsin have similar three-dimensional structures there is no amino acid sequence identity between them. Nevertheless, based on the argument that three-dimensional structure is more conserved than amino acid sequence, it has been suggested that the members of the chymotrypsin superfamily evolved by gene duplication of a single ancestral proteinase domain. The putative ancestral domain, obviously, could not have had the catalytic triad in present-day serine proteinases since the contemporary triad is derived from both domains. However, this is less of an obstacle to the gene-duplication hypothesis than it seems at first sight. The ancestral domain could have been a barrel structure similar to the second domain of chymotrypsin, which contains most of the essential features of the active site, including the reactive serine residue. We also now know from experiments with genetically engineered mutants in which the triad has been abolished that the catalytic triad is not absolutely essential for catalytic activity. As we will see later, these mutants retain some proteinase activity. It is, therefore, quite possible that there was a single ancestral gene specifying a single domain with some catalytic activity. This activity could then have been enhanced by a gene-duplication event followed by mutation and evolution leading to the catalytic triad of today. The fact that the active-site residues that comprise the catalytic triad of chymotrypsin and its relatives are clustered in the same two loop regions of domains 1 and 2 supports such an evolutionary history.

### *Different side chains in the substrate specificity pocket confer preferential cleavage*

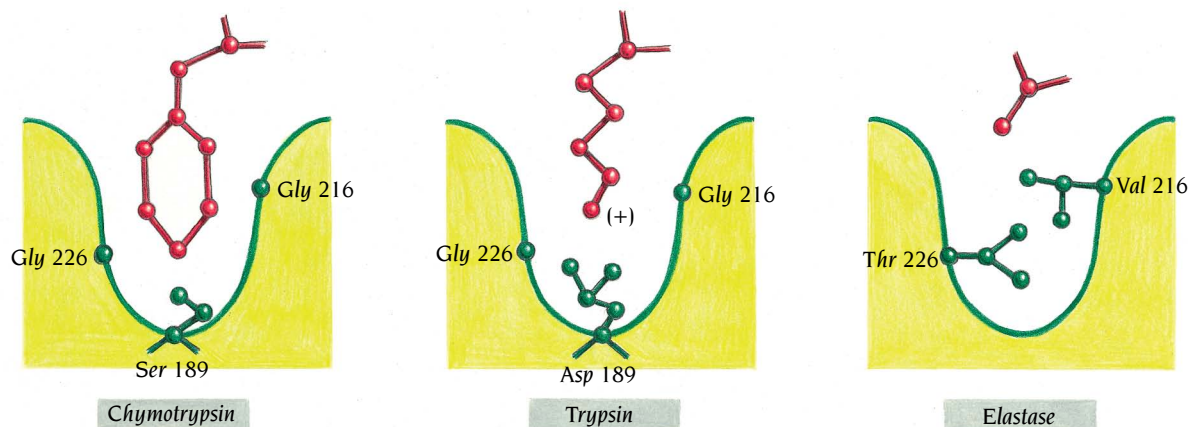
The serine proteinases all have the same substrate, namely, polypeptide chains of proteins. However, different members of the family preferentially cleave polypeptide chains at sites adjacent to different amino acid residues. The structural basis for this preference lies in the side chains that line the substrate specificity pocket in the different enzymes.

This is nicely illustrated by members of the chymotrypsin superfamily: the enzymes chymotrypsin, trypsin, and elastase have very similar three-dimensional structures but different specificity. They preferentially cleave adjacent to bulky aromatic side chains, positively charged side chains, and small uncharged side chains, respectively. Three residues, numbers 189, 216, and 226, are responsible for these preferences (Figure 11.11). Residues 216



**Figure 11.10** Topological diagram of the two domains of chymotrypsin, illustrating that the essential active-site residues are part of the same two loop regions (3–4 and 5–6, red) of the two domains. These residues form the catalytic triad, the oxyanion hole (green), and the substrate binding regions (yellow and blue) including essential residues in the specificity pocket.





**Figure 11.11** Schematic diagrams of the specificity pockets of chymotrypsin, trypsin and elastase, illustrating the preference for a side chain adjacent to the scissile bond in polypeptide substrates. Chymotrypsin prefers aromatic side chains and trypsin prefers positively charged side chains that can interact with Asp 189 at the bottom of the specificity pocket. The pocket is blocked in elastase, which therefore prefers small uncharged side chains.

and 226 line the sides of the pocket. In trypsin and chymotrypsin these are both glycine residues that allow side chains of the substrate to penetrate into the interior of the specificity pocket. In elastase they are Val and Thr, respectively, that fill up most of the pocket with hydrophobic groups (Figure 11.11). Consequently, elastase does not cleave adjacent to large or charged side chains but adjacent to small uncharged side chains instead.

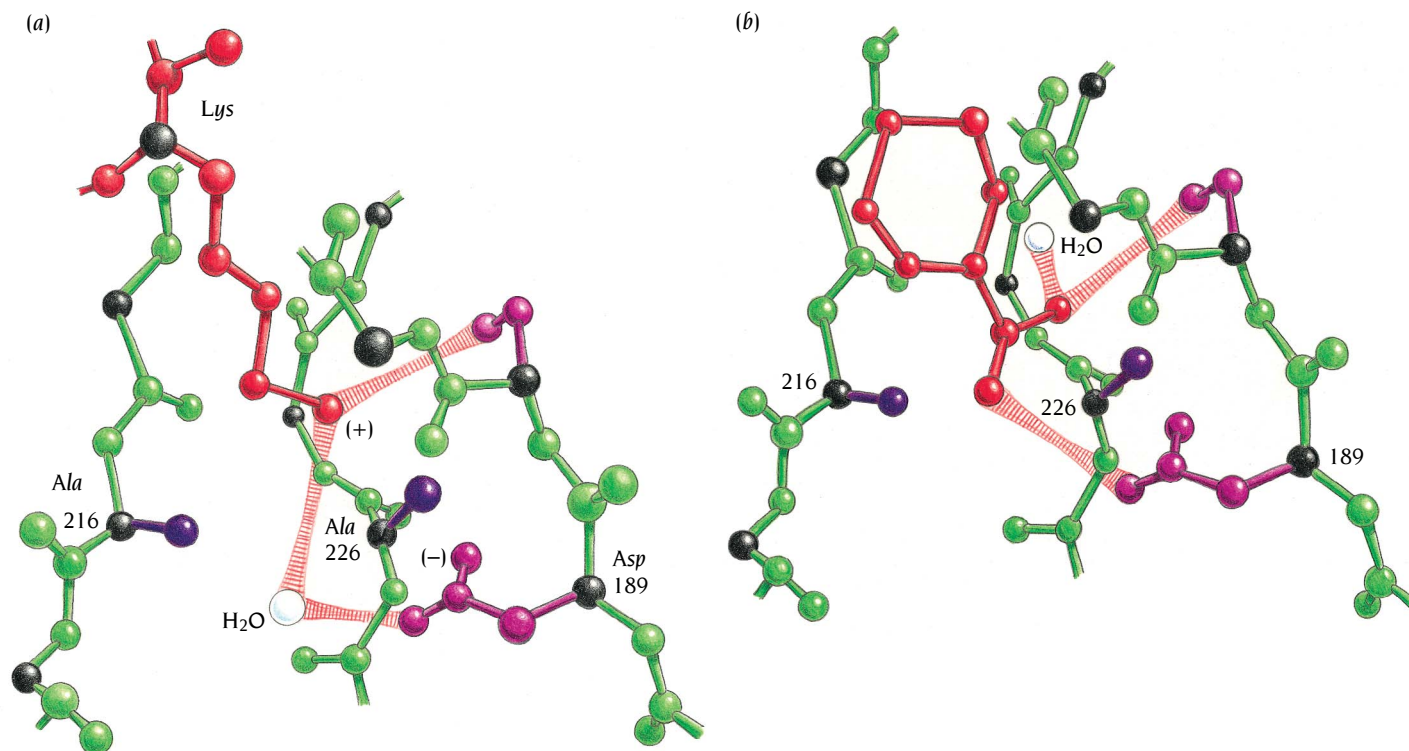
Residue 189 is at the bottom of the specificity pocket. In trypsin the Asp residue at this position interacts with the positively charged side chains Lys or Arg of a substrate. This accounts for the preference of trypsin to cleave adjacent to these residues. In chymotrypsin there is a Ser residue at position 189, which does not interfere with the binding of the substrate. Bulky aromatic groups are therefore preferred by chymotrypsin since such side chains fill up the mainly hydrophobic specificity pocket. It has now become clear, however, from site-directed mutagenesis experiments that this simple picture does not tell the whole story.

### Engineered mutations in the substrate specificity pocket change the rate of catalysis

How would substrate preference be changed if the glycine residues in trypsin at positions 216 and 226 were changed to alanine rather than to the more bulky valine and threonine groups that are present in elastase? This question was addressed by the groups of Charles Craik, William Rutter, and Robert Fletterick in San Francisco, who have made and studied three such trypsin mutants: one in which Ala is substituted for Gly at 216, one in which the same substitution is made at Gly 226, and a third containing both substitutions.

Model building shows that both Arg- and Lys-containing substrates should be accommodated by the substrate specificity pocket after these Gly to Ala changes but that some details of the binding mode at the bottom of the pocket would be altered. The Ala 226 substitution would introduce a methyl group in the region where the end of the substrate's side chain binds (Figure 11.12) and would therefore be expected to accommodate Lys better than Arg, since the latter has a longer and more bulky side chain. Based on these steric arguments alone, one would therefore predict that the  $K_m$  for an Arg-containing substrate would be larger (less favorable binding) and that the  $K_m$  for Lys would be essentially unaltered. The specificity constant,  $k_{cat}/K_m$ , would decrease more for an Arg-containing substrate than for one with Lys.

Model building also predicts that the Ala 216 mutant would displace a water molecule at the bottom of the specificity pocket that in the wild type enzyme binds to the  $NH_3^+$  group of the substrate Lys side chain (Figure 11.12). The extra  $CH_3$  group of this mutant is not expected to disturb the binding of the Arg side chain. One would therefore expect that the  $K_m$  for Lys



**Figure 11.12** Schematic diagram of the specificity pocket of mutant trypsin with Ala (purple) at positions 216 and 226. (a) The position of a bound Lys side chain (red) in this pocket as observed in the structure of a complex between trypsin (green) and a peptide inhibitor. The  $\text{NH}_3^+$  group of the Lys side chain interacts with the  $\text{COO}^-$  group of Asp 189 through a water molecule. (b) No structure is available for an Arg side chain in the substrate specificity pocket of trypsin. It is assumed that the complex of trypsin (green) with benzamide (red) is a good model for arginine binding in this pocket. One  $\text{NH}_2$  group of benzamide interacts directly with the  $\text{COO}^-$  group of Asp 189 and the second  $\text{NH}_2$  group interacts with a water molecule and the OH group of Ser 190. (Adapted from C.S. Craik et al., *Science* 228: 291–297, 1985.)

substrates would increase and therefore  $k_{\text{cat}}/K_m$  would decrease more for Lys- than for Arg-containing substrates. For the double mutant where both Gly 216 and Gly 226 are changed to Ala, one would predict an increase in the  $K_m$  values for both Lys- and Arg-containing substrates.

The experimentally determined  $k_{\text{cat}}$  and  $K_m$  values for the wild-type enzyme and the mutants are shown in Table 11.1. The dramatic kinetic effects of these mutations are best illustrated with the Arg substrate. The three mutants have roughly similar  $K_m$  values 15–35 times higher than for the wild type, but the  $k_{\text{cat}}$  values decrease by factors of 10 to about 1000. The mutants were designed to change the specificity, but by far the largest changes occur in the catalytic rates. Apparently, these mutations affect the structure of the enzyme in additional ways, possibly by causing conformational changes outside the specificity pocket, so that the stabilization of the transition state is reduced and consequently the activation energies for the reactions are different.

The changes in the specificity constants, on the other hand, were as expected from the predictions. The ratio of the  $k_{\text{cat}}/K_m$  values for the Arg and Lys substrates (last column in Table 11.1) gives a measure of the relative specificities. This ratio decreases for the Ala 226 mutant and increases for the Ala 216 mutant as predicted. However, the changes in these values depend not

**Table 11.1** Kinetic data for wild-type and mutant trypsins

| Enzyme               | Arg              |       |                      | Lys              |       |                      | $(k_{\text{cat}}/K_m)_{\text{Arg}}$<br>$(k_{\text{cat}}/K_m)_{\text{Lys}}$ |
|----------------------|------------------|-------|----------------------|------------------|-------|----------------------|----------------------------------------------------------------------------|
|                      | $k_{\text{cat}}$ | $K_m$ | $k_{\text{cat}}/K_m$ | $k_{\text{cat}}$ | $K_m$ | $k_{\text{cat}}/K_m$ |                                                                            |
| Wild type            | 1                | 1     | 1                    | 0.9              | 10    | 0.1                  | 10                                                                         |
| Gly 216, Gly 226→Ala | 0.001            | 15    | 0.0001               | 0.0005           | 25    | 0.00002              | 25                                                                         |
| Gly 226→Ala          | 0.01             | 35    | 0.0003               | 0.1              | 250   | 0.0005               | 0.5                                                                        |
| Gly 216→Ala          | 0.7              | 30    | 0.02                 | 0.2              | 280   | 0.001                | 20                                                                         |

The substrates used were D-Val-Leu-Arg-amino fluorocoumarin (Arg) and D-Val-Leu-Lys-amino fluorocoumarin (Lys). For clarity the  $K_m$  and  $k_{\text{cat}}$  values have been normalized to those of the wild-type enzyme for the Arg substrate.

only on changes in the  $K_m$  values, which reflect binding of substrate, but even more on changes in the  $k_{cat}$  values, which reflect catalytic rate. It can, therefore, be argued that the agreement with prediction is fortuitous.

The simple lesson to be learnt from these experiments is that critical amino acid residues can have pleiotropic roles in determining a protein's structure and therefore its function.

### *The Asp 189-Lys mutation in trypsin causes unexpected changes in substrate specificity*

Asp 189 at the bottom of the substrate specificity pocket interacts with Lys and Arg side chains of the substrate, and this is the basis for the preferred cleavage sites of trypsin (see Figures 11.11 and 11.12). It is almost trivial to infer, from these observations, that a replacement of Asp 189 with Lys would produce a mutant that would prefer to cleave substrates adjacent to negatively charged residues, especially Asp. On a computer display, similar Asp-Lys interactions between enzyme and substrate can be modeled within the substrate specificity pocket but reversed compared with the wild-type enzyme.

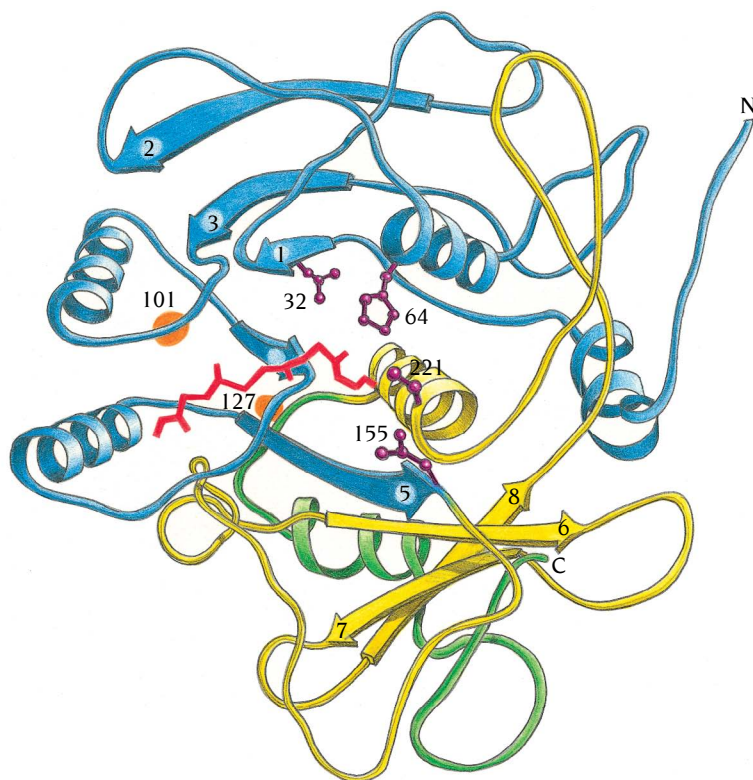
The results of experiments in which the mutation was made were, however, a complete surprise. The Asp 189-Lys mutant was totally inactive with both Asp and Glu substrates. It was, as expected, also inactive toward Lys and Arg substrates. The mutant was, however, catalytically active with Phe and Tyr substrates, with the same low turnover number as wild-type trypsin. On the other hand, it showed a more than 5000-fold increase in  $k_{cat}/K_m$  with Leu substrates over wild type. The three-dimensional structure of this interesting mutant has not yet been determined, but the structure of a related mutant Asp 189-His shows the histidine side chain in an unexpected position, buried inside the protein.

As these experiments with engineered mutants of trypsin prove, we still have far too little knowledge of the functional effects of single point mutations to be able to make accurate and comprehensive predictions of the properties of a point-mutant enzyme, even in the case of such well-characterized enzymes as the serine proteinases. Predictions of the properties of mutations using computer modeling are not infallible. Once produced, the mutant enzymes often exhibit properties that are entirely surprising, but they may be correspondingly informative.

### *The structure of the serine proteinase subtilisin is of the $\alpha/\beta$ type*

Subtilisins are a group of serine proteinases that are produced by different species of bacilli. These enzymes are of considerable commercial interest because they are added to the detergents in washing powder to facilitate removal of proteinaceous stains. Numerous attempts have therefore recently been made to change by protein engineering such properties of the subtilisin molecule as its thermal stability, pH optimum, and specificity. In fact, in 1988 subtilisin mutants were the subject of the first US patent granted for an engineered protein.

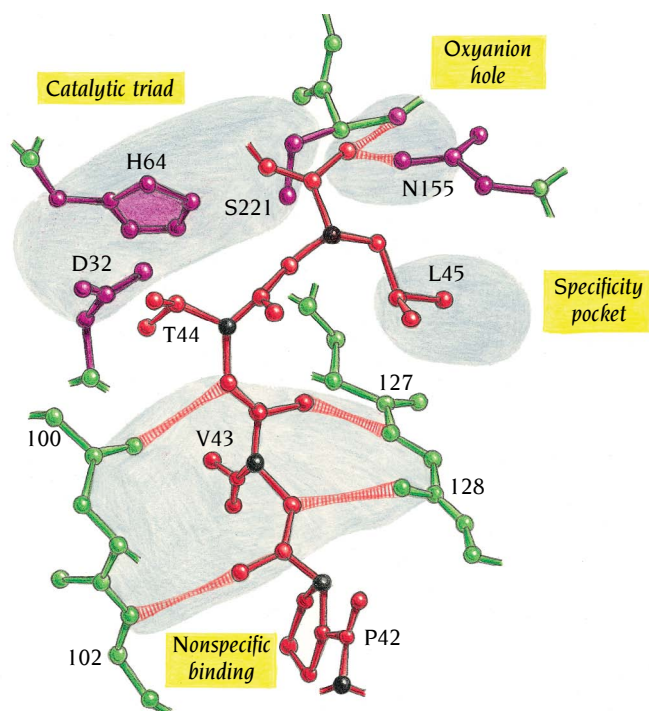
The subtilisin molecule is a single polypeptide chain of 275 amino acids with no similarities in the amino acid sequence to chymotrypsin. The three-dimensional structure of subtilisin BPN' from *Bacillus amyloliquefaciens* was determined in 1969 by the group of Joseph Kraut in San Diego, California, and that of subtilisin Novo in 1971 by the group of Jan Drenth in Groningen, The Netherlands. The main feature of the subtilisin structure is a region of five parallel  $\beta$  strands (blue in Figure 11.13) surrounded by four helices, two on each side of the parallel  $\beta$  sheet. This  $\alpha/\beta$  structure is thus quite different from the double antiparallel  $\beta$ -barrel structure of chymotrypsin (compare with Figure 11.7).



**Figure 11.13** Schematic diagram of the three-dimensional structure of subtilisin viewed down the central parallel  $\beta$  sheet. The N-terminal region that contains the  $\alpha/\beta$  structure is blue. It is followed by a yellow region, which ends with the fourth  $\alpha$  helix of the  $\alpha/\beta$  structure. The C-terminal part is green. The catalytic triad Asp 32, His 64, and Ser 221 as well as Asn 155, which forms part of the oxyanion hole are shown in purple. The main chain of part of a polypeptide inhibitor is shown in red. Main-chain residues around 101 and 127 (orange circles) form the nonspecific binding regions of peptide substrates.

### *The active sites of subtilisin and chymotrypsin are similar*

The active site of subtilisin is outside the carboxy ends of the central  $\beta$  strands analogous to the position of the binding sites in other  $\alpha/\beta$  proteins as discussed in Chapter 4. Details of this active site are surprisingly similar to those of chymotrypsin, in spite of the completely different folds of the two enzymes (Figures 11.14 and 11.9). A catalytic triad is present that comprises residues Asp 32, His 64 and the reactive Ser 221. The negatively charged oxygen atom of the tetrahedral transition state binds in an oxyanion hole,



**Figure 11.14** Schematic diagram of the active site of subtilisin. A region (residues 42–45) of a bound polypeptide inhibitor, eglin, is shown in red. The four essential features of the active site—the catalytic triad, the oxyanion hole, the specificity pocket, and the region for non-specific binding of substrate—are highlighted in yellow. Important hydrogen bonds between enzyme and inhibitor are striped. This figure should be compared to Figure 11.9, which shows the same features for chymotrypsin. (Adapted from W. Bode et al., *EMBO J.* 5: 813–818, 1986.)



forming hydrogen bonds with the side-chain amide group of Asn 155 and the main-chain nitrogen atom of Ser 221. Peptide substrates and inhibitors bind nonspecifically by forming a small antiparallel pleated sheet, which in subtilisin comprises three  $\beta$  strands (Figure 11.14). There is also a hydrophobic specificity pocket adjacent to the scissile bond.

All the four essential features of the active site of chymotrypsin are thus also present in subtilisin. Furthermore, these features are spatially arranged in the same way in the two enzymes, even though different framework structures bring different loop regions into position in the active site. This is a classical example of convergent evolution at the molecular level.

### *A structural anomaly in subtilisin has functional consequences*

There is one anomalous and puzzling feature of the subtilisin structure. We mentioned in Chapter 4 that virtually all  $\beta$ - $\alpha$ - $\beta$  motifs were of the same hand, they were right-handed. Subtilisin contains the one exception to this general rule, which is illustrated in the topology diagram Figure 11.15. There are three  $\beta$ - $\alpha$ - $\beta$  motifs in subtilisin,  $\beta_2$ - $\alpha_B$ - $\beta_3$ ,  $\beta_3$ - $\alpha_C$ - $\beta_4$ , and  $\beta_4$ - $\alpha_D$ - $\beta_5$ . If these motifs were of the same hand, the three  $\alpha$  helices  $\alpha_B$ ,  $\alpha_C$ , and  $\alpha_D$  would be on the same side of the  $\beta$  sheet. However,  $\alpha_B$  is beneath the sheet in the topology diagram in contrast to the other two helices because  $\beta_2$ - $\alpha_B$ - $\beta_3$  is left-handed. Why has this exception to the general rule of right-handed  $\beta$ - $\alpha$ - $\beta$  motifs evolved?

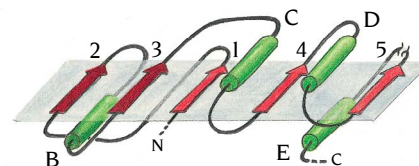
The answer is quite clear. His 64, which is part of the catalytic triad, is in the first turn of helix  $\alpha_B$  (Figure 11.13). This helix would be on the other side of the  $\beta$  sheet, far removed from the active site if the motif  $\beta_2$ - $\alpha_B$ - $\beta_3$  were right-handed. Therefore, to produce a proper catalytic triad of Asp 32, His 64, and Ser 221, helix  $\alpha_B$  must be on the same side of the  $\beta$  sheet as Ser 221; consequently, the motif has evolved to be left-handed.

### *Transition-state stabilization in subtilisin is dissected by protein engineering*

Two essential features are required to stabilize the covalent tetrahedral transition state in serine proteinases—the oxyanion hole, which provides hydrogen bonds to the negatively charged oxygen atom in the transition state, and the histidine residue of the catalytic triad, which provides a positive charge. The charge on this histidine is, in turn, stabilized by the aspartic acid side chain of the catalytic triad (Figure 11.6). The histidine residue also plays a second role in the catalytic mechanism by accepting a proton from the reactive serine residue and then donating that proton to the nitrogen atom of the leaving group. The effects on the catalytic rate of the different side chains involved in the catalytic triad and the oxyanion hole have been examined by P. Carter, J.A. Wells, and D. Estell at Genentech, USA, by analyses of mutants of subtilisin with one or several of these side chains have been changed.

### *Catalysis occurs without a catalytic triad*

By changing Ser 221 in subtilisin to Ala the reaction rate (both  $k_{\text{cat}}$  and  $k_{\text{cat}}/K_m$ ) is reduced by a factor of about  $10^6$  compared with the wild-type enzyme. The  $K_m$  value and, by inference, the initial binding of substrate are essentially unchanged. This mutation prevents formation of the covalent bond with the substrate and therefore abolishes the reaction mechanism outlined in Figure 11.5. When the Ser 221 to Ala mutant is further mutated by changes of His 64 to Ala or Asp 32 to Ala or both, as expected there is no effect on the catalytic reaction rate, since the reaction mechanism that involves the catalytic triad is no longer in operation. However, the enzyme still has an appreciable catalytic effect; peptide hydrolysis is still about  $10^3$ – $10^4$  times the nonenzymatic rate. Whatever the reaction mechanism



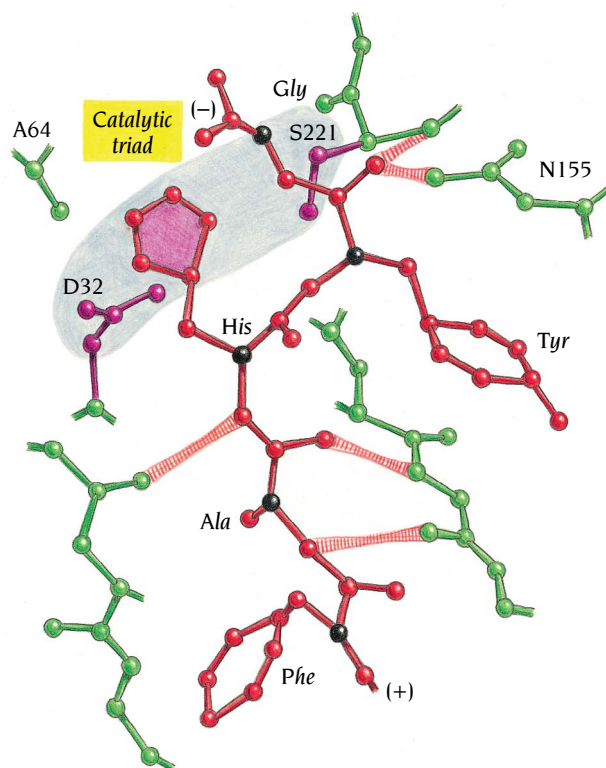
**Figure 11.15** Topology diagram of the  $\alpha/\beta$  region of subtilisin illustrating that  $\beta_2$ - $\alpha_B$ - $\beta_3$  has a different hand than the other  $\beta$ - $\alpha$ - $\beta$  units.

used by these mutants, it is apparent that the remaining parts of the active site must bind more tightly to the substrate in its transition state than in its initial state, thereby giving a higher reaction rate than in the absence of enzyme.

### **Substrate molecules provide catalytic groups in substrate-assisted catalysis**

The single mutation His 64→Ala decreases the reaction rate of subtilisin for most substrates by the same factor (approx.  $10^6$ ) as the mutation of Ser 221. This histidine (His 64), therefore, seems to be as important as Ser 221 for the formation of a covalent tetrahedral intermediate. However, model building suggested that it might be possible at least partly to compensate for the loss of this histidine in the catalytic triad of the mutant protein with a histidine side chain from a peptide substrate (Figure 11.16). Experiments confirmed this prediction and showed that the mutant His 64→Ala catalyzes hydrolysis of a peptide substrate about 400 times faster when the peptide has histidine at the appropriate position in its sequence. However, the rate is still several orders of magnitude below the rate of the wild-type enzyme, presumably because of the slightly different position and orientation of the histidine side chain. Nevertheless, the principle of **substrate-assisted catalysis** has been demonstrated: an essential group that is lacked by a mutant enzyme can be replaced by a similar group from the substrate. One consequence of substrate-assisted catalysis is that the mutant enzyme is highly specific for substrates containing the essential group. The His 64→Ala mutant of subtilisin, for example, has a specificity factor (ratios of  $k_{\text{cat}}/K_m$ ) of about 200 for substrates containing histidine.

The single mutation Asp 32→Ala reduces the catalytic reaction rate by a factor of about  $10^4$  compared with wild type. This rate reduction reflects the role of Asp 32 in stabilizing the positive charge that His 64 acquires in the transition state. A similar reduction of  $k_{\text{cat}}$  and  $k_{\text{cat}}/K_m$  ( $2.5 \times 10^3$ ) is obtained for the single mutant Asn 155→Thr. Asn 155 provides one of the two hydrogen bonds to the substrate transition state in the oxyanion hole of subtilisin.



**Figure 11.16** Substrate-assisted catalysis. Schematic diagram from model building of a substrate,  $\text{NH}_2\text{-Phe-Ala-His-Tyr-Gly-COOH}$  (red), bound to the subtilisin mutant His 64→Ala. The diagram illustrates that the His residue of the substrate can occupy roughly the same position in this mutant as His 64 in wild-type subtilisin (see Figure 11.14) and thereby partly restore the catalytic triad.

Model building shows that the OH group of Thr in the mutant is too far away to provide such a hydrogen bond. The loss of this feature of the stabilization of the transition state thus reduces the rate by more than a thousandfold.

The subtilisin mutants described here illustrate the power of protein engineering as a tool to allow us to identify the specific roles of side chains in the catalytic mechanisms of enzymes. In Chapter 17 we shall discuss the utility of protein engineering in other contexts, such as design of novel proteins and the elucidation of the energetics of ligand binding to proteins.

## Conclusion

Enzymes increase the rate of chemical reactions by decreasing the activation energy of the reactions. This is achieved primarily by the enzyme preferentially binding to the transition state of the substrate. Catalytic groups of the enzyme are required to achieve a specific reaction path for the conversion of substrate to product.

Serine proteinases such as chymotrypsin and subtilisin catalyze the cleavage of peptide bonds. Four features essential for catalysis are present in the three-dimensional structures of all serine proteinases: a catalytic triad, an oxyanion binding site, a substrate specificity pocket, and a nonspecific binding site for polypeptide substrates. These four features, in a very similar arrangement, are present in both chymotrypsin and subtilisin even though they are achieved in the two enzymes in completely different ways by quite different three-dimensional structures. Chymotrypsin is built up from two  $\beta$ -barrel domains, whereas the subtilisin structure is of the  $\alpha/\beta$  type. These two enzymes provide an example of convergent evolution where completely different loop regions, attached to different framework structures, form similar active sites.

The catalytic triad consists of the side chains of Asp, His, and Ser close to each other. The Ser residue is reactive and forms a covalent bond with the substrate, thereby providing a specific pathway for the reaction. His has a dual role: first, it accepts a proton from Ser to facilitate formation of the covalent bond; and, second, it stabilizes the negatively charged transition state. The proton is subsequently transferred to the N atom of the leaving group. Mutations of either of these two residues decrease the catalytic rate by a factor of  $10^6$  because they abolish the specific reaction pathway. Asp, by stabilizing the positive charge of His, contributes a rate enhancement of  $10^4$ .

The oxyanion binding site stabilizes the transition state by forming two hydrogen bonds to a negatively charged oxygen atom of the substrate. Mutations that prevent formation of one of these bonds in subtilisin decrease the rate by a factor of about  $10^3$ .

Mutations in the specificity pocket of trypsin, designed to change the substrate preference of the enzyme, also have drastic effects on the catalytic rate. These mutants demonstrate that the substrate specificity of an enzyme and its catalytic rate enhancement are tightly linked to each other because both are affected by the difference in binding strength between the transition state of the substrate and its normal state.

## Selected readings

### General

- Blow, D.M. Structure and mechanism of chymotrypsin. *Acc. Chem. Res.* 9: 145–152, 1976.
- Fersht, A. *Enzyme Structure and Mechanism*, 2nd ed. New York: W.H. Freeman, 1984.
- Huber, R., Bode, W. Structural basis of the activation and action of trypsin. *Acc. Chem. Res.* 11: 114–122, 1978.
- James, M.N.G. An x-ray crystallographic approach to enzyme structure and function. *Can. J. Biochem.* 58: 251–270, 1980.
- Jencks, W.P. Binding energy, specificity, and enzymatic catalysis: the Circe effect. *Adv. Enzymol.* 43: 219–410, 1975.
- Knowles, J.R. Tinkering with enzymes: what are we learning? *Science* 236: 1252–1258, 1987.
- Kraut, J. How do enzymes work? *Science* 242: 533–540, 1988.
- Kraut, J. Serine proteases: structure and mechanism of catalysis. *Annu. Rev. Biochem.* 46: 331–358, 1977.
- Neurath, H. Evolution of proteolytic enzymes. *Science* 224: 350–357, 1984.
- Pauling, L. Nature of forces between large molecules of biological interest. *Nature* 161: 707–709, 1948.
- Steitz, T.A., Shulman, R.G. Crystallographic and NMR studies of the serine proteases. *Annu. Rev. Biochem. Biophys.* 11: 419–444, 1982.
- Stroud, R.M. A family of protein-cutting proteins. *Sci. Am.* 231(1): 74–88, 1974.
- Walsh, C. *Enzymatic Reaction Mechanisms*. New York: W.H. Freeman, 1979.
- Warshel, A., et al. How do serine proteases really work? *Biochemistry* 28:3629–3637, 1989.
- Wells, J.A., et al. On the evolution of specificity and catalysis in subtilisin. *Cold Spring Harbor Symp. Quant. Biol.* 52: 647–652, 1987.
- Carter, P., Wells, J.A. Engineering enzyme specificity by “substrate-assisted catalysis.” *Science* 237: 394–399, 1987.
- Craik, C.S., et al. Redesigning trypsin: alteration of substrate specificity. *Science* 228: 291–297, 1985.
- Craik, C.S., et al. The catalytic role of the active site aspartic acid in serine proteases. *Science* 237: 909–913, 1987.
- Cunningham, B.C., Wells, J.A. Improvement in the alkaline stability of subtilisin using an efficient random mutagenesis and screening procedure. *Prot. Eng.* 1: 319–325, 1987.
- Drenth, J., et al. Subtilisin novo. The three-dimensional structure and its comparison with subtilisin BPN. *Eur. J. Biochem.* 26: 177–181, 1972.
- Estell, D.A., Graycar, T.P., Wells, J.A. Engineering an enzyme by site-directed mutagenesis to be resistant to chemical oxidation. *J. Biol. Chem.* 260: 6518–6521, 1985.
- Fehlhammer, H., Bode, W., Huber, R. Crystal structure of bovine trypsinogen at 1.8 Å resolution. II. Crystallographic refinement, refined crystal structure and comparison with bovine trypsin. *J. Mol. Biol.* 111: 415–438, 1977.
- Fujinaga, M., et al. Crystal and molecular structures of the complex of  $\alpha$ -chymotrypsin with its inhibitor turkey ovomucoid third domain at 1.8 Å resolution. *J. Mol. Biol.* 195: 397–418, 1987.
- Graf, L., et al. Selective alteration of substrate specificity by replacement of aspartic acid 189 with lysine in the binding pocket of trypsin. *Biochemistry* 26: 2616–2623, 1987.
- Grütter, M.G., et al. Crystal structure of the thrombin-hirudin complex: a novel mode of serine protease inhibition. *EMBO J.* 9: 2361–2365, 1990.
- James, M.N.G., et al. Structures of product and inhibitor complexes of *Streptomyces griseus* protease A at 1.8 Å resolution. A model for serine protease catalysis. *J. Mol. Biol.* 144: 43–88, 1980.

### Specific structures

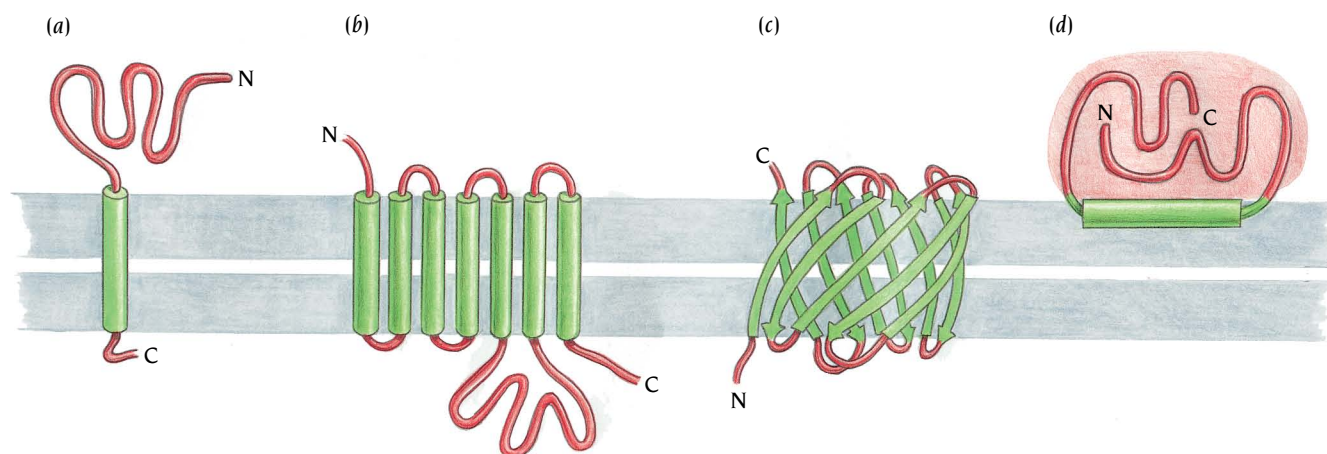
- Bode, W., et al. Refined 1.2 Å crystal structure of the complex formed between subtilisin Carlsberg and the inhibitor eglin c. Molecular structure of eglin and its detailed interaction with subtilisin. *EMBO J.* 5: 813–818, 1986.
- Bode, W., et al. The refined 1.9 Å crystal structure of human  $\alpha$ -thrombin: interaction with D-Phe-Pro-Arg chloromethylketone and significance of the Tyr-Pro-Pro-Trp insertion segment. *EMBO J.* 8: 3467–3475, 1989.
- Bryan, P., et al. Site-directed mutagenesis and the role of the oxyanion hole in subtilisin. *Proc. Natl. Acad. Sci. USA* 83: 3743–3745, 1986.
- Carter, P., Wells, J.A. Dissecting the catalytic triad of a serine protease. *Nature* 332: 564–568, 1988.
- Krieger, M., Kay, L.M., Stroud, R.M. Structure and specific binding of trypsin: comparison of inhibited derivatives and a model for substrate binding. *J. Mol. Biol.* 83: 209–230, 1974.
- Matthews, B.W., Sigler, P.B., Henderson, R., Blow, D.M. Three-dimensional structure of tosyl- $\alpha$ -chymotrypsin. *Nature* 214: 652–656, 1967.
- McLachlan, A.D. Gene duplications in the structural evolution of chymotrypsin. *J. Mol. Biol.* 128: 49–79, 1979.
- Poulos, T.L., et al. Polypeptide halomethyl ketones bind to serine proteases as analogs of the tetrahedral intermediate. *J. Biol. Chem.* 251: 1097–1103, 1976.
- Read, R.J., James, M.N.G. Refined crystal structure of *Streptomyces griseus* trypsin at 1.7 Å resolution. *J. Mol. Biol.* 200: 523–551, 1988.



- Rühlman, A., et al. Structure of the complex formed by bovine trypsin and bovine pancreatic trypsin inhibitor. *J. Mol. Biol.* 77: 417–436, 1973.
- Shotton, D.M., Watson, H.C. Three-dimensional structure of tosyl-elastase. *Nature* 225: 811–816, 1970.
- Sigler, P.B., et al. Structure of crystalline  $\alpha$ -chymotrypsin II. A preliminary report including a hypothesis for the activation mechanism. *J. Mol. Biol.* 35: 143–164, 1968.
- Smith, S.O., et al. Crystal versus solution structures of enzymes: NMR spectroscopy of a crystalline serine protease. *Science* 244: 961–964, 1989.
- Sprang, S., et al. The three-dimensional structure of Asn<sup>102</sup> mutant of trypsin: role of Asp<sup>102</sup> in serine protease catalysis. *Science* 237: 905–909, 1987.
- Thomas, P.G., Russel, A.J., Fersht, A. Tailoring the pH dependence of enzyme catalysis using protein engineering. *Nature* 318: 375–376, 1985.
- Tsukada, H., Blow, D.M. Structure of  $\alpha$ -chymotrypsin refined at 1.68 Å resolution. *J. Mol. Biol.* 184: 703–711, 1985.
- Wang, D., Bode, W., Huber, R. Bovine chymotrypsinogen A. X-ray crystal structure analysis and refinement of a new crystal form at 1.8 Å resolution. *J. Mol. Biol.* 185: 595–624, 1985.
- Wells, J.A., et al. Designing substrate specificity by protein engineering of electrostatic interactions. *Proc. Natl. Acad. Sci. USA* 84: 1219–1223, 1987.
- Wells, J.A., et al. Recruitment of substrate-specificity properties from one enzyme into a related one by protein engineering. *Proc. Natl. Acad. Sci. USA* 84: 5167–5171, 1987.
- Wright, C.S., Alden, R.A., Kraut, J. Structure of subtilisin BPN' at 2.5 Å resolution. *Nature* 221: 235–242, 1969.

Cells and organelles within them are bounded by membranes, which are extremely thin (4.5 nm) films of lipids and protein molecules. The lipids form a bilayered sheet structure that is hydrophilic on its two outer surfaces and hydrophobic in between. Protein molecules are embedded in this layer, and in the simplest case they are arranged with three distinct regions: one hydrophobic transmembrane segment and two hydrophilic regions, one on each side of the membrane. Those proteins whose polypeptide chain traverses the membrane only once usually form functional globular domains on at least one side of the membrane (Figure 12.1a). Often these can be cleaved off by proteolytic enzymes. The hemagglutinin and neuraminidase of influenza virus (discussed in Chapter 5), G-proteins and receptors (discussed in Chapter 13), and HLA proteins (discussed in Chapter 15) are examples of such cleavage products that can be handled as functional soluble globular domains. The polypeptide chain of other transmembrane proteins passes through the membrane several times, usually as  $\alpha$  helices but in some cases as  $\beta$  strands (Figure 12.1b,c). In these cases the hydrophilic regions on either side of the membrane are the termini of the chain and the loops between the membrane-spanning parts. Proteolytic cleavage of these hydrophilic regions produces a number of fragments, and function is not preserved. Some proteins do not traverse the membrane but are instead attached to one side either through  $\alpha$  helices that lie parallel to the membrane surface (Figure 12.1d) or by fatty acids, covalently linked to the protein, that intercalate in the lipid bilayer of the membrane.

**Figure 12.1** Four different ways in which protein molecules may be bound to a membrane. Membrane-bound regions are green and regions outside the membrane are red. Alpha-helices are drawn as cylinders and  $\beta$  strands as arrows. From left to right are (a) a protein whose polypeptide chain traverses the membrane once as an  $\alpha$  helix, (b) a protein that forms several transmembrane  $\alpha$  helices connected by hydrophilic loop regions, (c) a protein with several  $\beta$  strands that form a channel through the membrane, and (d) a protein that is anchored to the membrane by one  $\alpha$  helix parallel to the plane of the membrane.



A biological membrane functions basically as a permeability barrier that establishes discrete compartments and prevents the random mixing of the contents of one compartment with those of another. However, biological membranes are more than passive containers. The embedded proteins serve as highly active mediators between the cell and its environment or the interior of an organelle and the cytosol. They catalyze specific transport of metabolites and ions across the membrane barriers. They convert the energy of sunlight into chemical and electrical energy, and they couple the flow of electrons to the synthesis of ATP. Furthermore, they act as signal receptors and transduce signals across the membrane. The signals can be, for example, neurotransmitters, growth factors, hormones, light or chemotactic stimuli. The transmembrane proteins of the plasma membrane are also involved in cell–cell recognition.

In this chapter we describe some examples of structures of membrane-bound proteins known to high resolution, and outline how the elucidation of these structures has contributed to understanding the specific function of these proteins, as well as some general principles for the construction of membrane-bound proteins. In Chapter 13 we describe some examples of the domain organization of receptor families and their associated proteins involved in signal transduction through the membrane.

### *Membrane proteins are difficult to crystallize*

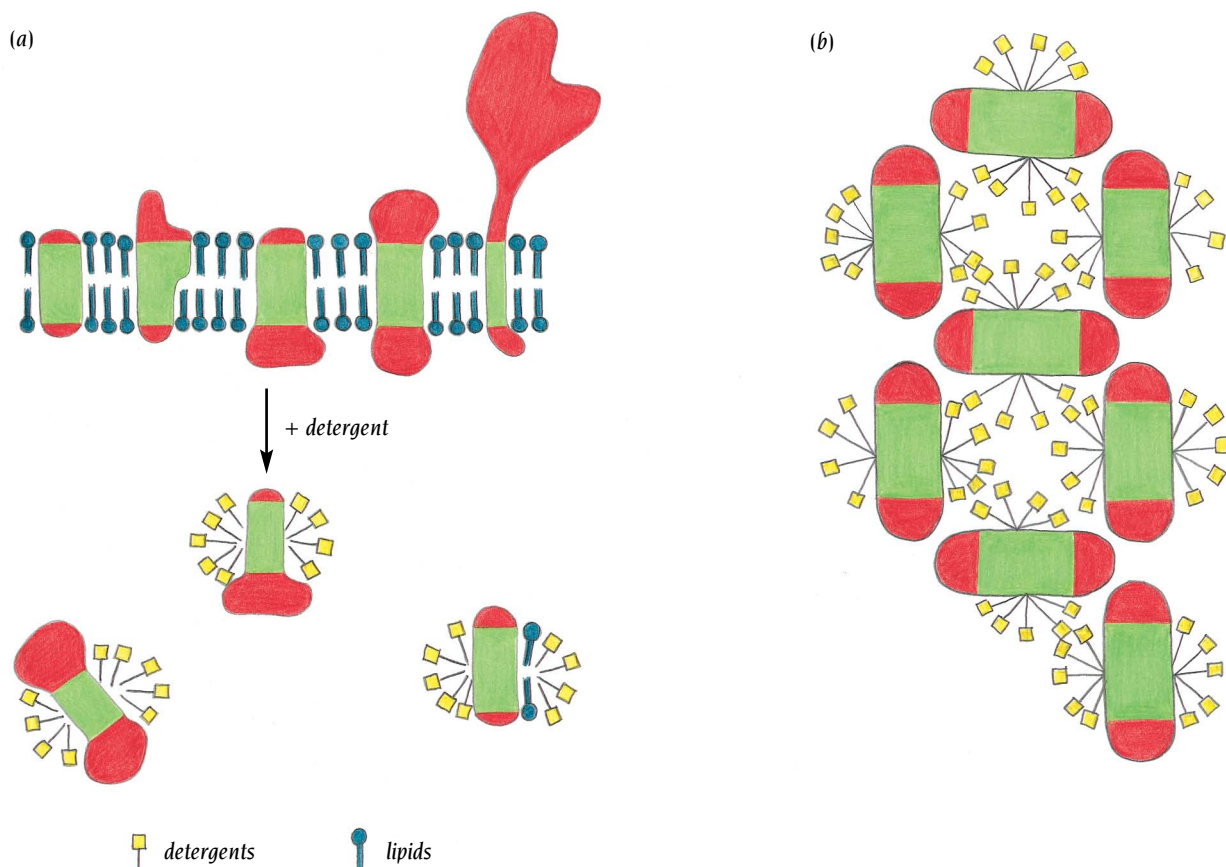
Membrane proteins, which have both hydrophobic and hydrophilic regions on their surfaces, are not soluble in aqueous buffer solutions and denature in organic solvents. However, if detergents, such as octylglucoside, are added to an aqueous solution, these proteins can be solubilized and purified in their native conformation. The hydrophobic part of the detergent molecules binds to the protein's hydrophobic surfaces, while the detergents' polar head-groups face the solution (Figure 12.2a). In this way the protein–detergent complex acquires an essentially hydrophilic surface with the hydrophobic parts buried inside the complex.

Such solubilized protein–detergent complexes are the starting material for purification and crystallization. For some proteins, the addition of small amphipathic molecules to the detergent-solubilized protein promotes crystallization, probably by facilitating proper packing interactions between the molecules in all three dimensions in a crystal (Figure 12.2b). Therefore, many different amphipathic molecules are added in separate crystallization experiments until, by trial and error, the correct one is found.

Despite considerable efforts very few membrane proteins have yielded crystals that diffract x-rays to high resolution. In fact, only about a dozen such proteins are currently known, among which are porins (which are outer membrane proteins from bacteria), the enzymes cytochrome c oxidase and prostaglandin synthase, and the light-harvesting complexes and photosynthetic reaction centers involved in photosynthesis. In contrast, many other membrane proteins have yielded small crystals that diffract poorly, or not at all, using conventional x-ray sources. However, using the most advanced synchrotron sources (see Chapter 18) it is now possible to determine x-ray structures from protein crystals as small as 20  $\mu\text{m}$  wide which will permit more membrane protein structures to be elucidated.

### *Novel crystallization methods are being developed*

These difficulties have prompted a search for novel techniques for crystallization of membrane proteins. Two approaches have given promising results; one using antibodies to solubilize the proteins and the second using continuous lipidic phases as crystallization media. Complexes with specific antibodies have larger polar surfaces than the membrane protein itself and are therefore likely to form crystals more easily in an aqueous environment. A recent example of an antibody–membrane protein complex utilized an  $F_v$



fragment (see Chapter 15) to crystallize a bacterial cytochrome c oxidase. In these crystals the major packing interactions are formed by the polar surfaces of the complex.

A continuous lipidic cubic phase is obtained by mixing a long-chain lipid such as monoolein with a small amount of water. The result is a highly viscous state where the lipids are packed in curved continuous bilayers extending in three dimensions and which are interpenetrated by communicating aqueous channels. Crystallization of incorporated proteins starts inside the lipid phase and growth is achieved by lateral diffusion of the protein molecules to the nucleation sites. This system has recently been used to obtain three-dimensional crystals  $20 \times 20 \times 8 \mu\text{m}$  in size of the membrane protein bacteriorhodopsin, which diffracted to  $2 \text{ \AA}$  resolution using a microfocus beam at the European Synchrotron Radiation Facility.

### Two-dimensional crystals of membrane proteins can be studied by electron microscopy

The first really useful information about the structure of membrane proteins came not from x-ray crystallography but from high-resolution electron microscopy of **two-dimensional crystals**. Two-dimensional crystals can be thought of as crystalline membranes in which the membrane protein is arranged on a two-dimensional lattice. Naturally abundant membrane proteins sometimes form two-dimensional crystals *in vivo* or in isolated native membranes, in particular when some components such as lipids or other proteins are selectively extracted. A different way of making two-dimensional crystals is by reconstitution of detergent-solubilized membrane proteins into bilayers, which provide a natural, membrane-like environment for the protein. When the detergent is removed from a lipid-protein detergent mixture by dialysis or absorption, the hydrophobic effect causes the hydrophobic fatty acid tails of the lipids to associate with each other, and

**Figure 12.2** (a) Schematic drawing of membrane proteins in a typical membrane and their solubilization by detergents. The hydrophilic surfaces of the membrane proteins are indicated by red. (b) A membrane protein crystallized with detergents bound to its hydrophobic protein surface. The hydrophilic surfaces of the proteins and the symbols for detergents are as in (a). (Adapted from H. Michel, *Trends Biochem. Sci.* 8: 56–59, 1983.)

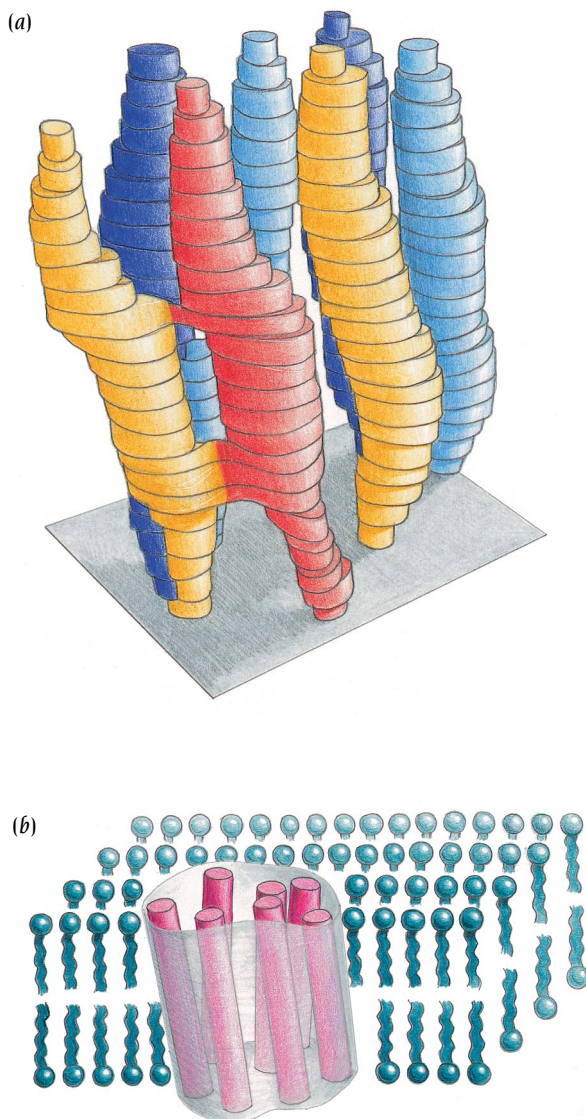


with the hydrophobic surface of the membrane protein. In this way, the protein is incorporated into lipid sheets or vesicles. In favorable conditions it can then form a crystalline lattice.

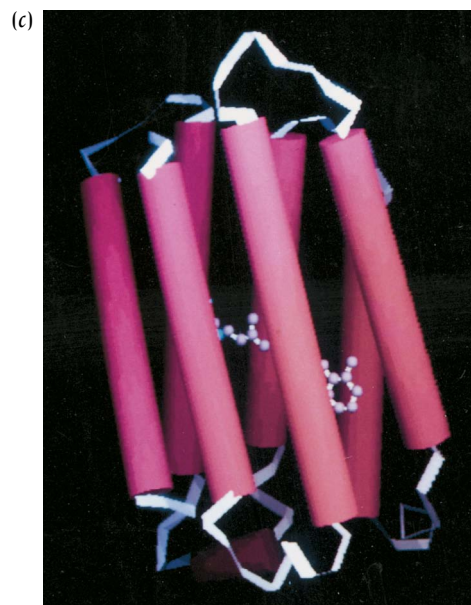
Given the difficulty of obtaining three-dimensional crystals of membrane proteins, it is not surprising that the electron microscope technique is now widely used to study large membrane-bound complexes such as the acetylcholine receptor, rhodopsin, ion pumps, gap junctions, water channels and light-harvesting complexes, which crystallize in two dimensions.

### *Bacteriorhodopsin contains seven transmembrane $\alpha$ helices*

The purple membrane of *Halobacterium halobium* contains ordered sheets of **bacteriorhodopsin**, a protein of 248 amino acid residues which binds retinal, the same photosensitive pigment that is used to capture light in our eyes. Bacteriorhodopsin uses the energy of light to pump protons across the membrane. Richard Henderson and Nigel Unwin at LMB, Cambridge, UK, pioneered high-resolution three-dimensional reconstruction of tilted low-dose electron microscopy images using such two-dimensional crystals. The 7-Å model of bacteriorhodopsin (Figure 12.3a) that they obtained in this way in 1975 provided the first glimpse of how membrane proteins are constructed. The fundamental observation that this protein has a number of **trans-membrane  $\alpha$  helices** (Figure 12.3b) has had a great impact on subsequent theories and experiments on membrane proteins; it also provided the first



**Figure 12.3** Two-dimensional crystals of the protein bacteriorhodopsin were used to pioneer three-dimensional high-resolution structure determination from electron micrographs. An electron density map to 7 Å resolution (a) was obtained and interpreted in terms of seven transmembrane helices (b). In 1990 the resolution was extended to 3 Å, which confirmed the presence of the seven  $\alpha$  helices (c). This structure also showed how these helices were connected by loop regions and where the retinal molecule was bound to bacteriorhodopsin. (Courtesy of R. Henderson.)



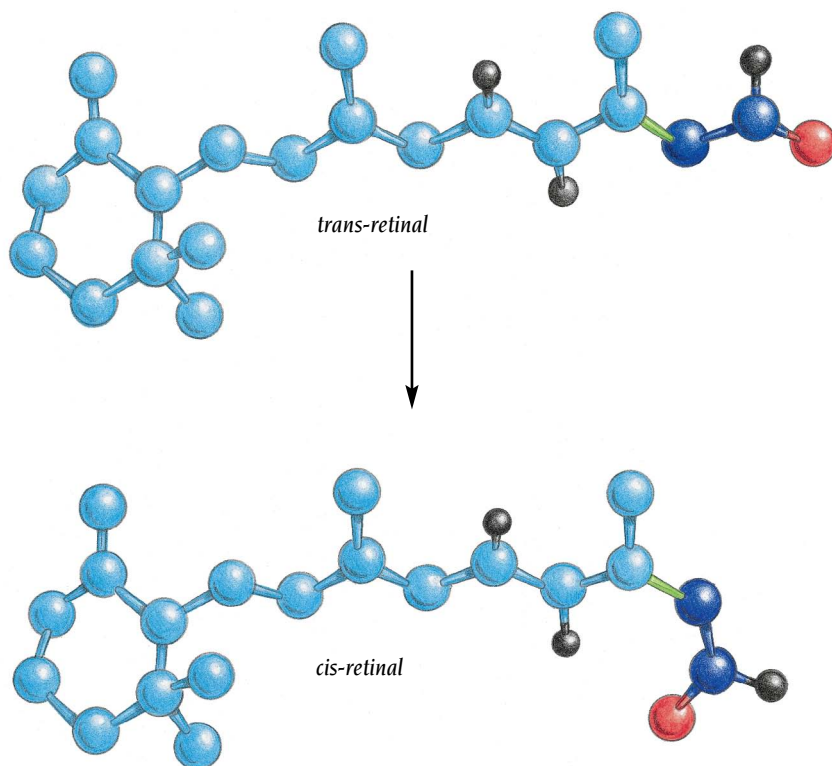
experimental evidence behind the now extensively used methods to predict transmembrane helices from amino acid sequences.

This electron microscopy reconstruction has since been extended to high resolution (3 Å) where the connections between the helices and the bound retinal molecule are visible together with the seven helices (Figure 12.3c). The helices are tilted by about 20° with respect to the plane of the membrane. This is the first example of a high-resolution three-dimensional protein structure determination using electron microscopy. The structure has subsequently been confirmed by x-ray crystallographic studies to 2 Å resolution.

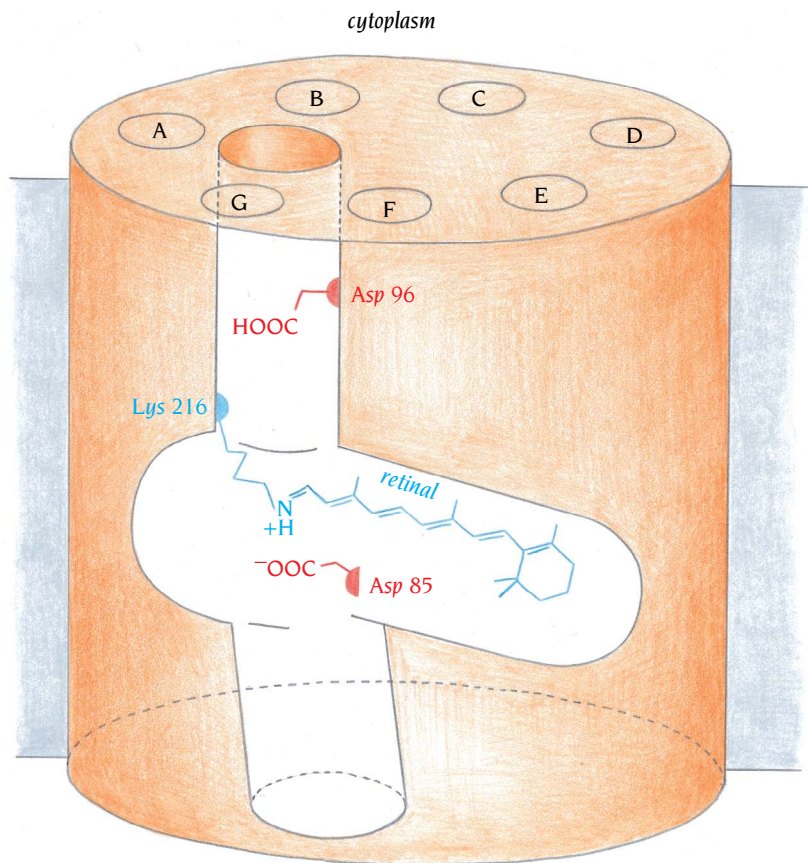
### *Bacteriorhodopsin is a light-driven proton pump*

*Halobacteria* have the simplest biological system for the conversion of light to chemical energy. Under conditions of low oxygen tension and intense illumination the cells synthesize bacteriorhodopsin. When the bound retinal absorbs a photon it undergoes an isomerization from *trans* to *cis* (Figure 12.4) and, as a consequence, protons are pumped from the cytosol to the extracellular space, creating a proton gradient. This gradient is used to generate ATP and to transport ions and molecules across the membrane. The mechanism by which bacteriorhodopsin acts as a **proton pump** has been studied by many biophysical methods over several decades and the results, in conjunction with Henderson's structural studies, have given the following simplified scheme for proton pumping.

Retinal is bound in a pocket of bacteriorhodopsin about equidistant from the two sides of the membrane (Figure 12.5). The pigment forms a Schiff base with a lysine residue, Lys 216; in other words, it is covalently linked to the nitrogen atom of the lysine side chain that is protonated and therefore has a positive charge (see Figure 12.5). This positive charge is positioned in a channel of the protein that extends across the membrane and through which protons are pumped from the cytosolic to the extracellular side. The channel is narrow on the cytosolic side and lined with hydrophobic residues with the exception of Asp 96, which has been shown by studies of mutant proteins to be essential for proton pumping. In contrast, the channel is wide and hydrophilic on the extracellular side and includes a second essential acidic residue, Asp 85.



**Figure 12.4** The light-absorbing pigment retinal undergoes a conformational change called isomerization, when it absorbs light. One part of the molecule (dark blue and red) rotates 180° around a double bond between two carbon atoms (green). The geometry of the molecule is changed by this rotation from a *trans* form to a *cis* form. Carbon atoms are blue, hydrogen atoms gray and the oxygen atom red.



**Figure 12.5** Schematic diagram of the bacteriorhodopsin molecule illustrating the relation between the proton channel and bound retinal in its *trans* form. A to E are the seven transmembrane helices. Retinal is covalently bound to a lysine residue. The relative positions of two Asp residues, which are important for proton transfer, are also shown. (Adapted from R. Henderson et al., *J. Mol. Biol.* 213: 899–929, 1990.)

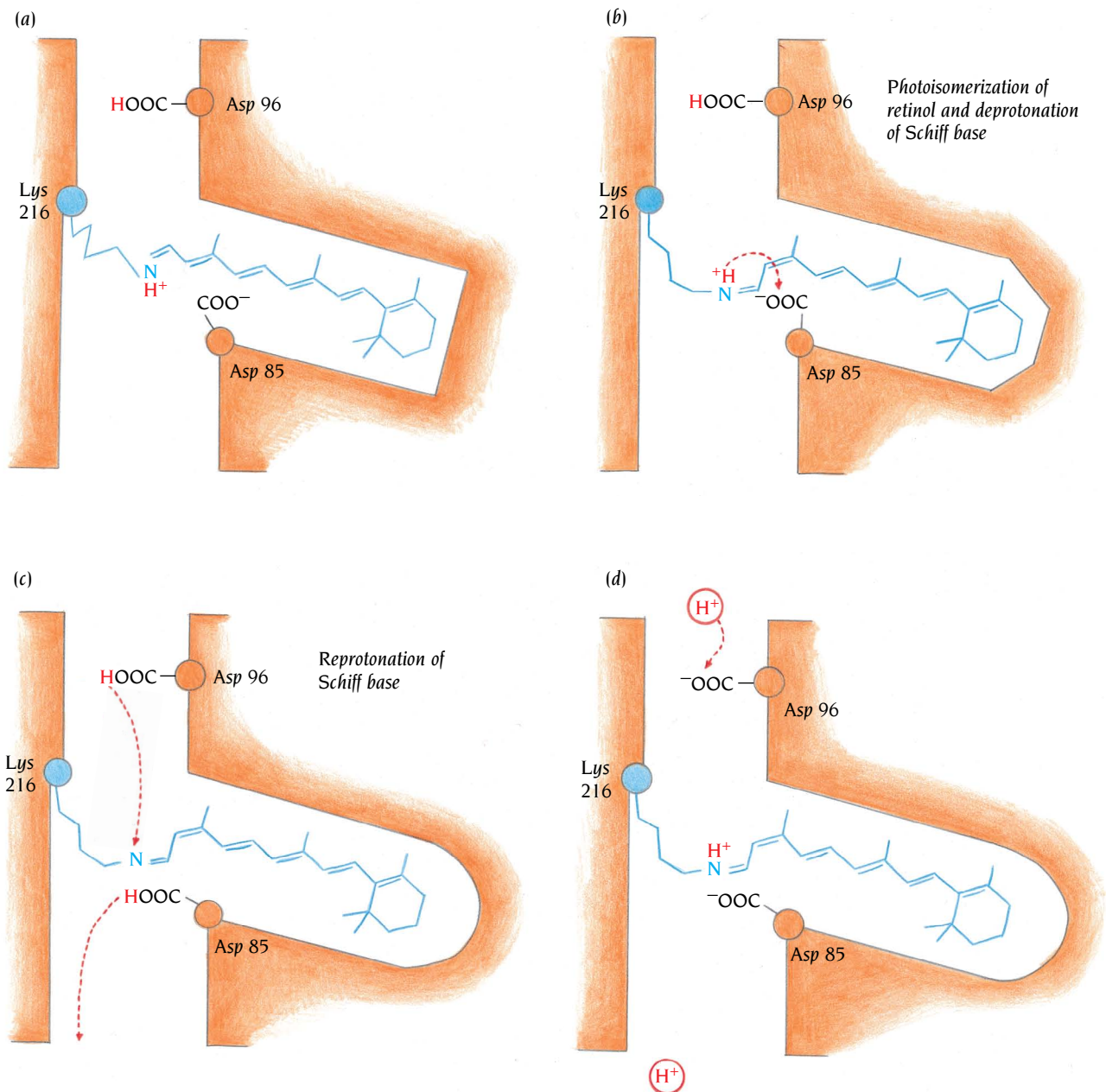
In the unisomerized *trans* state of retinal, Asp 85 is close to the positive charge of the Schiff base (Figure 12.6a). The structural change of the retinal molecule due to the *trans* to *cis* photoisomerization causes the Schiff base to change its position relative to Asp 85, which induces transfer of the Schiff base proton to the aspartate group (Figure 12.6b). Once the Schiff base–Asp 85 ion pair is converted to a neutral pair by this proton transfer the protein undergoes a conformational change from the T state to the relaxed R state (see Chapter 6). X-ray diffraction and electron microscopy studies have shown that this conformational change involves a reorganization of some of the transmembrane helices that bind retinal, with the consequence that the Schiff base is moved from the extracellular part of the channel to the cytoplasmic part, away from Asp 85 towards Asp 96. Asp 85 then delivers a proton through the hydrophilic part of the channel to the extracellular space and Asp 96 reprotonates the Schiff base, which subsequently reverts to the *trans* state and the protein changes its conformation back to the T-state ready for another cycle of photoisomerization-induced proton transfer (Figure 12.6c,d). In short, light causes a chemical change at the active site that alters the conformation of the protein, which in turn drives protons from the cytosolic side of the membrane to the extracellular side.

### Porins form transmembrane channels by $\beta$ strands

Gram-negative bacteria are surrounded by two membranes, an inner plasma membrane and an outer membrane. These are separated by a periplasmic space. Most plasma membrane proteins contain long, continuous sequences of about 20 hydrophobic residues that are typical of transmembrane  $\alpha$ helices such as those found in bacteriorhodopsin. In contrast, most outer membrane proteins do not show such sequence patterns.

This enigma was resolved in 1990 when the x-ray structure of an outer membrane protein, **porin**, showed that the transmembrane regions were  $\beta$





strands and not  $\alpha$  helices. Sequence comparisons have since shown that most if not all bacterial outer membrane porins have transmembrane  $\beta$  strands. Porins are among the most abundant proteins in bacteria. Each *Escherichia coli* cell contains about 100,000 copies of porin molecules in its outer membrane. Each porin forms an open water-filled channel that allows passive diffusion of nutrients and waste products across the outer membrane. These channels are restricted in size, and this excludes larger, potentially toxic compounds from entering the cell.

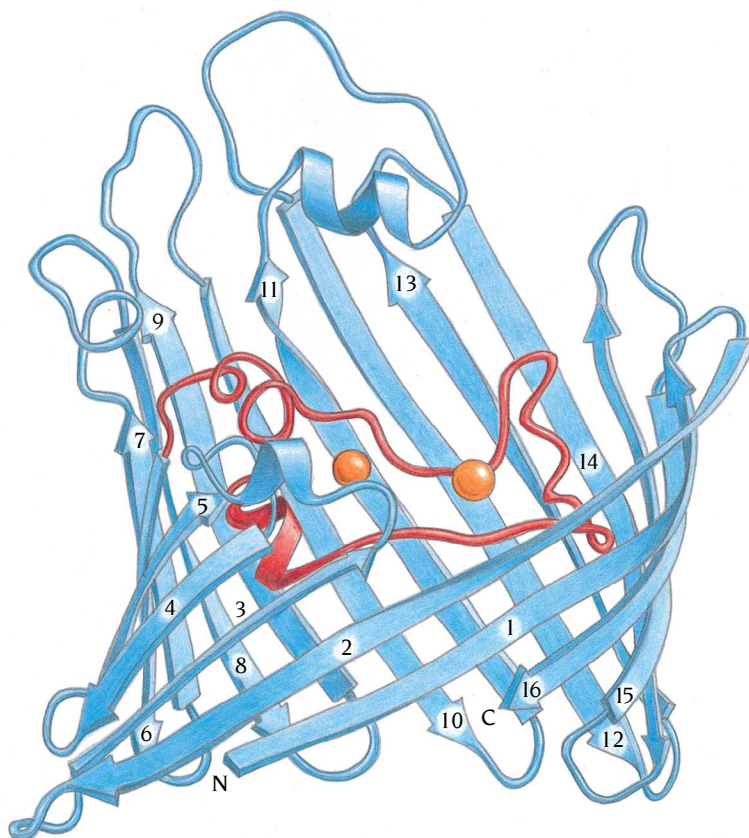
### Porin channels are made by up and down $\beta$ barrels

The first x-ray structure of a porin was determined by the group of Georg Schulz and Wolfram Welte at Freiburg University, Germany, who succeeded in growing crystals of a porin from *Rhodobacter capsulatus* that diffracted to 1.8 Å resolution. Since then the x-ray structures of several other porin molecules have been determined and found to be very similar to the *R. capsulatus* porin despite having no significant sequence identity.

Each subunit of the trimeric porin molecule from *R. capsulatus* folds into a 16-stranded up and down antiparallel  $\beta$  barrel in which all  $\beta$  strands form

**Figure 12.6** Schematic diagram illustrating the proton movements in the photocycle of bacteriorhodopsin. The protein adopts two main conformational states, tense (T) and relaxed (R). The T state binds *trans*-retinal tightly and the R state binds *cis*-retinal. (a) Structure of bacteriorhodopsin in the T state with *trans*-retinal bound to Lys 216 via a Schiff base. (b) A proton is transferred from the Schiff base to Asp 85 following isomerization of retinal and a conformational change of the protein. (c) Structure of bacteriorhodopsin in the R state with *cis*-retinal bound. A proton is transferred from Asp 96 to the Schiff base and from Asp 85 to the extracellular space. (d) A proton is transferred from the cytoplasm to Asp 96. (Adapted from R. Henderson et al., *J. Mol. Biol.* 213: 899–929, 1990.)





**Figure 12.7** Ribbon diagram of one subunit of porin from *Rhodobacter capsulatus* viewed from within the plane of the membrane. Sixteen  $\beta$  strands form an antiparallel  $\beta$  barrel that traverses the membrane. The loops at the top of the picture are extracellular whereas the short turns at the bottom face the periplasm. The long loop between  $\beta$  strands 5 and 6 (red) constricts the channel of the barrel. Two calcium atoms are shown as orange circles. (Adapted from M.S. Weiss and G.E. Schulz, *J. Mol. Biol.* 227: 493–509, 1992.)

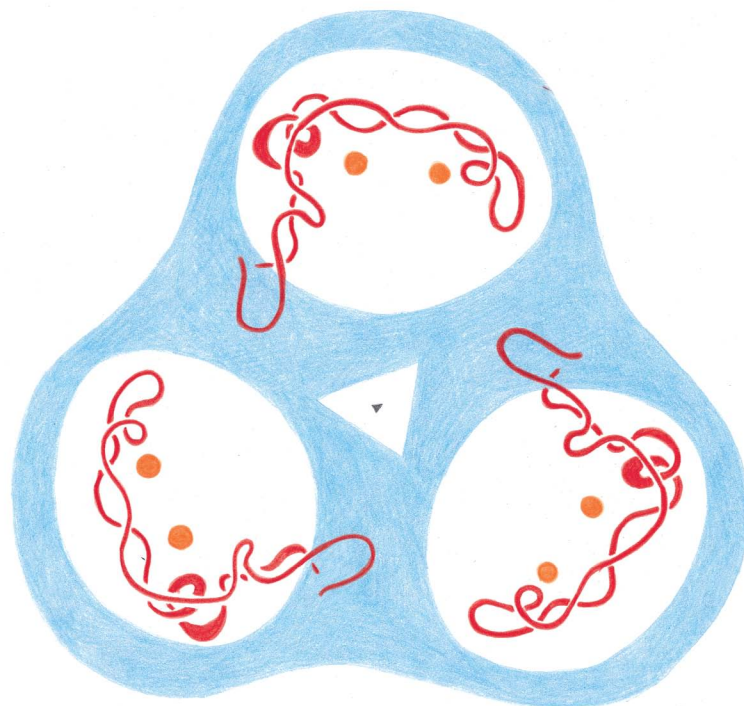
hydrogen bonds to their neighbors (Figure 12.7). All other known porin barrels also contain 16  $\beta$  strands except a maltoporin from *E. coli*, which contains 18  $\beta$  strands. In contrast to the  $\beta$  barrels we have previously discussed, which have a tightly packed hydrophobic core, the porin barrels contain a central channel because of the large number of  $\beta$  strands. The channel is, however, partially blocked by a long loop region between  $\beta$  strands 5 and 6 that projects into the channel. This arrangement creates a narrow region in the middle of the channel, called the eyelet, about 9 Å long and 8 Å in diameter, which defines the size of solute molecules that can traverse the channel (see Figure 12.8).

The eyelet is lined almost exclusively with positively and negatively charged groups that are arranged on opposite sides of the channel, causing a transversal electric field across the pore. One His, two Lys and three Arg residues form the positive side and four Glu and seven Asp residues are on the negative side. The large local surplus of negative charges is partially compensated by two bound calcium atoms. This asymmetric arrangement of charges no doubt contributes to the selection of molecules that can pass through the channel.

Since the outside of the barrel faces hydrophobic lipids of the membrane and the inside forms the solvent-exposed channel, one would expect the  $\beta$  strands to contain alternating hydrophobic and hydrophilic side chains. This requirement is not strict, however, because internal residues can be hydrophobic if they are in contact with hydrophobic residues from loop regions. The prediction of transmembrane  $\beta$  strands from amino acid sequences is therefore more difficult and less reliable than the prediction of transmembrane  $\alpha$  helices.

### *Each porin molecule has three channels*

The complete porin molecule is a stable trimer of three identical subunits, three each with a functional channel (Figure 12.8). About one-third of the



**Figure 12.8** Schematic diagram of the trimeric porin molecule viewed from the extracellular space. Blue regions illustrate the walls of the three porin barrels, the loop regions that constrict the channel are red and the calcium atoms are orange.

barrel's outer surface is involved in subunit interactions with the other two subunits, comprising polar interactions from loop regions and hydrophobic interactions from side chains of the  $\beta$ strands. The bottom part of the trimer facing the periplasmic space has a flat and smooth surface made up of the short loop regions at this end of the three  $\beta$ barrels (as shown in Figure 12.7). In contrast the upper part has long loop regions and is funnel-shaped, with the channels of the three barrels providing three outlets from the single funnel. The inner sides of the funnel are lined with hydrophilic residues that are in contact with solvent from the extracellular space.

The journey of an extracellular solute molecule through the channel may now be depicted in the following way. Large molecules are prevented from entering by the size of the funnel of the trimer. Further screening occurs at the entrance of the channel in the individual barrels. Finally, a molecule small enough to enter the central channel then encounters the eyelet, where the charged side chains determine the size limitations and the ion selectivity of the pore. After this narrow passage the molecule is effectively released into the periplasmic space. It should, however, be borne in mind that this picture describes only one state of the channel. Triggers such as an electric potential or a change in osmotic pressure can modify the structure of the channel and therefore its permeability, but as yet we have no structural information on such changes.

The outer surface of the trimeric porin molecule shows a pronounced partitioning with respect to hydrophobicity. Polar side chains of the loop regions are abundant at the top of the trimer, followed by a hydrophobic band with a width of 25 Å that encircles the molecule. This band presumably forms the region that is embedded in the core of the outer membrane, which has a thickness of about 25 Å. The top and the bottom of this hydrophobic band are largely composed of aromatic residues, Phe and Tyr, whereas the central region is composed of small to medium-sized aliphatic residues. This suggests that aromatic rings are energetically favored at the interface between the inner lipid part of the membrane and the hydrophilic regions facing the extracellular and periplasmic spaces. A similar distribution of aromatic and aliphatic residues is also present in other membrane proteins such as the photosynthetic reaction center and bacteriorhodopsin.

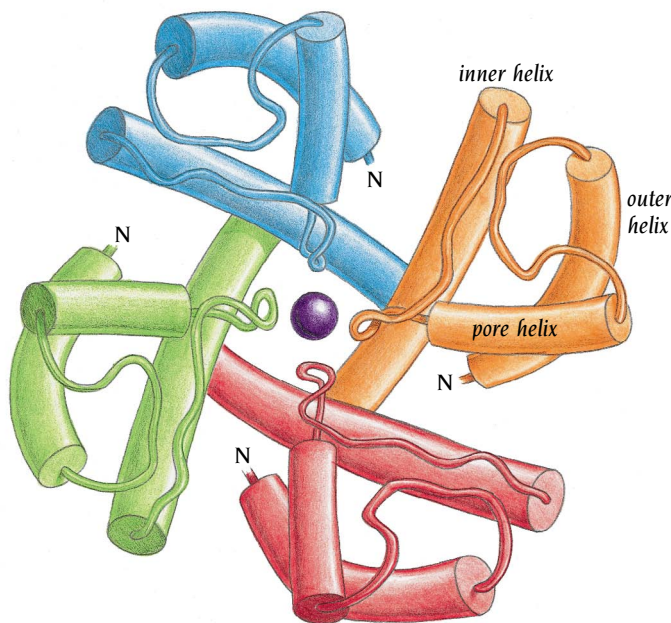
## ***Ion channels combine ion selectivity with high levels of ion conductance***

Outer membrane channel-forming proteins such as porins, which have relatively large and permissive pores, would have disastrous effects if they directly connected the inside of a cell to the extracellular space. Therefore, most channel proteins in the plasma membranes of plant and animal cells have narrow and highly selective pores that are concerned specifically with the transport of inorganic ions, and so are referred to as **ion channels**. The function of such channels is to allow specific inorganic ions, mainly  $K^+$ ,  $Na^+$ ,  $Ca^{2+}$  and  $Cl^-$  to diffuse rapidly across the lipid bilayer and therefore balance differences in electric charge between the two sides of the membrane, the membrane potential. The membrane potential of resting cells is determined largely by  $K^+$ , which is actively pumped into the cell by an ATP-driven  $Na^+$ ,  $K^+$  pump, but which can also move freely in or out through  **$K^+$  leak channels** in the plasma membrane.

The  $K^+$  leak channels are highly selective for  $K^+$  ions by a factor of 10,000 over  $Na^+$ . Two aspects of ion conduction by these  $K^+$  channels have intrigued biophysicists. First, what is the chemical basis for this selectivity since both  $K^+$  and  $Na^+$  are featureless spheres? Steric occlusion cannot account for the selectivity since  $Na^+$  is smaller than  $K^+$  (0.95 Å and 1.35 Å radii respectively). Second, how can  $K^+$  channels be so selective and at the same time exhibit a throughput of  $10^8$  ions per second, which approaches the diffusion-limited rate? The selectivity implies strong interactions between  $K^+$  and the pore and intuitively one would assume that the off velocity of  $K^+$  binding would be low, and consequently the rate of release would be low. A recent x-ray structure determination of a  $K^+$  channel from the bacterium *Streptomyces lividans* by the group of Roderick MacKinnon at Rockefeller University, New York, has revealed the structural basis for combining ion selectivity with a high rate of ion conduction in these and other ion channels.

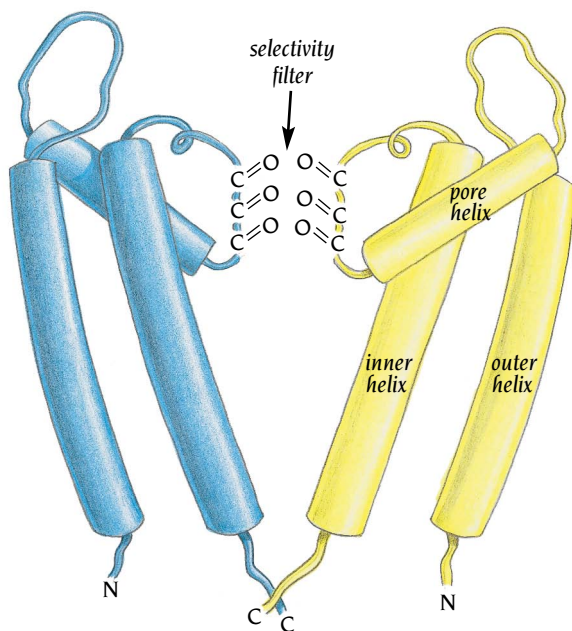
## ***The $K^+$ channel is a tetrameric molecule with one ion pore in the interface between the four subunits***

The polypeptide chain of the bacterial  $K^+$  channel comprises 158 residues folded into two transmembrane helices, a pore helix and a cytoplasmic tail of 33 residues that was removed before crystallization. Four subunits



**Figure 12.9** Schematic diagram of the structure of a potassium channel viewed perpendicular to the plane of the membrane. The molecule is tetrameric with a hole in the middle that forms the ion pore (purple). Each subunit forms two transmembrane helices, the inner and the outer helix. The pore helix and loop regions build up the ion pore in combination with the inner helix. (Adapted from S.A. Doyle et al., *Science* 280: 69–77, 1998.)





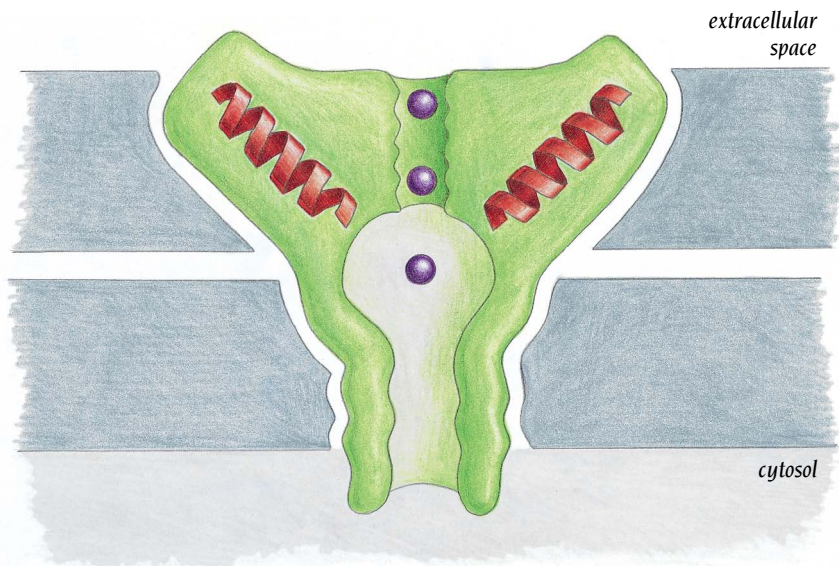
**Figure 12.10** Diagram showing two subunits of the  $K^+$  channel, illustrating the way the selectivity filter is formed. Main-chain atoms line the walls of this narrow passage with carbonyl oxygen atoms pointing into the pore, forming binding sites for  $K^+$  ions. (Adapted from D.A. Doyle et al., *Science* 280: 69–77, 1998.)

arranged around a central fourfold symmetry axis form the  $K^+$  channel molecule (Figure 12.9). The subunits pack together in such a way that there is a hole in the center which forms the ion pore through the membrane.

The C-terminal transmembrane helix, the inner helix, faces the central pore while the N-terminal helix, the outer helix, faces the lipid membrane. The four inner helices of the molecule are tilted and kinked so that the subunits open like petals of a flower towards the outside of the cell (Figure 12.10). The open petals house the region of the polypeptide chain between the two transmembrane helices. This segment of about 30 residues contains an additional helix, the pore helix, and loop regions which form the outer part of the ion channel. One of these loop regions with its counterparts from the three other subunits forms the narrow selectivity filter that is responsible for ion selectivity. The central and inner parts of the ion channel are lined by residues from the four inner helices.

### *The ion pore has a narrow ion selectivity filter*

The overall length of the ion pore is 45 Å and its diameter varies along its length (Figure 12.11). As expected for a  $K^+$  channel, there is a surplus of



**Figure 12.11** Schematic diagram of the ion pore of the  $K^+$  channel. From the cytosolic side the pore begins as a water-filled channel that opens up into a water-filled cavity near the middle of the membrane. A narrow passage, the selectivity filter, links this cavity to the external solution. Three potassium ions (purple spheres) bind in the pore. The pore helices (red) are oriented such that their carboxyl end (with a negative dipole moment) is oriented towards the center of the cavity to provide a compensating dipole charge to the  $K^+$  ions. (Adapted from D.A. Doyle et al., *Science* 280: 69–77, 1998.)



negative charges at both ends of the pore, which attract positively charged ions. From the cytosolic side, the pore begins as a channel 18 Å long, which opens into a wider cavity of about 10 Å diameter near the middle of the membrane. A narrow passage, the selectivity filter, links this cavity to the external solution. Main-chain atoms from all four subunits line the walls of this passage with carbonyl oxygen atoms pointing into the channel (see Figure 12.10). Three metal-binding sites have been identified in the ion pore (green in Figure 12.11), two within the selectivity filter and one in the cavity.

The structure of the selectivity filter has two essential features. First, the main-chain atoms create a stack of sequential oxygen rings along the passage, providing several closely spaced binding sites of the required dimensions for coordinating naked, dehydrated  $K^+$  ions. The  $K^+$  thus have only a small distance to diffuse from one site to the next within the selectivity filter. Second, the side chains of the residues that provide these binding sites point away from the channel and pack against the side chains from the pore helices. This packing firmly fixes the positions of the main-chain atoms, including the oxygen atoms that bind  $K^+$ . Since the side chains involved in these packing interactions are invariant in all known  $K^+$  channels it is reasonable to assume that the carbonyl oxygen atoms are fixed in positions with the correct dimensions to provide strong binding sites for  $K^+$ . The resolution of the structure determination is, however, too low to establish details of these binding sites.

On the basis of the structure, MacKinnon has suggested a plausible mechanism for the ion selectivity and conductivity of the channel. When an ion, which in solution has a water hydration shell, enters the selectivity filter it dehydrates. Binding to the carbonyl oxygen atoms in the filter compensates the energetic cost of dehydration. The dimensions of the binding sites are such that a  $K^+$  ion fits in the filter precisely so that the energetic costs and gains are well balanced, but the firm packing of the side chains prevents the carbonyl oxygen atoms from approaching close enough to compensate for the cost of dehydration of a  $Na^+$  ion.

In the crystal, the selectivity filter has two  $K^+$  ions, one bound at each end of the filter, separated by a distance of 7.5 Å. This is the same distance as the average distance between  $K^+$  ions in 4 M KCl. There is thus a high local concentration of  $K^+$  ions in the filter, implying that the filter attracts and concentrates  $K^+$  ions. However, since there are no negative ions within the filter to balance these positive charges there is also a repulsive force between the two  $K^+$  ions. Since in the crystal there is no concentration gradient of  $K^+$  ions across the channel there is no net flow of the ions. When, however, channel molecules are embedded in cell membranes across which there is a  $K^+$  ion concentration gradient, the ions flow through the channel. They are forced through by a combination of the mutual repulsion of the ions in the channel and the membrane potential. Within the filter the ions cascade from one carbonyl atom to the next.

All  $K^+$  channels are tetrameric molecules. There are two closely related varieties of subunits for  $K^+$  channels, those containing two membrane-spanning helices and those containing six. However, residues that build up the ion channel, including the pore helix and the inner helix, show a strong sequence similarity among all  $K^+$  channels. Consequently, the structural features and the mechanism for ion selectivity and conductance described for the bacterial  $K^+$  channel in all probability also apply for  $K^+$  channels in plant and animal cells.

### *The bacterial photosynthetic reaction center is built up from four different polypeptide chains and many pigments*

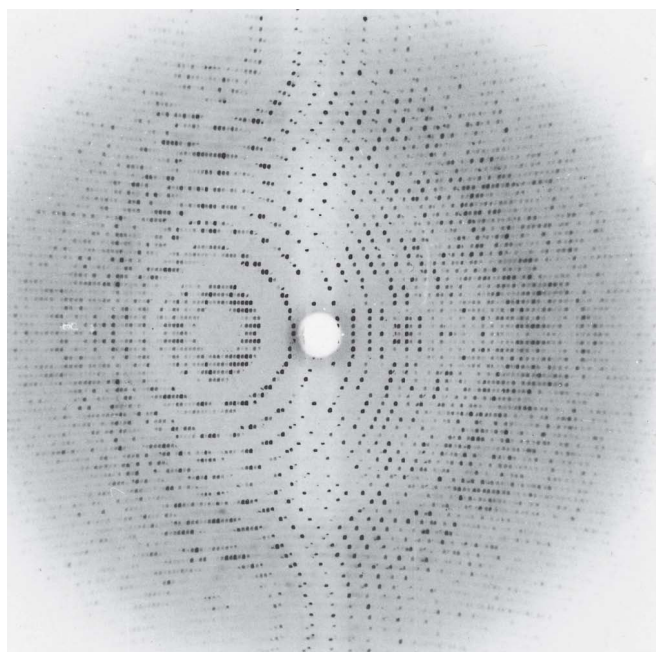
The crystallographic world was stunned when at a meeting in Erice, Sicily, in 1982, Hartmut Michel of the Max-Planck Institute in Martinsried, Germany, displayed the x-ray diagram shown in Figure 12.12. Not only was this the first x-ray picture to high resolution of a membrane protein, but the crystal was

formed not from a small protein of trivial function but from a large complex of polypeptide chains that represents a class of proteins having a function of central importance for life on earth. The protein complex was a **photosynthetic reaction center** from the photosynthetic purple bacterium *Rhodospseudomonas viridis*, which converts the energy of captured sunlight into electrical and chemical energy in the first steps of photosynthesis by pumping protons from one side of a membrane to the other. The structure has subsequently been resolved to 2.5 Å by H. Michel in collaboration with Hans Deisenhofer and Robert Huber at the same institute.

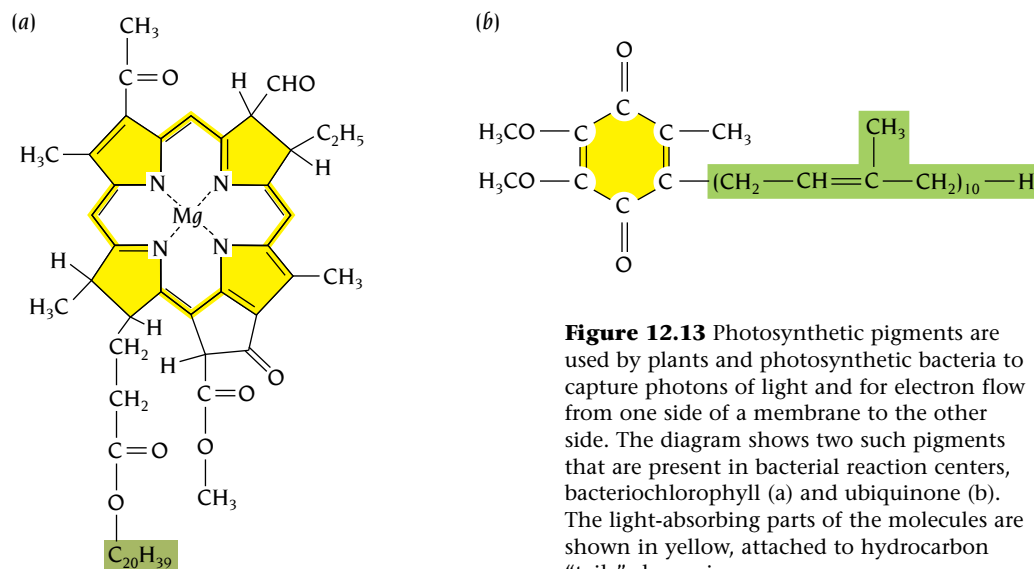
The interiors of rhodospseudomonad bacteria are filled with photosynthetic vesicles, which are hollow, membrane-enveloped spheres. The photosynthetic reaction centers are embedded in the membrane of these vesicles. One end of the protein complex faces the inside of the vesicle, which is known as the periplasmic side; the other end faces the cytoplasm of the cell. Around each reaction center there are about 100 small membrane proteins, the antenna pigment protein molecules, which will be described later in this chapter. Each of these contains several bound chlorophyll molecules that catch photons over a wide area and funnel them to the reaction center. By this arrangement the reaction center can utilize about 300 times more photons than those that directly strike the special pair of chlorophyll molecules at the heart of the reaction center.

The reaction center is built up from four polypeptide chains, three of which are called L, M, and H because they were thought to have light, medium, and heavy molecular masses as deduced from their electrophoretic mobility on SDS-PAGE. Subsequent amino acid sequence determinations showed, however, that the H chain is in fact the smallest with 258 amino acids, followed by the L chain with 273 amino acids. The M chain is the largest polypeptide with 323 amino acids. This discrepancy between apparent relative masses and real molecular weights illustrates the uncertainty in deducing molecular masses of membrane-bound proteins from their mobility in electrophoretic gels.

The L and M subunits show about 25% sequence identity and are therefore homologous and evolutionarily related proteins. The H subunit, on the other hand, has a completely different sequence. The fourth subunit of the reaction center is a cytochrome that has 336 amino acids with a sequence that is not similar to any other known cytochrome sequence.



**Figure 12.12** X-ray diffraction pattern from crystals of a membrane-bound protein, the bacterial photosynthetic reaction center. (Courtesy of H. Michel.)



**Figure 12.13** Photosynthetic pigments are used by plants and photosynthetic bacteria to capture photons of light and for electron flow from one side of a membrane to the other side. The diagram shows two such pigments that are present in bacterial reaction centers, bacteriochlorophyll (a) and ubiquinone (b). The light-absorbing parts of the molecules are shown in yellow, attached to hydrocarbon “tails” shown in green.

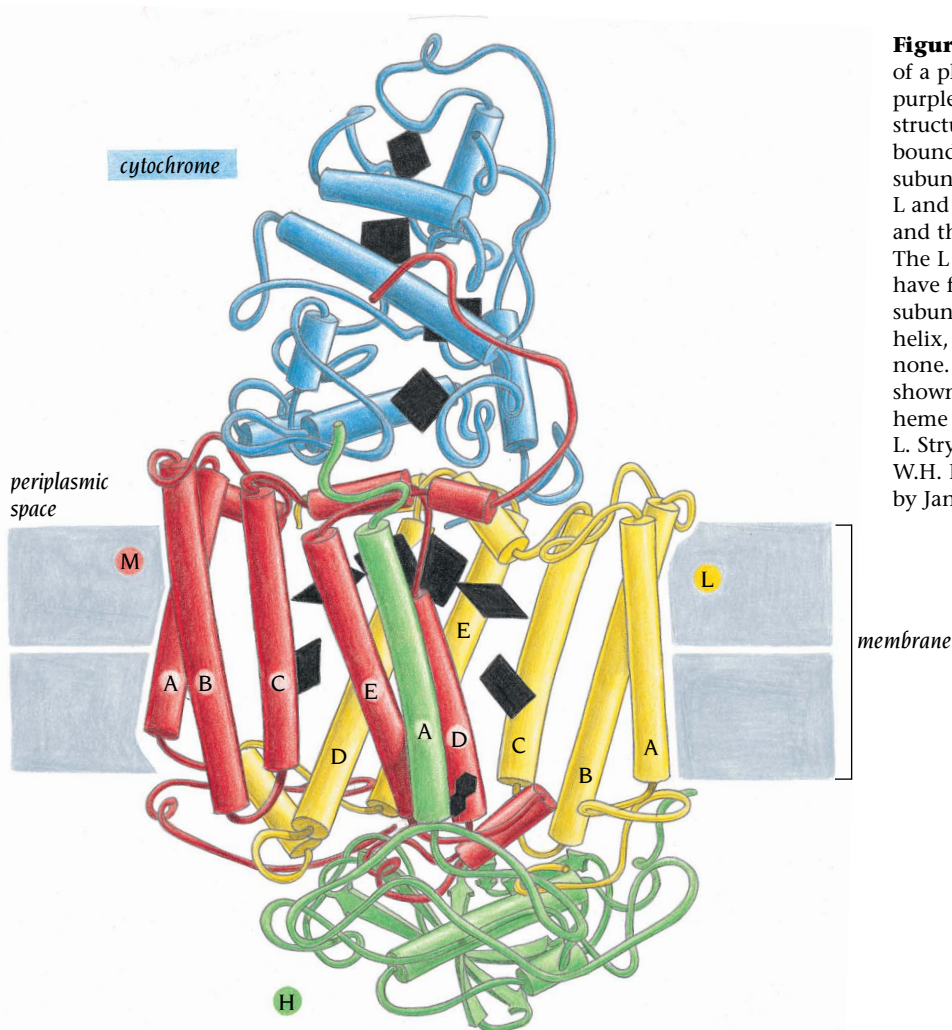
In addition to these polypeptide chains, the reaction center contains a number of pigments. There are four bacteriochlorophyll molecules (Figure 12.13a), two of which form the strongly interacting dimer called “the special pair.” Furthermore, there is one Fe atom, a carotenoid, two quinone molecules (Figure 12.13b) and two bacteriopheophytin molecules, which are chlorophyll molecules without the central  $Mg^{2+}$  atom. Finally, the cytochrome subunit has four bound heme groups. The crystal structure shows how the polypeptide chains bind these pigments into a functional unit allowing electrons to flow from one side of the membrane to the other.

### The L, M, and H subunits have transmembrane $\alpha$ helices

The L and the M subunits are firmly anchored in the membrane, each by five hydrophobic transmembrane  $\alpha$  helices (yellow and red, respectively, in Figure 12.14). The structures of the L and M subunits are quite similar as expected from their sequence similarity; they differ only in some of the loop regions. These loops, which connect the membrane-spanning helices, form rather flat hydrophilic regions on either side of the membrane to provide interaction areas with the H subunit (green in Figure 12.14) on the cytoplasmic side and with the cytochrome (blue in Figure 12.14) on the periplasmic side. The H subunit, in addition, has one transmembrane  $\alpha$  helix at the carboxy terminus of its polypeptide chain. The carboxy end of this chain is therefore on the same side of the membrane as the cytochrome. In total, eleven transmembrane  $\alpha$  helices attach the L, M, and H subunits to the membrane.

No region of the cytochrome penetrates the membrane; nevertheless, the cytochrome subunit is an integral part of this reaction center complex, held through protein–protein interactions similar to those in soluble globular multisubunit proteins. The protein–protein interactions that bind cytochrome in the reaction center of *Rhodospseudomonas viridis* are strong enough to survive the purification procedure. However, when the reaction center of *Rhodobacter sphaeroides* is isolated, the cytochrome is lost, even though the structures of the L, M, and H subunits are very similar in the two species.

Alpha helices D and E from the L and M subunits (Figure 12.14) form the core of the membrane-spanning part of the complex. These four helices are tightly packed against each other in a way quite similar to the four-helix bundle motif in water-soluble proteins. Each of these four helices provides a histidine side chain as ligand to the Fe atom, which is located between the helices close to the cytoplasm. The role of the Fe atom is probably to



**Figure 12.14** The three-dimensional structure of a photosynthetic reaction center of a purple bacterium was the first high-resolution structure to be obtained from a membrane-bound protein. The molecule contains four subunits: L, M, H, and a cytochrome. Subunits L and M bind the photosynthetic pigments, and the cytochrome binds four heme groups. The L (yellow) and the M (red) subunits each have five transmembrane  $\alpha$  helices A–E. The H subunit (green) has one such transmembrane helix, AH, and the cytochrome (blue) has none. Approximate membrane boundaries are shown. The photosynthetic pigments and the heme groups appear in black. (Adapted from L. Stryer, *Biochemistry*, 3rd ed. New York: W.H. Freeman, 1988, after a drawing provided by Jane Richardson.)

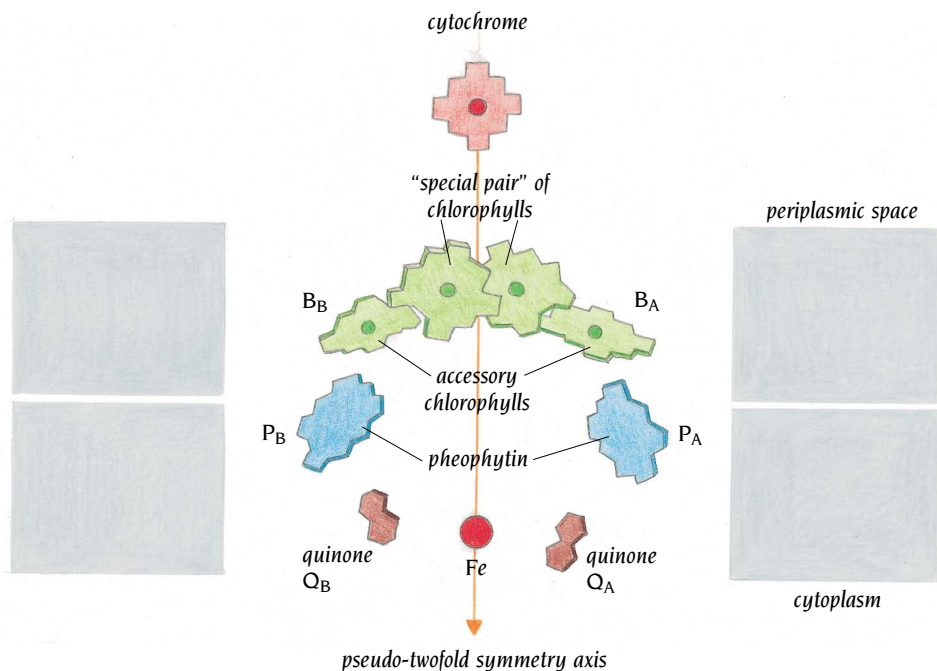
stabilize the structure of the four-helix bundle. Since its removal does not change the rate of electron flow through the system, the Fe atom cannot have a functional role in electron transfer. The remaining three helices of each subunit are arranged around the core in a manner that is not found in water-soluble proteins. Presumably, their positions are at least partly determined by the positions of the loop regions outside the membrane and not by close packing of the  $\alpha$  helices inside the membrane. It is interesting that none of the  $\alpha$  helices are perpendicular to the assumed plane of the membrane; instead, they are all tilted at angles of about  $20^\circ$  to  $25^\circ$ , similar to the tilt of the transmembrane helices in bacteriorhodopsin (see Figure 12.3) and other transmembrane proteins.

### *The photosynthetic pigments are bound to the L and M subunits*

The structurally similar L and M subunits are related by a pseudo-twofold symmetry axis through the core, between the helices of the four-helix bundle motif. The photosynthetic pigments are bound to these subunits, most of them to the transmembrane helices, and they are also related by the same twofold symmetry axis (Figure 12.15). The pigments are arranged so that they form two possible pathways for electron transfer across the membrane, one on each side of the symmetry axis.

This symmetry is important in bringing the two chlorophyll molecules of the “special pair” into close contact, giving them their unique function in initiating electron transfer. They are bound in a hydrophobic pocket close to the symmetry axis between the D and E transmembrane  $\alpha$  helices of both



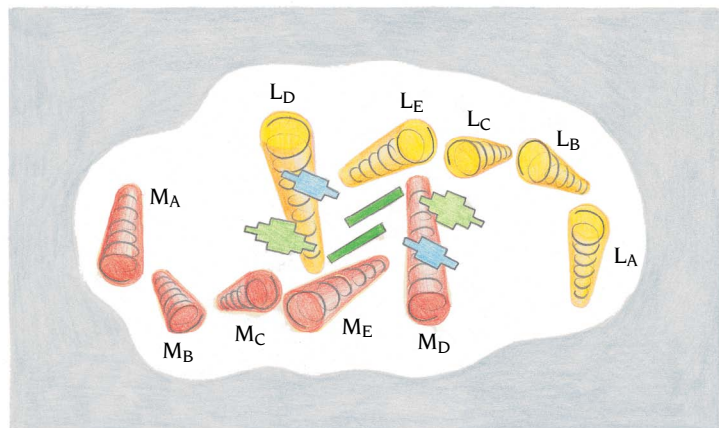


**Figure 12.15** Schematic arrangement of the photosynthetic pigments in the reaction center of *Rhodospseudomonas viridis*. The twofold symmetry axis that relates the L and the M subunits is aligned vertically in the plane of the paper. Electron transfer proceeds preferentially along the branch to the right. The periplasmic side of the membrane is near the top, and the cytoplasmic side is near the bottom of the structure. (From B. Furugren, courtesy of the Royal Swedish Academy of Science.)

subunits. The pyrrol ring systems of these chlorophyll molecules are parallel and interact closely with each other; in particular, two pyrrol rings, one from each molecule, are stacked on top of each other, 3 Å apart (Figure 12.15). The twofold symmetry axis passes between these stacked pyrrol rings.

This pair of chlorophyll molecules, which as we shall see accepts photons and thereby excites electrons, is close to the membrane surface on the periplasmic side. At the other side of the membrane the symmetry axis passes through the Fe atom. The remaining pigments are symmetrically arranged on each side of the symmetry axis (Figure 12.15). Two bacteriochlorophyll molecules, the accessory chlorophylls, make hydrophobic contacts with the special pair of chlorophylls on one side and with the pheophytin molecules on the other side. Both the accessory chlorophyll molecules and the pheophytin molecules are bound between transmembrane helices from both subunits in pockets lined by hydrophobic residues from the transmembrane helices (Figure 12.16).

The functional reaction center contains two quinone molecules. One of these,  $Q_B$  (Figure 12.15), is loosely bound and can be lost during purification. The reason for the difference in the strength of binding between  $Q_A$  and  $Q_B$  is unknown, but as we will see later, it probably reflects a functional asymmetry in the molecule as a whole.  $Q_A$  is positioned between the Fe atom and one of the pheophytin molecules (Figure 12.15). The polar-head group is outside the membrane, bound to a loop region, whereas the hydrophobic tail is



**Figure 12.16** View of the reaction center perpendicular to the membrane illustrating that the pigments are bound between the transmembrane helices. The five transmembrane-spanning  $\alpha$  helices of the L (yellow) and the M (red) subunits are shown as well as the chlorophyll (green) and pheophytin (blue) molecules.

bound to the pheophytin molecule and to hydrophobic side chains of transmembrane helices of the L subunit. If  $Q_B$  is lost during purification prior to crystallization, the corresponding binding site on the other side of the pseudotwofold axis is empty. Certain weedkillers that are inhibitors of photosynthesis in plants can bind in the crystal to this empty binding site.

### *Reaction centers convert light energy into electrical energy by electron flow through the membrane*

In photosynthesis light energy is converted to electrical energy by an electron flow that causes the separation of negatively and positively charged molecules. Many molecules can absorb photons and use the energy of this process to donate an electron to a nearby electron acceptor. The electron donor then becomes positively charged and the electron acceptor negatively charged. In most cases, however, the transfer of electrons back from the acceptor to the donor is as fast as the forward reaction and the absorbed energy is lost, usually as fluorescent radiation. The arrangement of photoreaction centers in both bacteria and green plants results in a very fast forward reaction and a slow back reaction; therefore, the electric charges induced by the absorbed light energy stay separated. This separation of charge represents a storage of energy because energy would be released if the charges were able to come together. This is the basic primary process of photosynthesis, the detailed mechanism of which we do not yet understand. However, by interpreting spectroscopic and genetic experiments in terms of the structure of the bacterial reaction center we may come to understand this process and be able to re-create this primary biological function—that of a solar-powered electrolytic battery.

In the bacterial reaction center the photons are absorbed by the special pair of chlorophyll molecules on the periplasmic side of the membrane (see Figure 12.14). Spectroscopic measurements have shown that when a photon is absorbed by the special pair of chlorophylls, an electron is moved from the special pair to one of the pheophytin molecules. The close association and the parallel orientation of the chlorophyll ring systems in the special pair facilitates the excitation of an electron so that it is easily released. This process is very fast; it occurs within 2 picoseconds. From the pheophytin the electron moves to a molecule of quinone,  $Q_A$ , in a slower process that takes about 200 picoseconds. The electron then passes through the protein, to the second quinone molecule,  $Q_B$ . This is a comparatively slow process, taking about 100 microseconds.

There are two pheophytin molecules, one on each side of the twofold axis, that in principle could accept the electrons (see Figure 12.15). However, only one pathway, on the right side of the symmetry axis that is shown in Figure 12.15, is used for electron transfer. Electrons do not pass through the chain of pigments on the left side, which appear to have no role in the charge separation. The best guess as to why these pigments are present is that the L and M chains have evolved from an ancestral chain that formed symmetrical homodimers in which both pigment chains were utilized. Presumably, the present-day reaction centers are more efficient for charge separation than the ancestral homodimers.  $Q_A$  is most stable when excited by two electrons, and if each electron arrived randomly at  $Q_A$  and  $Q_B$  the energy stored in  $Q_A$  after absorption of one electron might be dissipated before a second electron was absorbed.

One apparent discrepancy between the spectroscopic data and the crystal structure is that no spectroscopic signal has been measured for participation of the accessory chlorophyll molecule  $B_A$  in the electron transfer process. However, as seen in Figure 12.15, this chlorophyll molecule is between the special pair and the pheophytin molecule and provides an obvious link for electron transfer in two steps from the special pair through  $B_A$  to the pheophytin. This discrepancy has prompted recent, very rapid measurements of the electron transfer steps, still without any signal from  $B_A$ . This means either

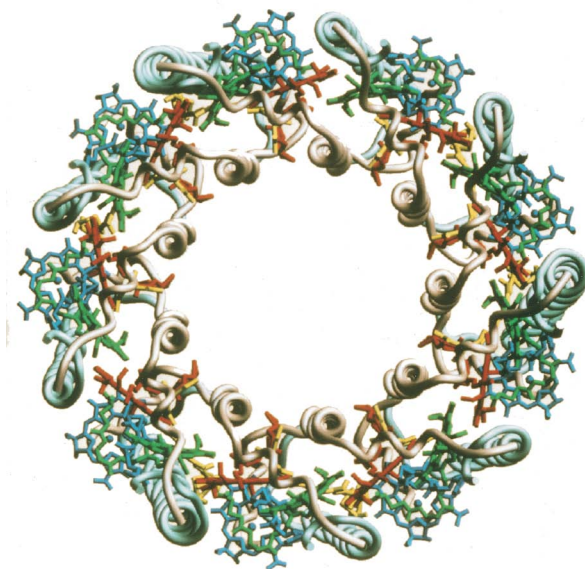
that the electron bypasses  $B_A$  and is transferred directly over the very long distance of about 25 Å from the special pair to pheophytin, or that the transfer through B is too rapid to detect with current technology, less than 0.01 picosecond. Neither of these conclusions is compatible with current theories for electron transfer. However, the components for electron transfer are embedded in the protein environment of subunits L and M with special properties that are not taken into account in the theories.

While this electron flow takes place, the cytochrome on the periplasmic side donates an electron to the special pair and thereby neutralizes it. Then the entire process occurs again: another photon strikes the special pair, and another electron travels the same route from the special pair on the periplasmic side of the membrane to the quinone,  $Q_B$ , on the cytosolic side, which now carries two extra electrons. This quinone is then released from the reaction center to participate in later stages of photosynthesis. The special pair is again neutralized by an electron from the cytochrome.

The charge separation that stores the energy of the photons is now complete: two positive charges have been left on the cytochrome side of the membrane, and two electrons have traveled through the membrane to a quinone molecule on the cytosolic side. In the photosynthetic reaction centers charge separation is remarkably efficient for capturing light energy. The forward electron transfer from the reaction center to  $Q_A$  is more than eight orders of magnitude faster than the back reaction. This large difference allows the reaction center to capture the energy of between 98 and 100% of the photons it absorbs. As a solar cell, it is extraordinarily efficient: the energy stored in separated charges is about half of the energy inherent in the photons. The rest of the energy is lost in other ways, some of which are the reactions that drive the electrons along the pathway of photosynthetic pigment molecules.

### **Antenna pigment proteins assemble into multimeric light-harvesting particles**

If the special pair of chlorophyll molecules of the reaction center was the only photon acceptor in the bacterial membrane, only a tiny fraction of the incoming sunlight would be captured and converted to chemical energy. All photosynthetic organisms have therefore evolved a system of **light-harvesting complexes**, which surrounds the reaction centers and increases the photon capturing area. The reaction centers receive practically all their light energy from such complexes. Detailed structural information is now available for the arrangement of the light-absorbing pigment molecules around the reaction center in photosynthetic bacteria. The pigments are firmly bound to small



**Figure 12.17** Computer-generated diagram of the structure of light-harvesting complex LH2 from *Rhodospseudomonas acidophila*. Nine  $\alpha$  chains (gray) and nine  $\beta$  chains (light blue) form two rings of transmembrane helices between which are bound nine carotenoids (yellow) and 27 bacteriochlorophyll molecules (red, green and dark blue). (Courtesy of M.Z. Papiz.)

hydrophobic protein molecules that are embedded in the membrane and which assemble into two types of multimeric complexes, called LH1 and LH2. The crystal structures of two LH2 complexes have been determined to high resolution, one from the purple bacterium *Rhodospseudomonas acidophila* by the group of Richard Cogdell and Neil Isaacs, Glasgow University, UK, and the other from *Rhodospirillum rubrum* by the group of Hartmut Michel at the Max-Planck Institute in Frankfurt, Germany. In addition, an 8.5-Å electron crystallography map of LH1 using two-dimensional crystals has given a broad outline of the structure of that complex. Strangely, plants seem to have evolved a totally different system of light-harvesting complexes. Werner Kühlbrandt at EMBL, Heidelberg, has determined the structure of a plant light-harvesting complex by electron crystallography and shown that the structure is quite different from those of the purple bacteria. We will here discuss only the bacterial system.

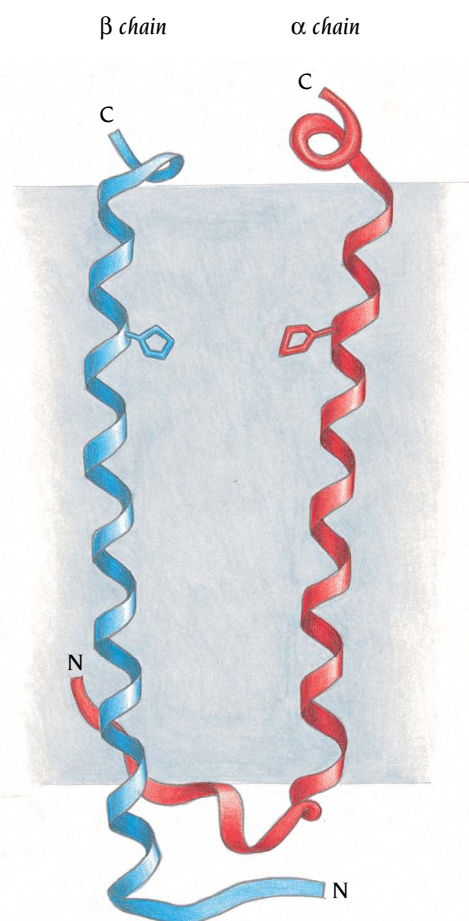
### Chlorophyll molecules form circular rings in the light-harvesting complex LH2

The structure of the LH2 complex of *R. acidophila* is both simple and elegant (Figure 12.17). It is a ring of nine identical units, each containing an  $\alpha$  and a  $\beta$  polypeptide of 53 and 41 residues, respectively, which both span the membrane once as  $\alpha$  helices (Figure 12.18). The two polypeptides bind a total of three chlorophyll molecules and two carotenoids. The nine heterodimeric units form a hollow cylinder with the  $\alpha$  chains forming the inner wall and the  $\beta$  chains the outer wall. The hole in the middle of the cylinder is empty, except for lipid molecules from the membrane.

Two of the chlorophyll molecules from each unit are in the space between the two walls and form a ring of 18 chlorophyll molecules near the periplasmic membrane surface (Figure 12.19a). The planar chlorophyll rings are oriented almost perpendicular to the plane of the membrane. Each magnesium atom of these chlorophyll molecules is bound to a histidine residue of either the  $\alpha$  or the  $\beta$  chain. The third chlorophyll molecule is bound between the  $\beta$  chains forming part of the outer wall and oriented parallel to the plane of the membrane (Figure 12.19b). The magnesium atom is bound to an oxygen atom of the formylated N-terminus of the  $\alpha$  chain and has a much more polar environment than the other two chlorophyll molecules. The third chlorophyll molecule therefore absorbs light at a considerably shorter wavelength than the other two molecules, with an absorption maximum at 800 nm compared with 850 nm. This arrangement allows the complex efficiently to capture a much broader energy band of the sunlight than would be possible if all the chlorophyll molecules had similar environments.

In contrast to bacteriorhodopsin or the reaction center, there is no direct contact within the membrane between the  $\alpha$  helices in this complex. The helices are held together through contacts mediated by the pigments and by contacts at the ends of the polypeptide chains outside the membrane.

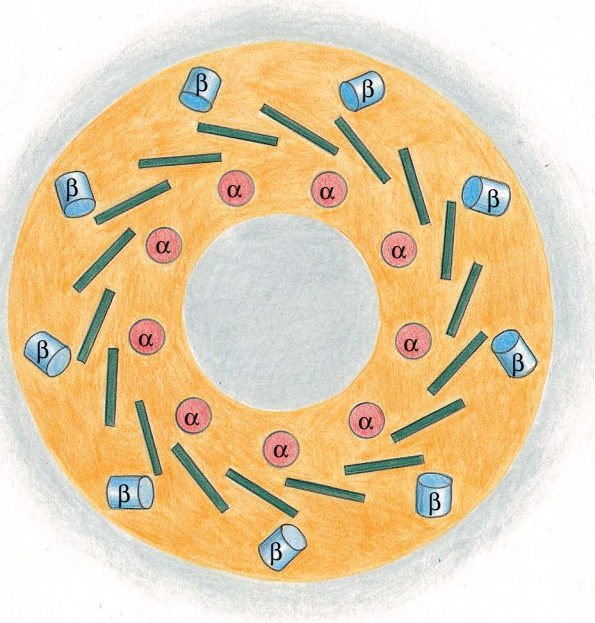
The LH2 complex of *R. rubrum* is very similar to that of *R. acidophila* except that the complex is built up into a ring of eight identical units instead of nine. In addition the third chlorophyll is coordinated to the O atom of an Asp residue of the  $\alpha$  chain. Since the  $\alpha$  and  $\beta$  chains of all bacterial light-harvesting complexes show some sequence similarity one can safely predict that they are all arranged in essentially the same way, except that the number of units forming the cylinder may differ. This prediction has been used to build a model of LH1 using the electron crystallographic map to 8.5 Å.



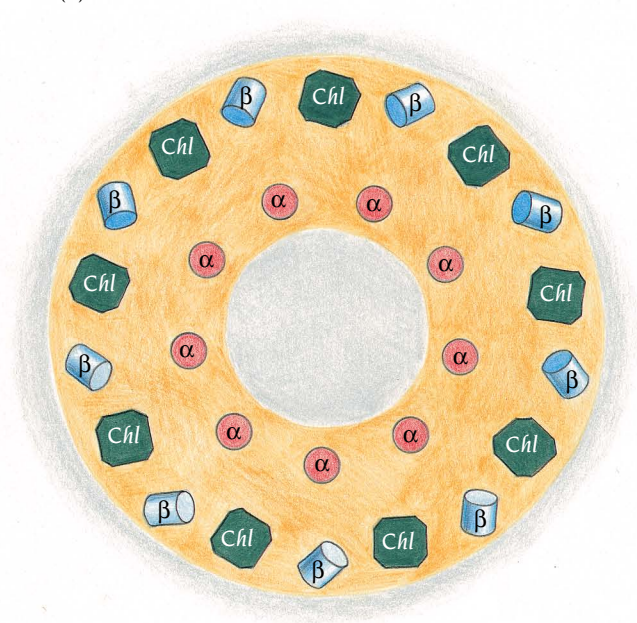
**Figure 12.18** Ribbon diagram showing the  $\alpha$  (red) and the  $\beta$  (blue) chains of the light-harvesting complex LH2. Each chain forms one transmembrane  $\alpha$  helix, which contains a histidine residue that binds to the Mg atom of one bacteriochlorophyll molecule. (Adapted from G. McDermott et al., *Nature* 374: 517–521, 1995.)



(a)



(b)



### *The reaction center is surrounded by a ring of 16 antenna proteins of the light-harvesting complex LH1*

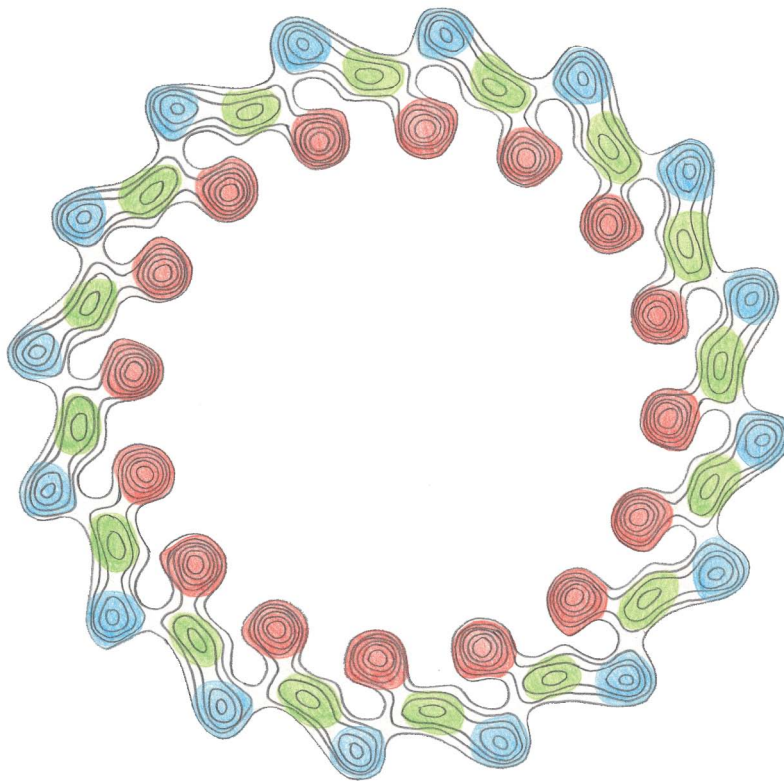
The light-harvesting complex LH1 is directly associated with the reaction center in purple bacteria and is therefore referred to as the core or inner antenna, whereas LH2 is known as the peripheral antenna. Both are built up from hydrophobic  $\alpha$  and  $\beta$  polypeptides of similar size and with low but significant sequence similarity. The two histidines that bind to chlorophyll with absorption maxima at 850 nm in the periplasmic ring of LH2 are also present in LH1, but the sequence involved in binding the third chlorophyll in LH2 is quite different in LH1. Not surprisingly, the chlorophyll molecules of the periplasmic ring are present in LH1 but the chlorophyll molecules with the 800 nm absorption maximum are absent.

The electron crystallographic map of LH1 at 8.5 Å resolution (Figure 12.20) shows a striking similarity in molecular design to LH2. The LH1 ring clearly consists of the same basic units as LH2. The  $\alpha$  and  $\beta$  polypeptides form an inner and an outer wall with the pigment molecules in between. However, the LH1 ring contains 16 rather than 9 units and has a hole 68 Å in diameter in the middle. By analogy with the structure of LH2 it can be assumed that LH1 has a ring of 32 chlorophyll molecules in the region between these two walls at the same distance from the periplasmic side of the membrane as in LH2. These chlorophyll molecules presumably have an even more hydrophobic environment than those in LH2 since they have an absorption maximum at 875 nm.

On the basis of these structural results it is now possible to derive the following schematic picture (Figure 12.21) of the photosynthetic apparatus in purple bacteria, and to begin to understand the design of this highly efficient apparatus for capturing photons from the sun and funneling them to the reaction center with minimum loss of energy.

Spectroscopic measurements show that the reaction center and LH1 are tightly associated and therefore it is assumed that the ring of pigments in LH1 surrounds the reaction center. Careful model building indicates that the hole in the middle of LH1 is large enough to accommodate the whole reaction center molecule. We do not know exactly how the LH2 complexes are arranged in the membrane around the LH1–reaction center complex, but at least some of them should be in contact with the outer rim of LH1 for efficient

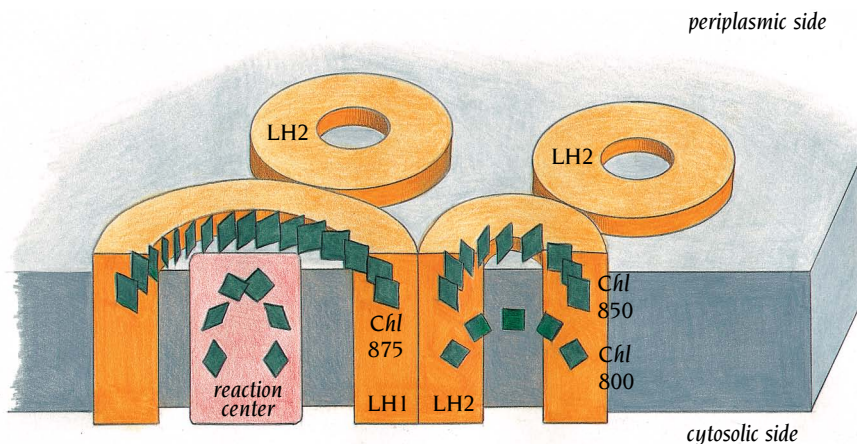
**Figure 12.19** Schematic diagrams illustrating the arrangement of bacteriochlorophyll molecules in the light-harvesting complex LH2, viewed from the periplasmic space. (a) Eighteen bacteriochlorophyll molecules (green) are bound between the two rings of  $\alpha$  (red) and  $\beta$  (blue) chains. The planes of these molecules are oriented perpendicular to the plane of the membrane and the molecules are bound close to the periplasmic space. (b) Nine bacteriochlorophyll molecules (green) are bound between the  $\beta$  chains (blue) with their planes oriented parallel to the plane of the membrane. These molecules are bound in the middle of the membrane.



**Figure 12.20** Electron density projection of the light-harvesting complex LH1 from *Rhodospirillum rubrum* determined by electron crystallography. On the basis of comparison with the LH2 complex the red regions can be interpreted as corresponding to the  $\alpha$  chains, the blue regions to the  $\beta$  chains and each green region to two bacteriochlorophyll molecules bound between the  $\alpha$  and the  $\beta$  chains. The ring of 16 units has a hole in the middle that is large enough to accommodate a complete reaction center molecule. (Adapted from Karrasch et al., *EMBO J.* 14: 631–638, 1995.)

energy transfer. There are 8 to 10 LH2 complexes for each reaction center and consequently around 300 energy-capturing chlorophyll molecules per reaction center.

In the photosynthetic membrane there is a downhill flow of energy from the light-harvesting proteins to the reaction centers. In purple bacteria, LH2 absorbs radiation at a shorter wavelength (higher energy) than LH1 and therefore delivers it to LH1, which in turn passes it on to the reaction center. A photon that is absorbed by the 800-nm chlorophylls in LH2 is rapidly transmitted to the energetically lower periplasmic ring of 850-nm chlorophyll in the same complex. Spectroscopic measurements have shown that the energy absorbed by a chlorophyll in the periplasmic rings spreads to the others, within 0.2 to 0.3 picoseconds. When a photon is absorbed by any of these chlorophylls it becomes in effect delocalized between the chlorophyll molecules of the ring. It can then easily jump to another chlorophyll in an adjacent complex, where it again becomes delocalized until it ends up at the reaction center, as schematically illustrated in Figure 12.22.



**Figure 12.21** Schematic diagram of the relative positions of bacteriochlorophylls (green) in the photosynthetic membrane complexes LH1, LH2, and the reaction center. The special pair of bacteriochlorophyll molecules in the reaction center is located at the same level within the membrane as the periplasmic bacteriochlorophyll molecules Chl 875 in LH1 and the Chl 850 in LH2. (Adapted from W. Kühlbrandt, *Structure* 3: 521–525, 1995.)